

(12) **United States Patent**
Topčagić et al.

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(54) **SURGE PROTECTIVE DEVICES**

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Related U.S. Application Data

(60) Provisional application No. 63/493,026, filed on Mar. 30, 2023, provisional application No. 63/375,588, filed on Sep. 14, 2022, provisional application No. 63/375,591, filed on Sep. 14, 2022.

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H01T 4/14 (2006.01)
H02H 9/06 (2006.01)
H01T 4/04 (2006.01)

(52) **U.S. Cl.**
CPC **H02H 9/06** (2013.01); **H01T 4/14** (2013.01); **H01T 4/04** (2013.01)

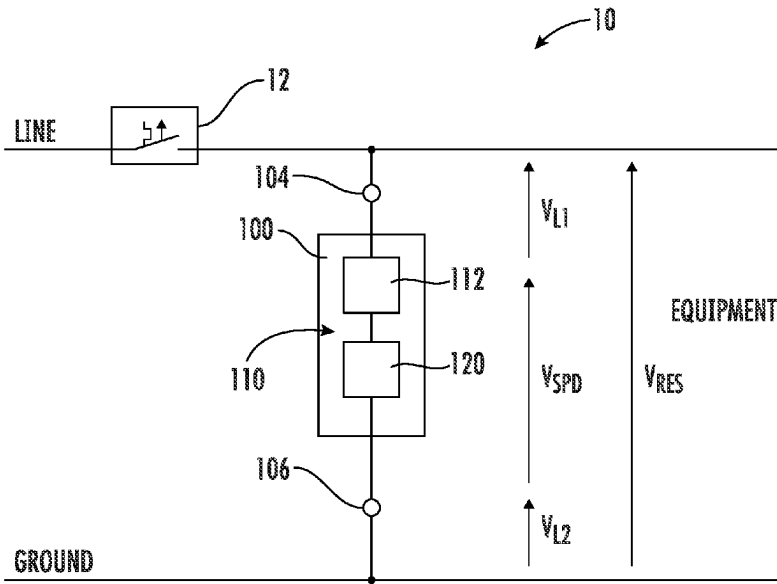
(58) **Field of Classification Search**
CPC H01T 4/14; H01T 4/10; H01T 4/16; H01T 4/06; H01T 2/02; H01T 1/15

See application file for complete search history.

(57) **ABSTRACT**

A surge protective device includes a first electrical terminal, a second electrical terminal, and an overvoltage protection circuit connected between the first electrical terminal and the second electrical terminal. The overvoltage protection circuit includes a spark gap assembly between the first electrical terminal and the second electrical terminal. The spark gap assembly includes a first spark gap (SG) electrode and a second SG electrode defining a spark gap therebetween, and a trigger circuit operative to ignite a main electric arc between the first and second SG electrodes across the spark gap. The trigger circuit includes a groove defined in the second SG electrode, and a trigger member disposed in the groove. The trigger member is operative to assist formation of a trigger arc.

24 Claims, 39 Drawing Sheets



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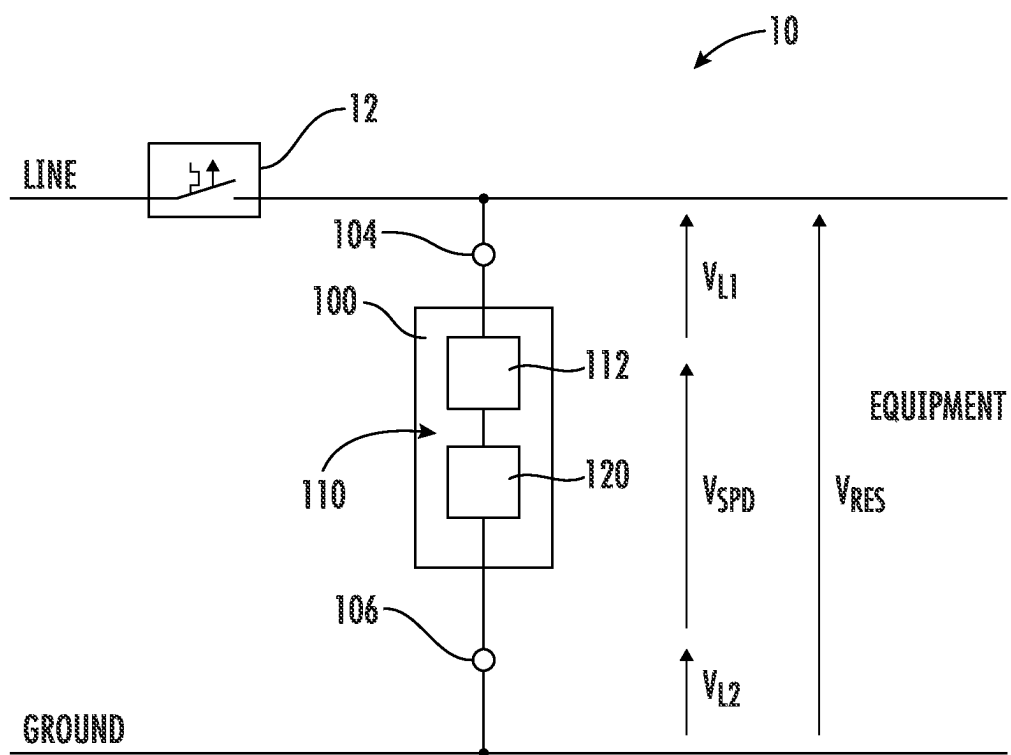


FIG. 1

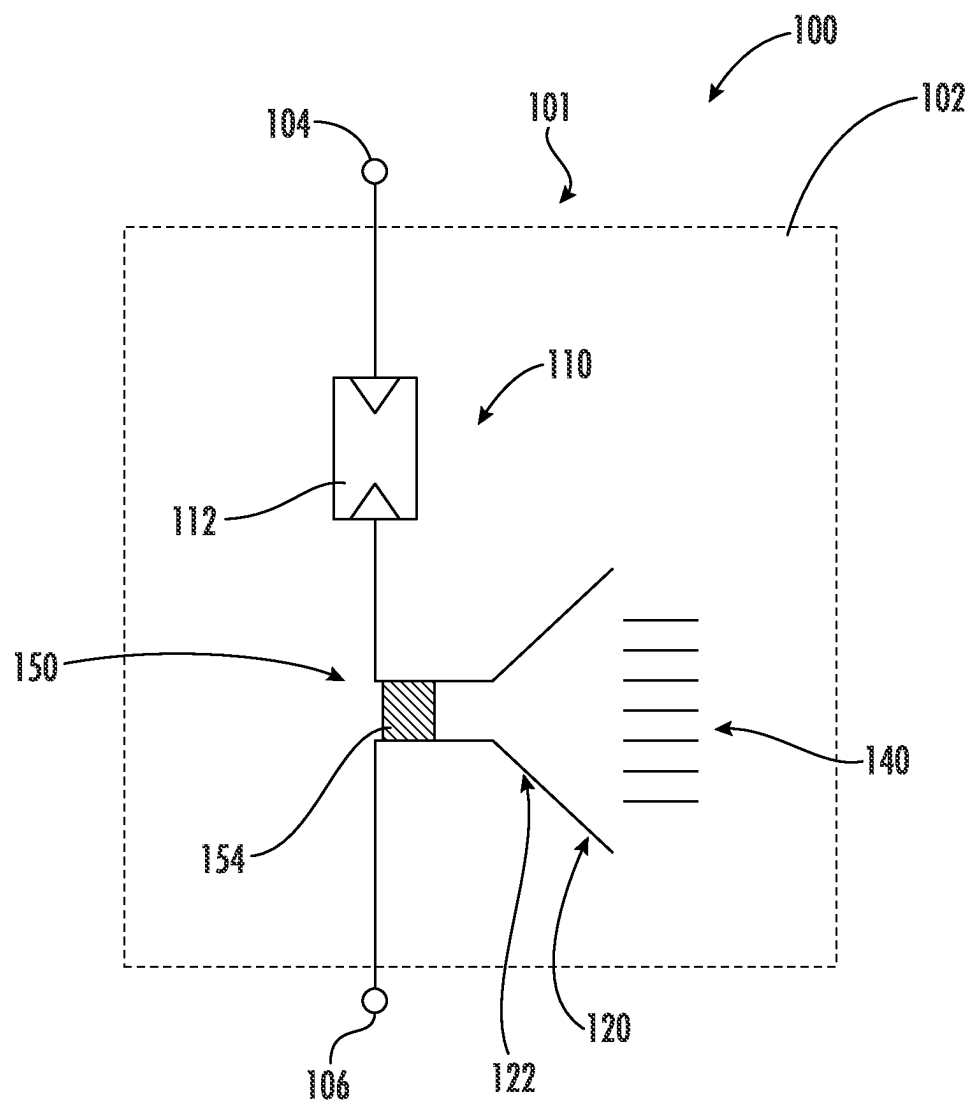


FIG. 2

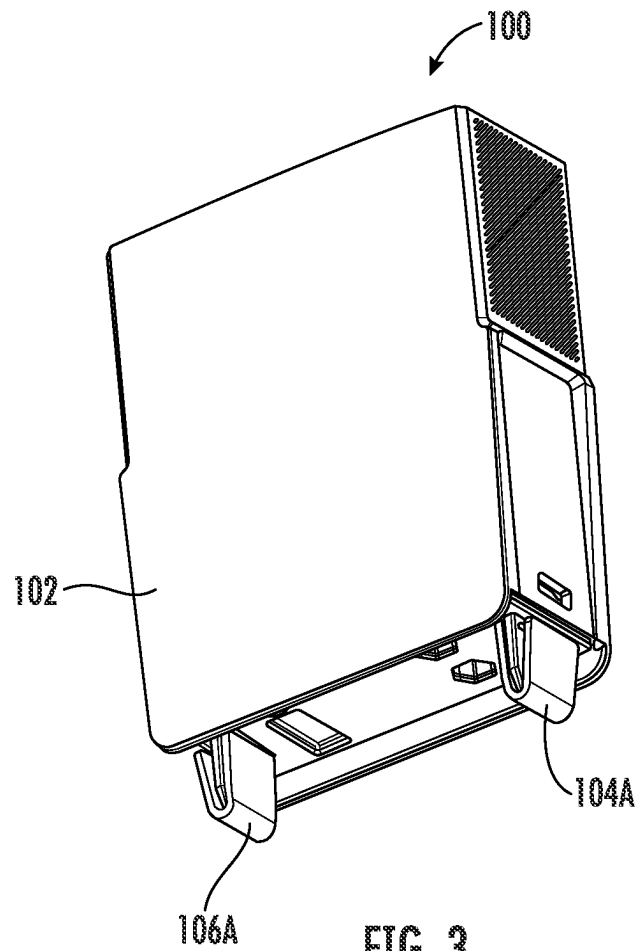


FIG. 3

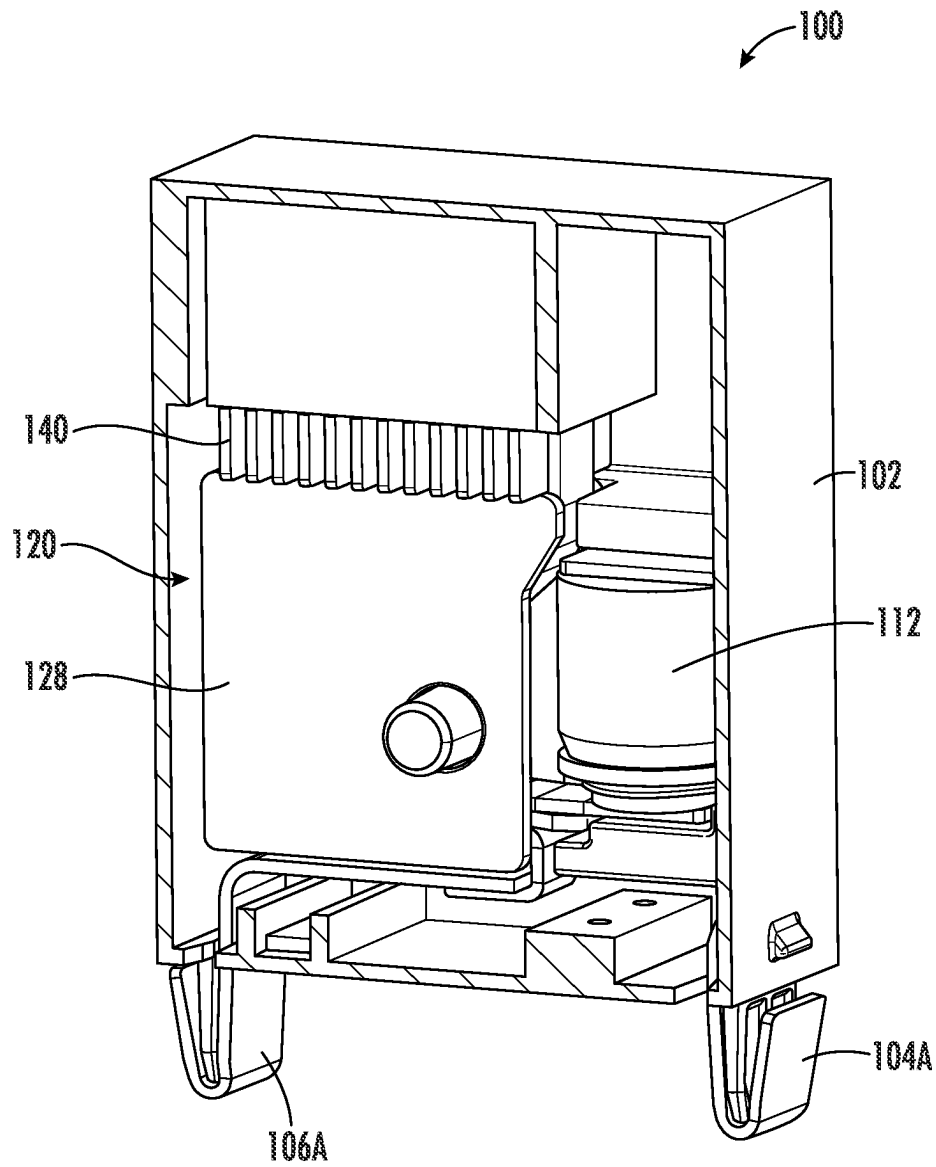
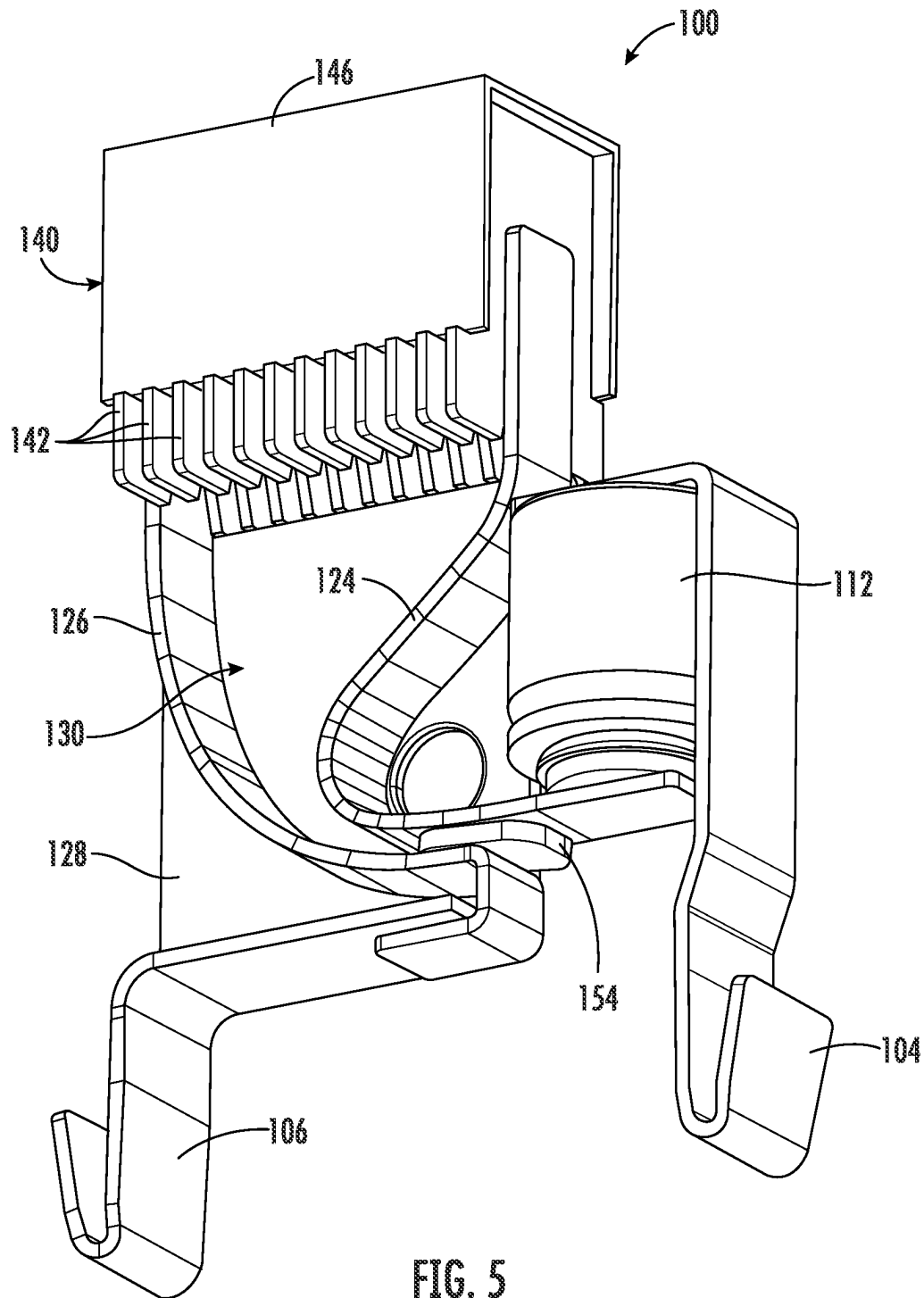
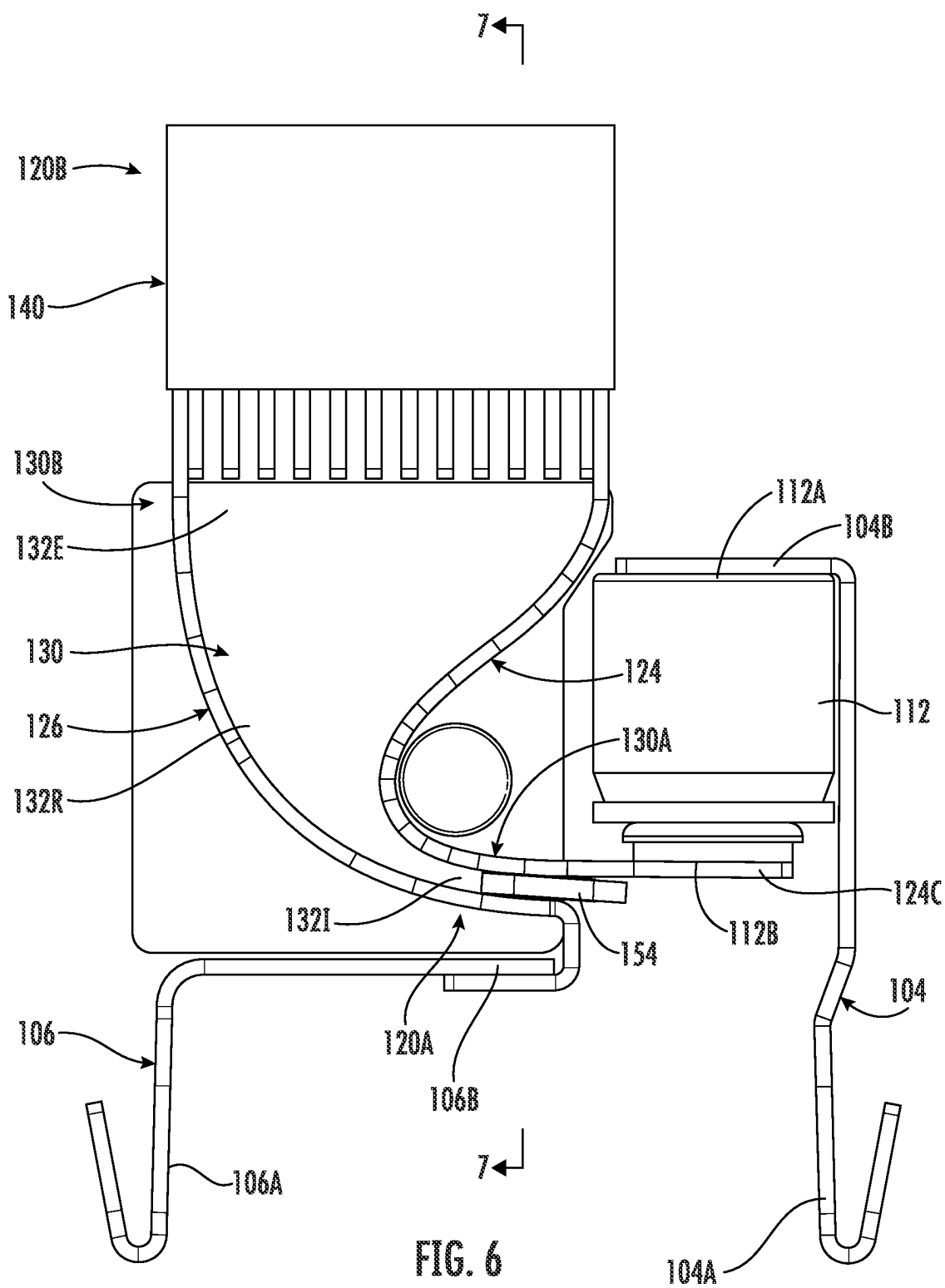


FIG. 4





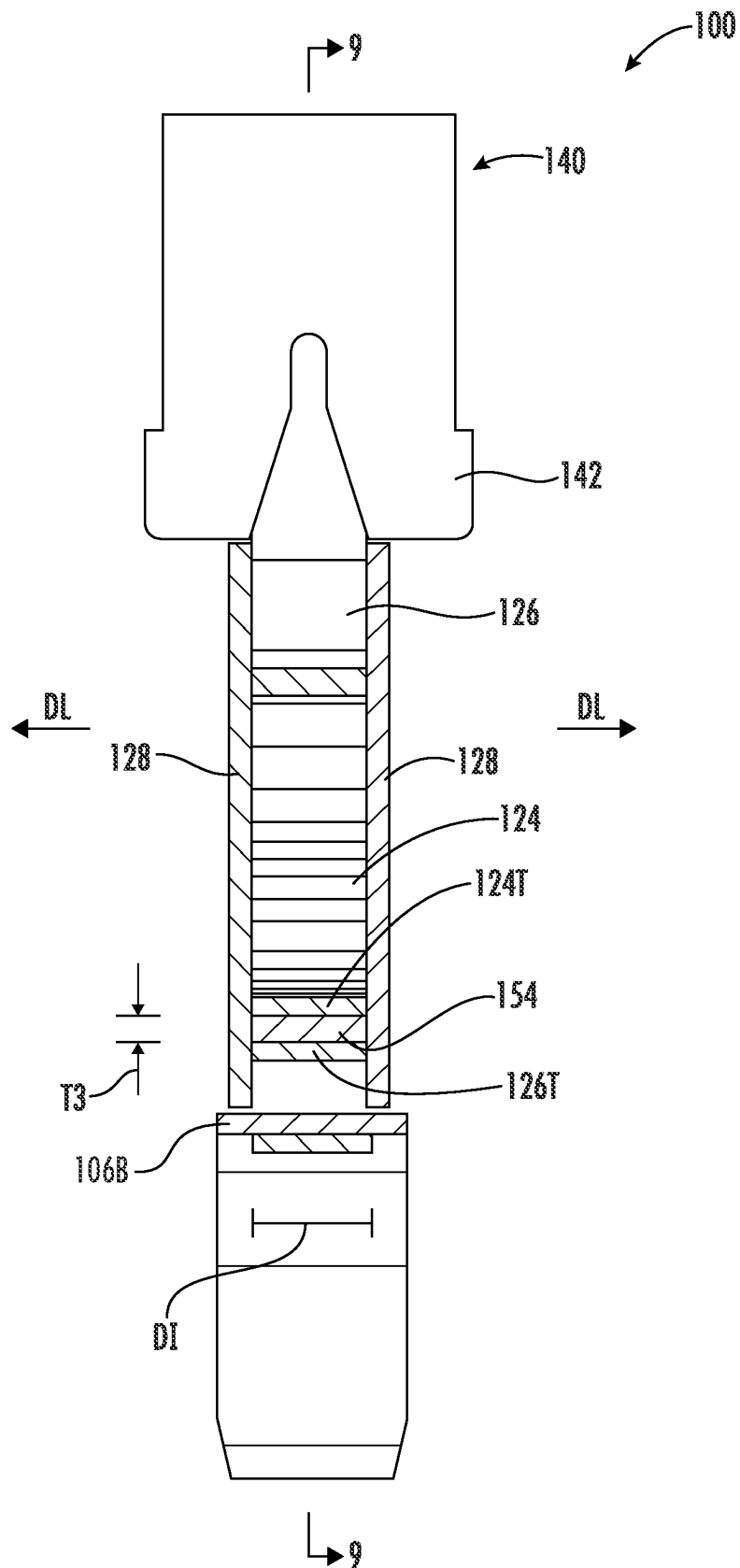


FIG. 7

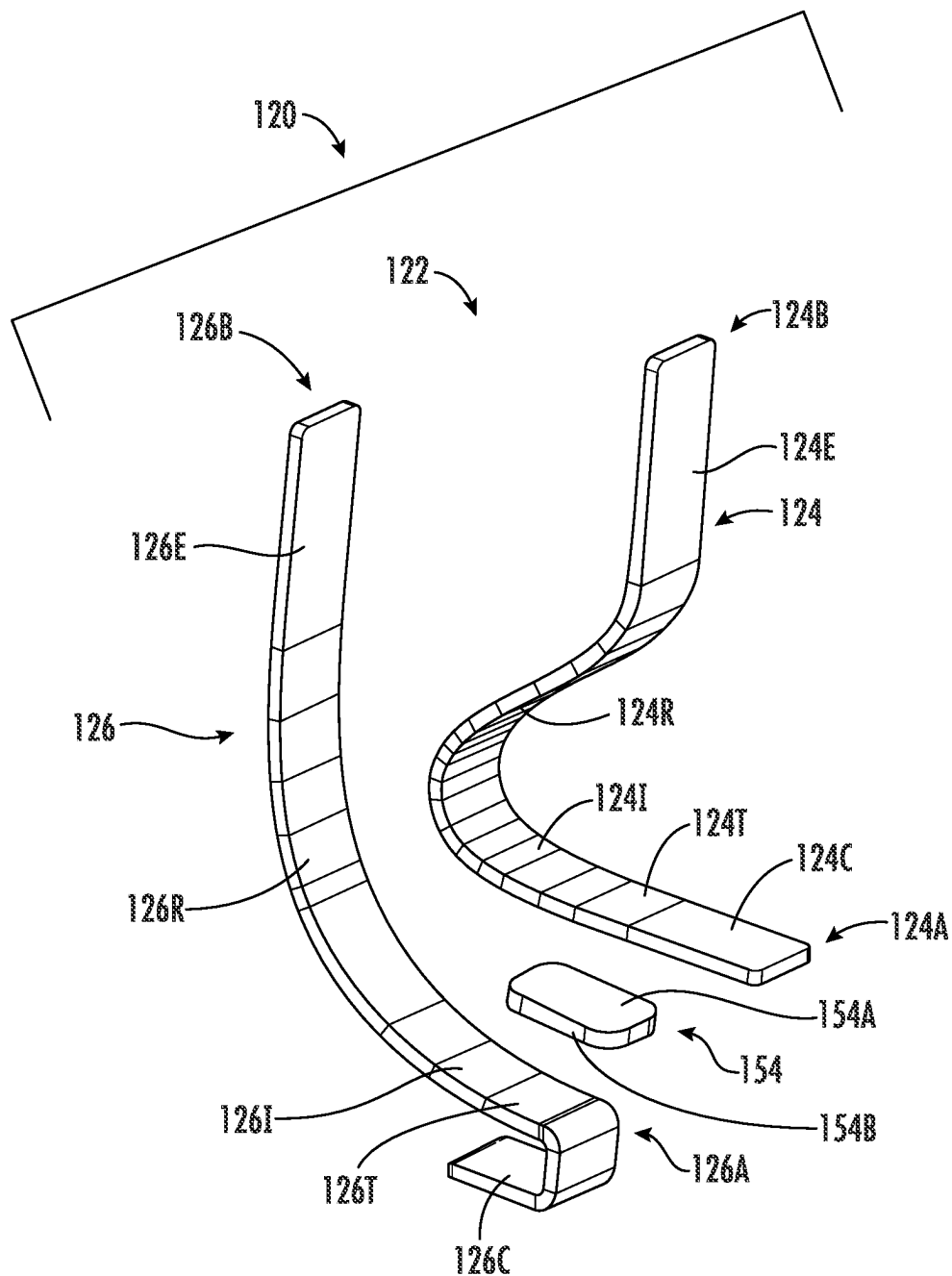
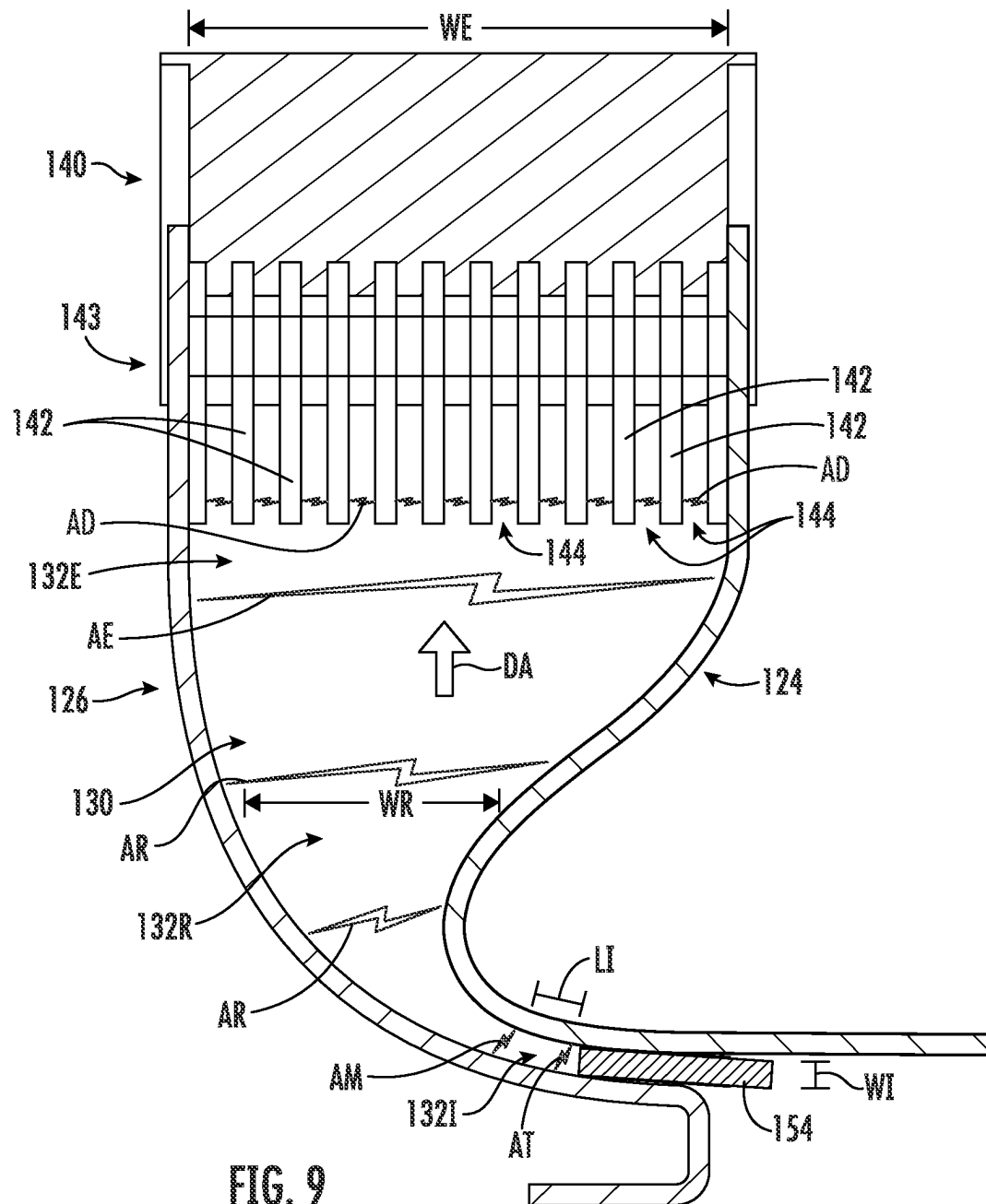


FIG. 8



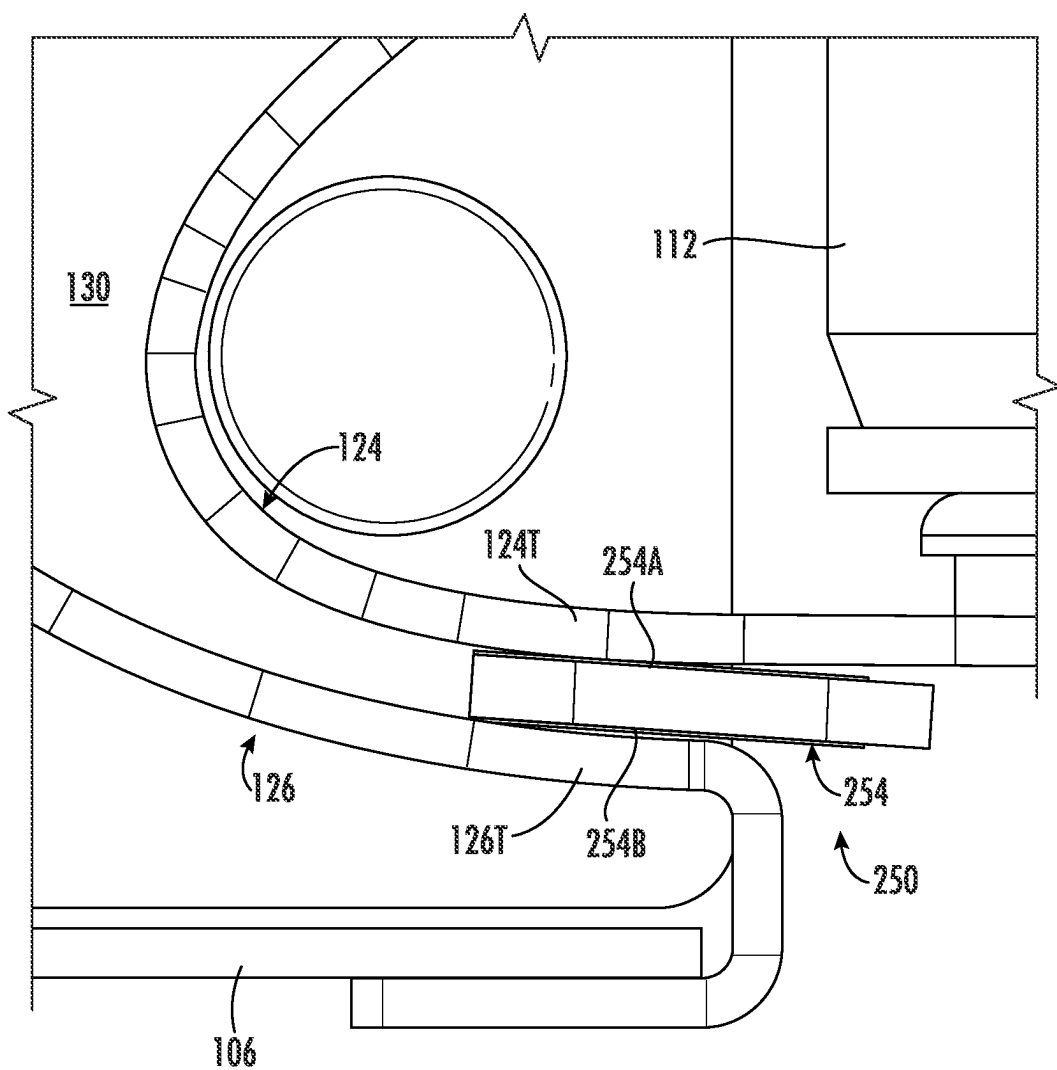


FIG. 10

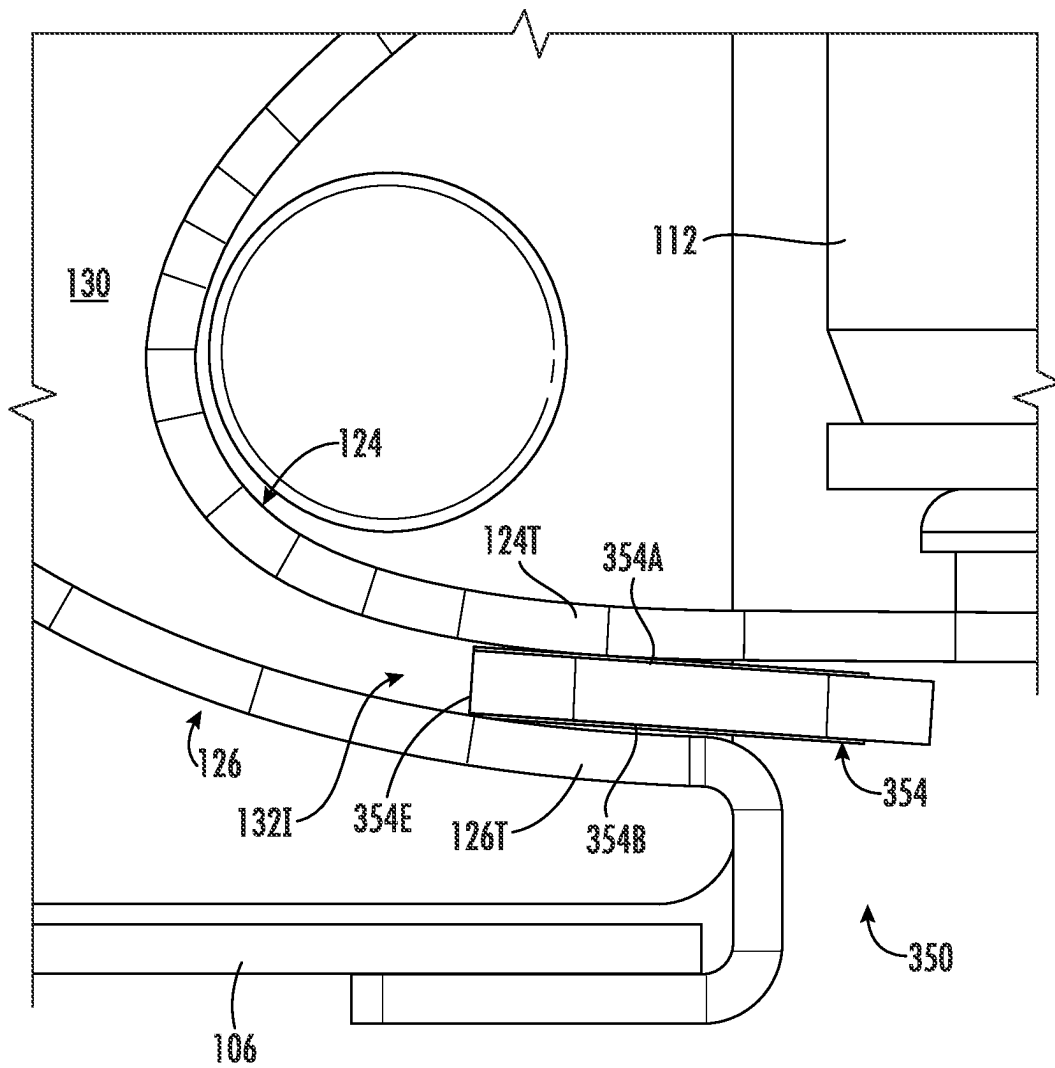


FIG. 11

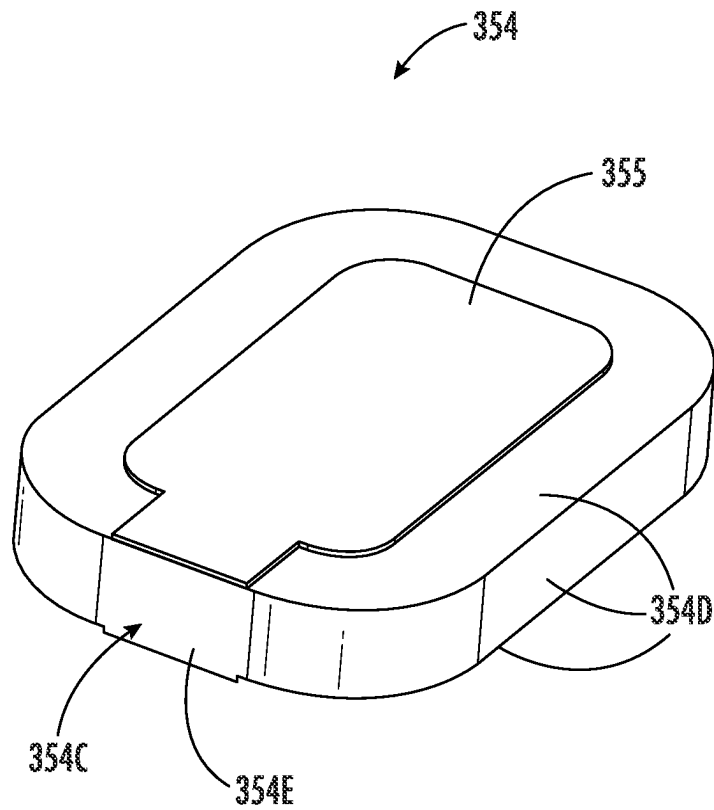
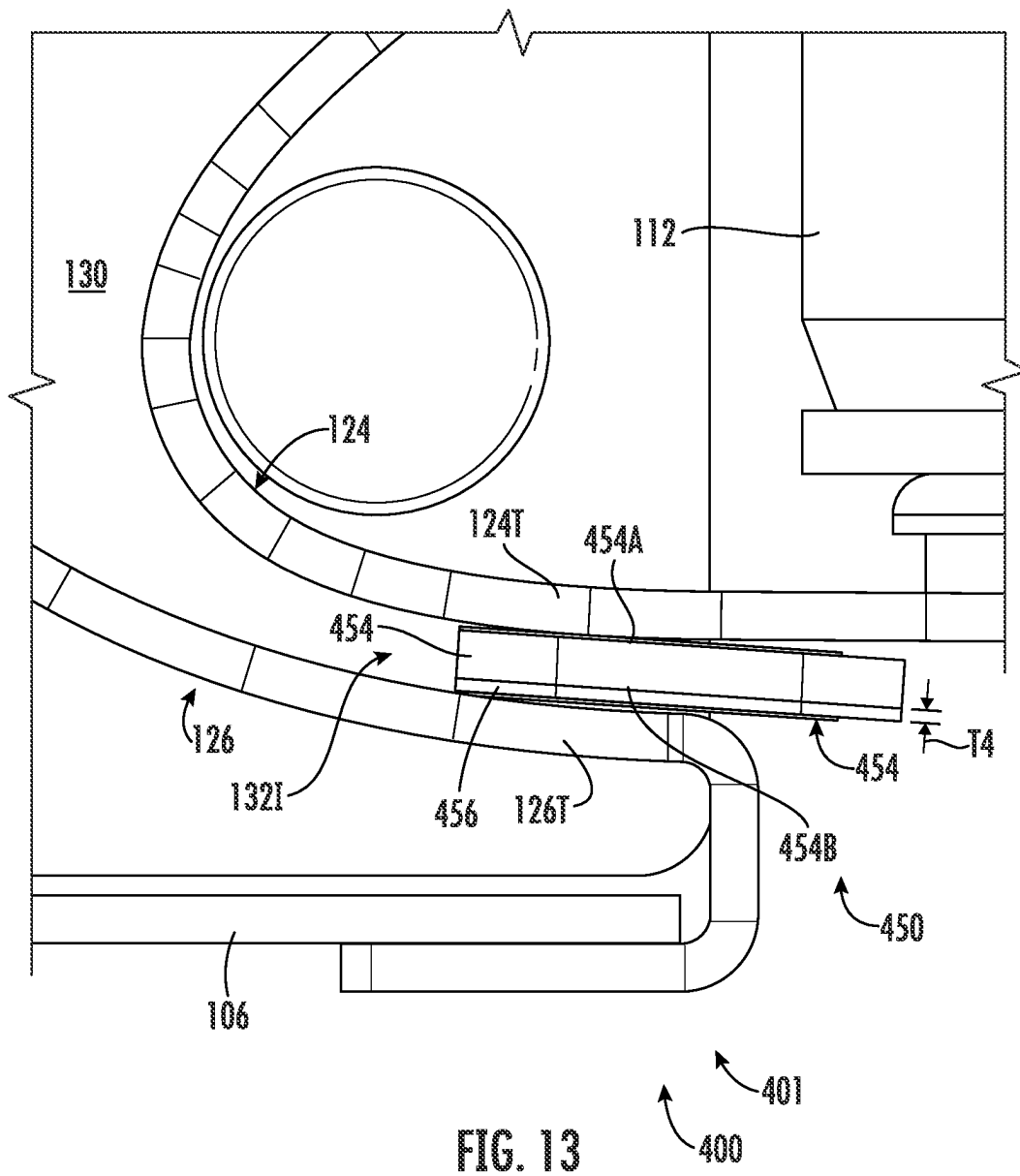


FIG. 12



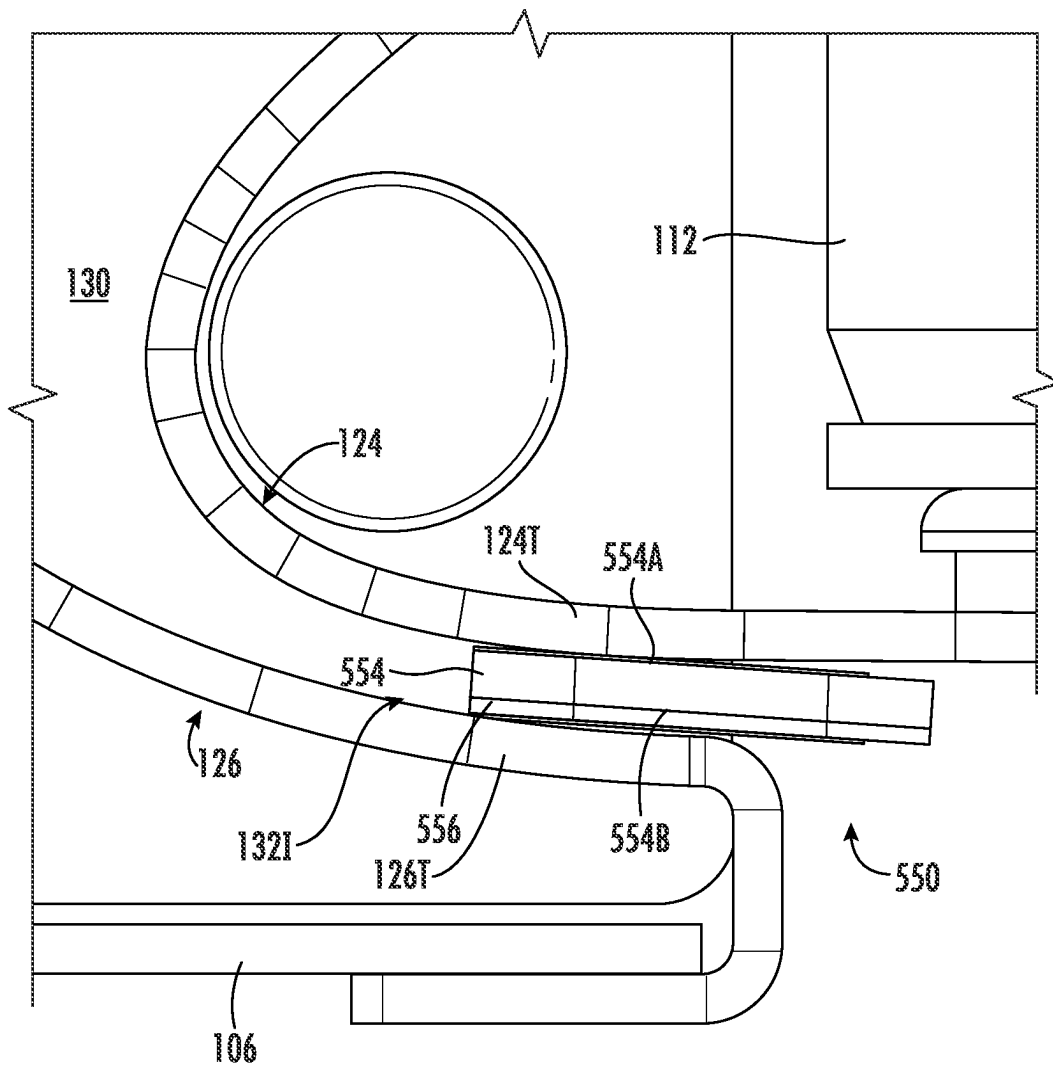
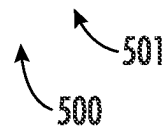
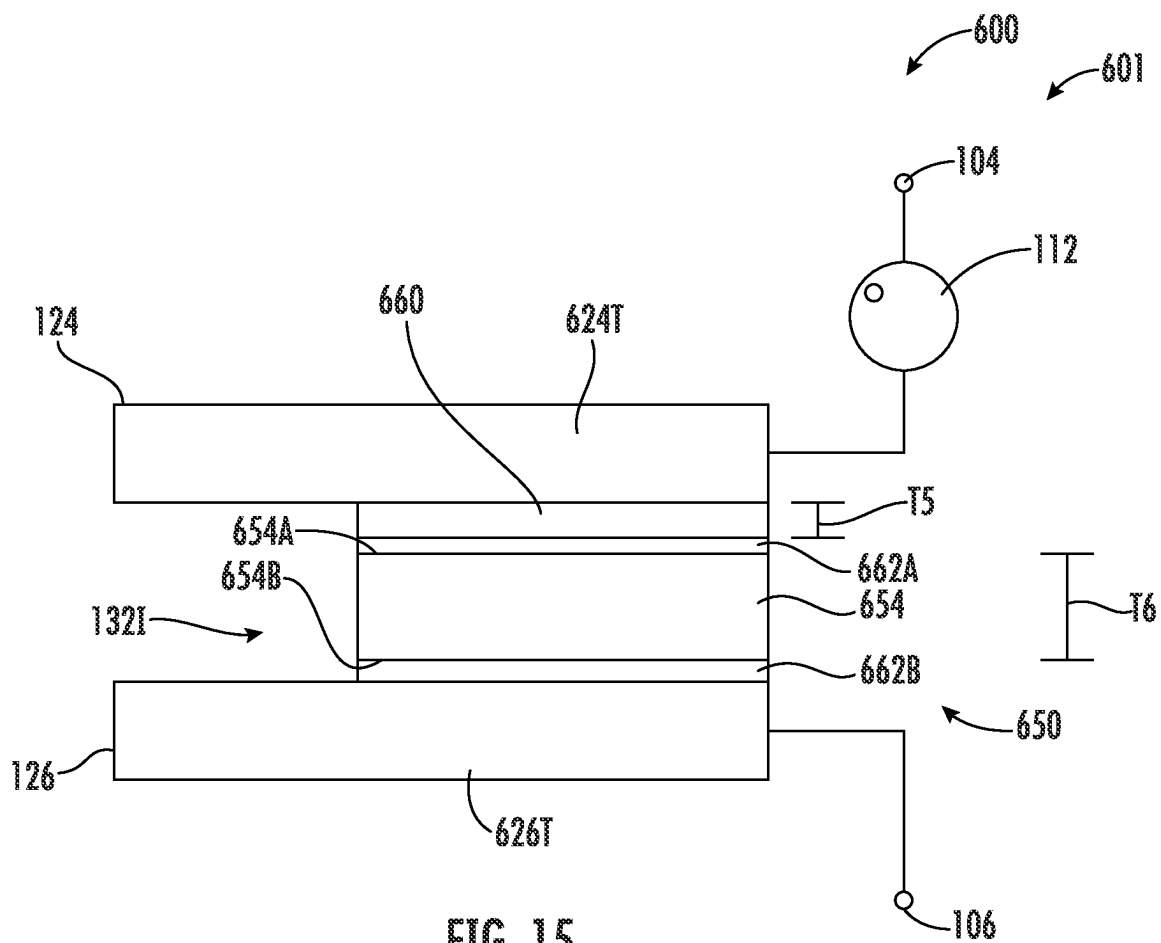


FIG. 14





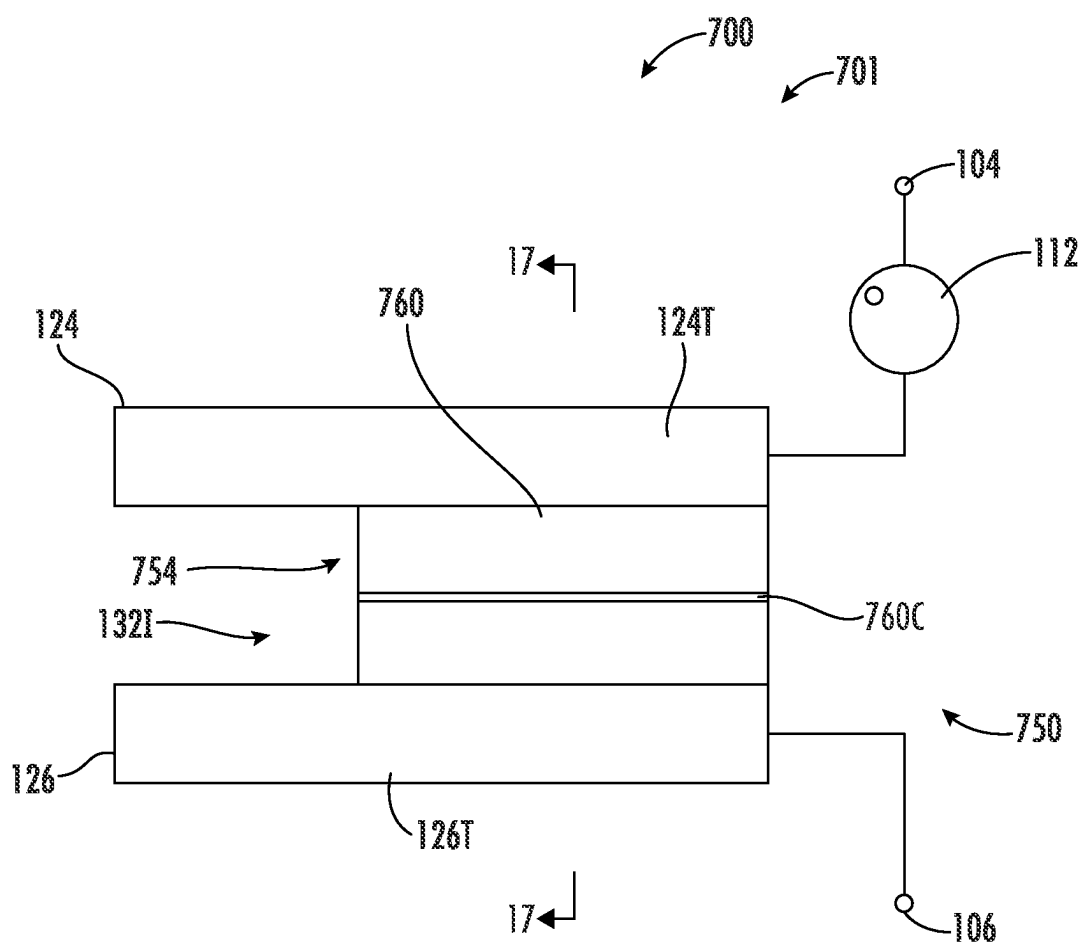


FIG. 16

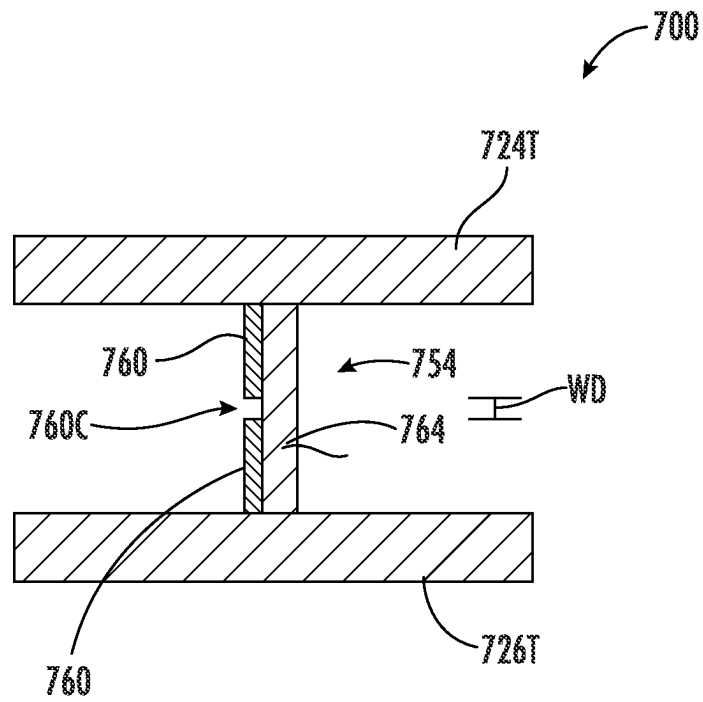


FIG. 17

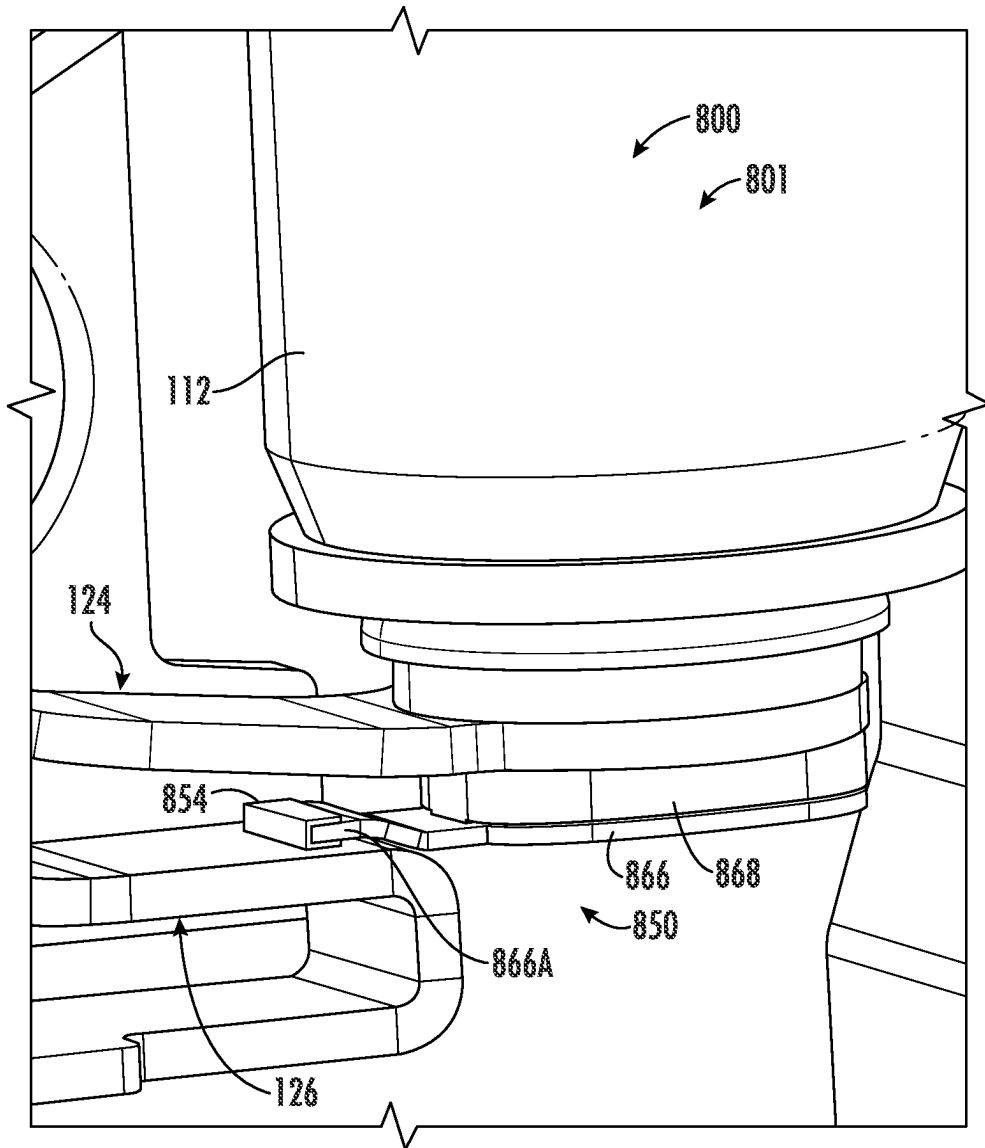
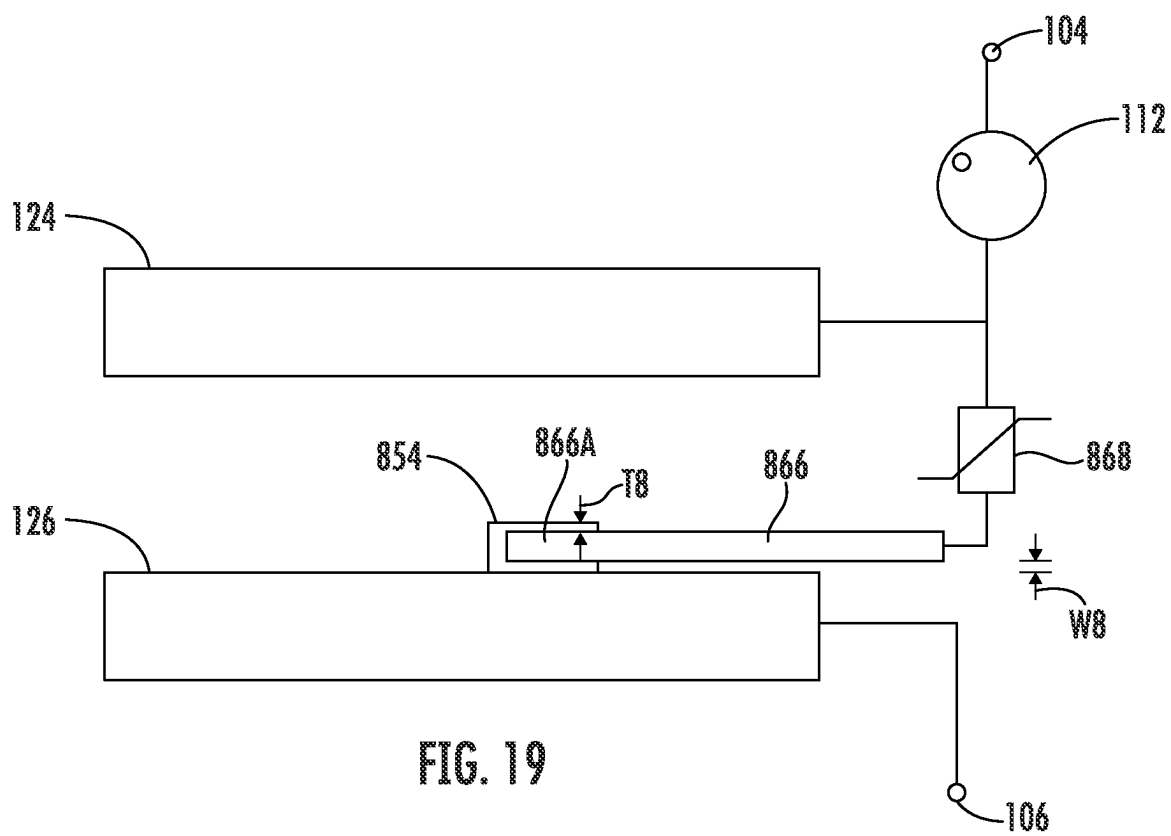
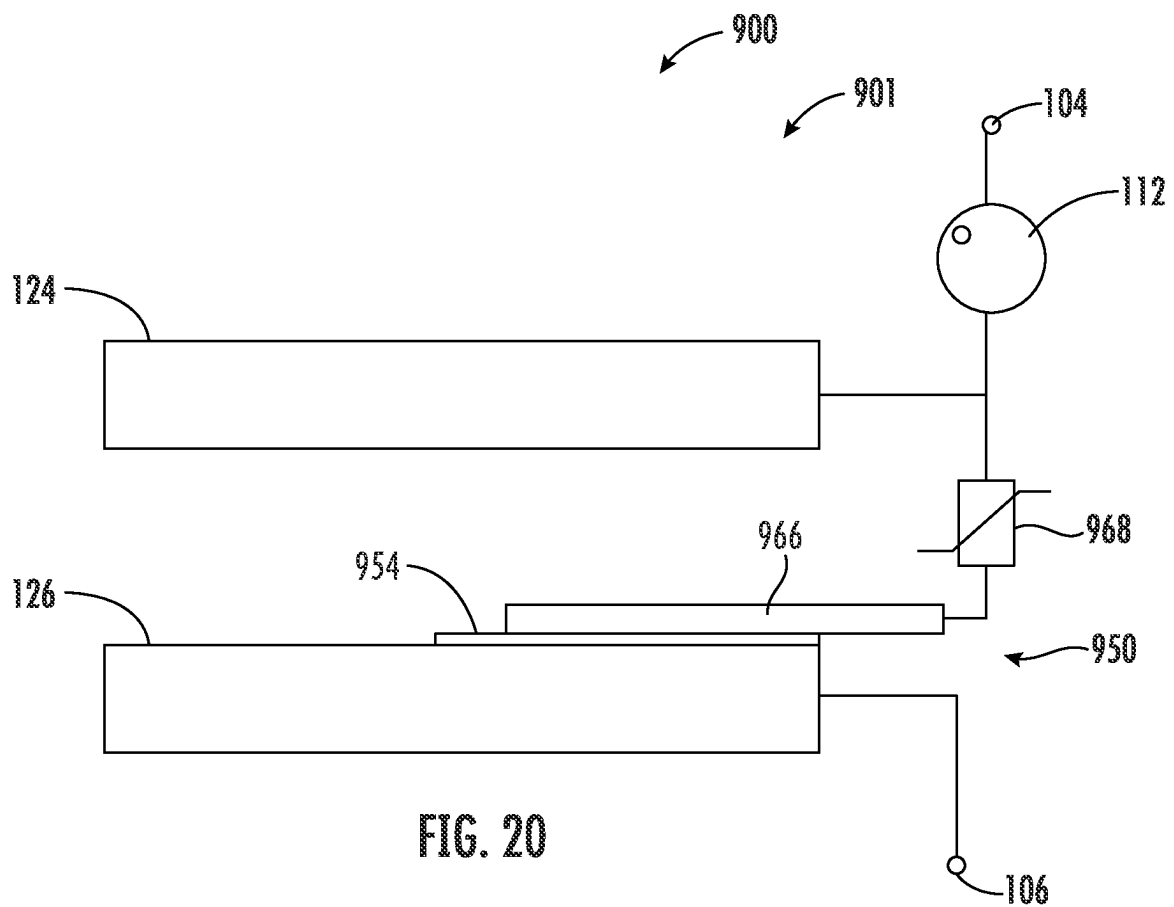
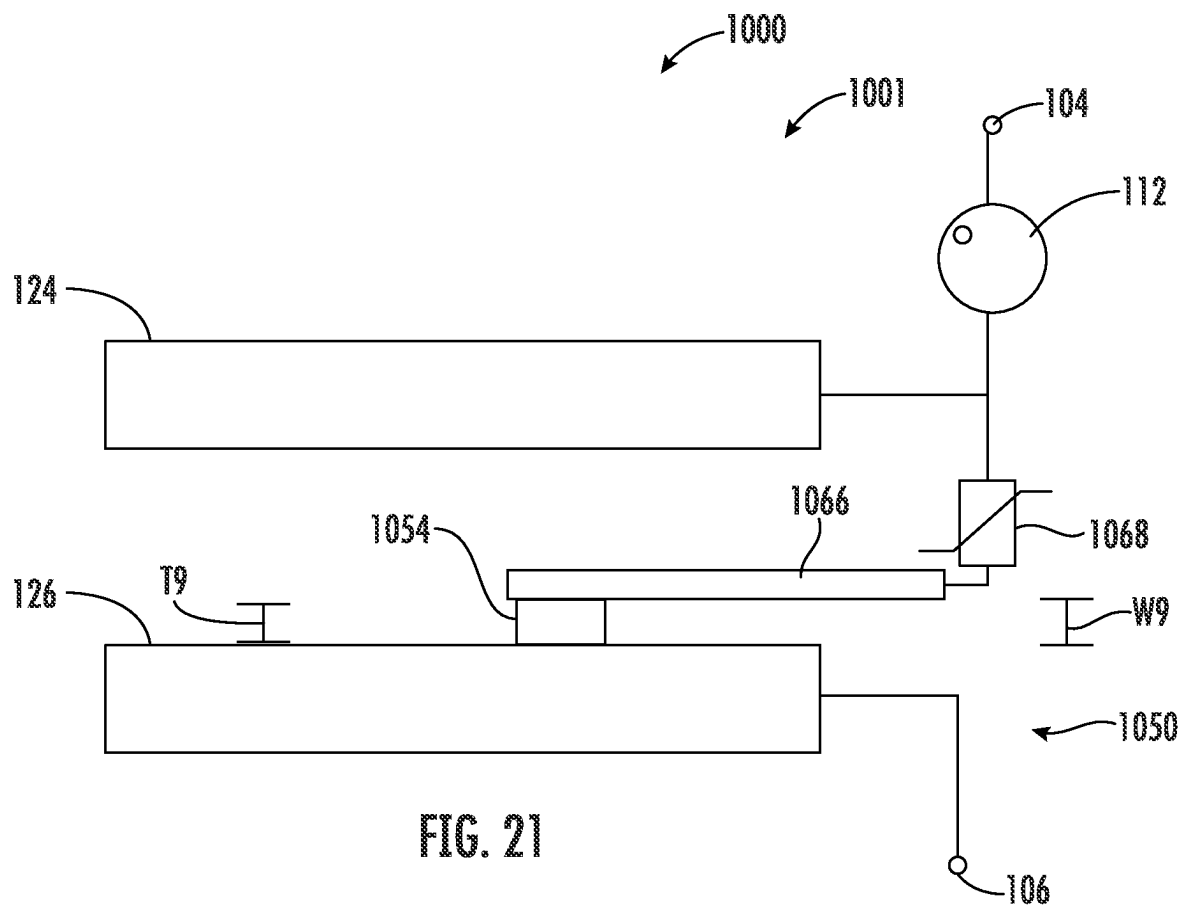
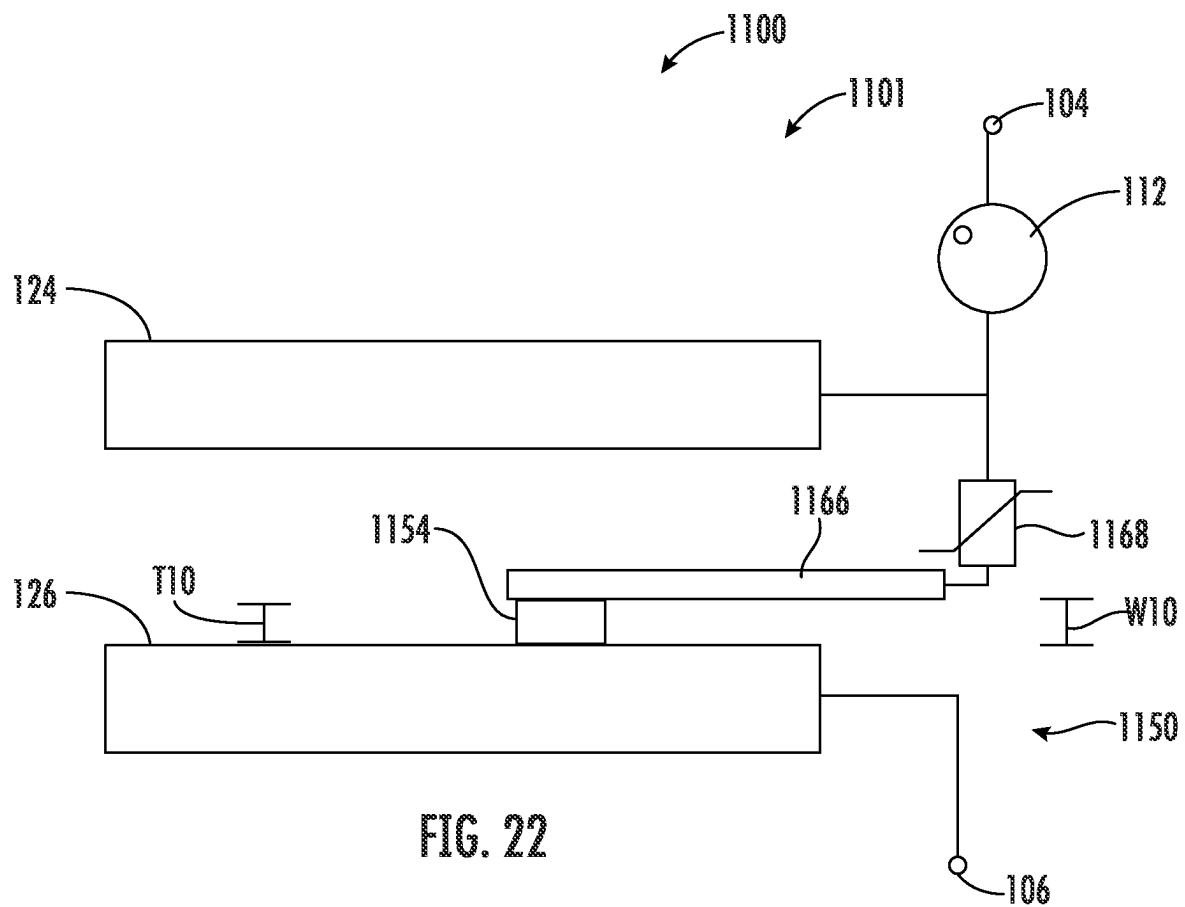


FIG. 18









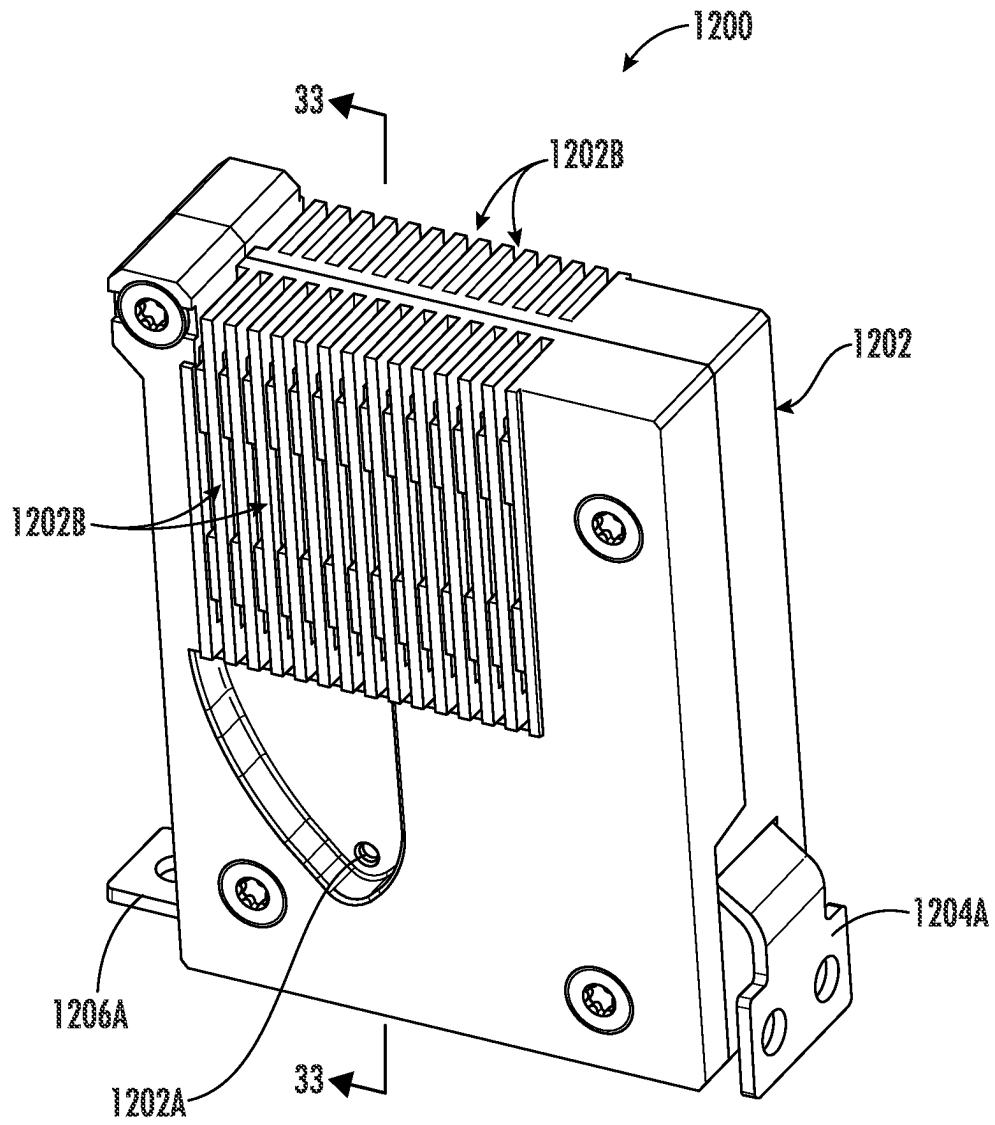


FIG. 23

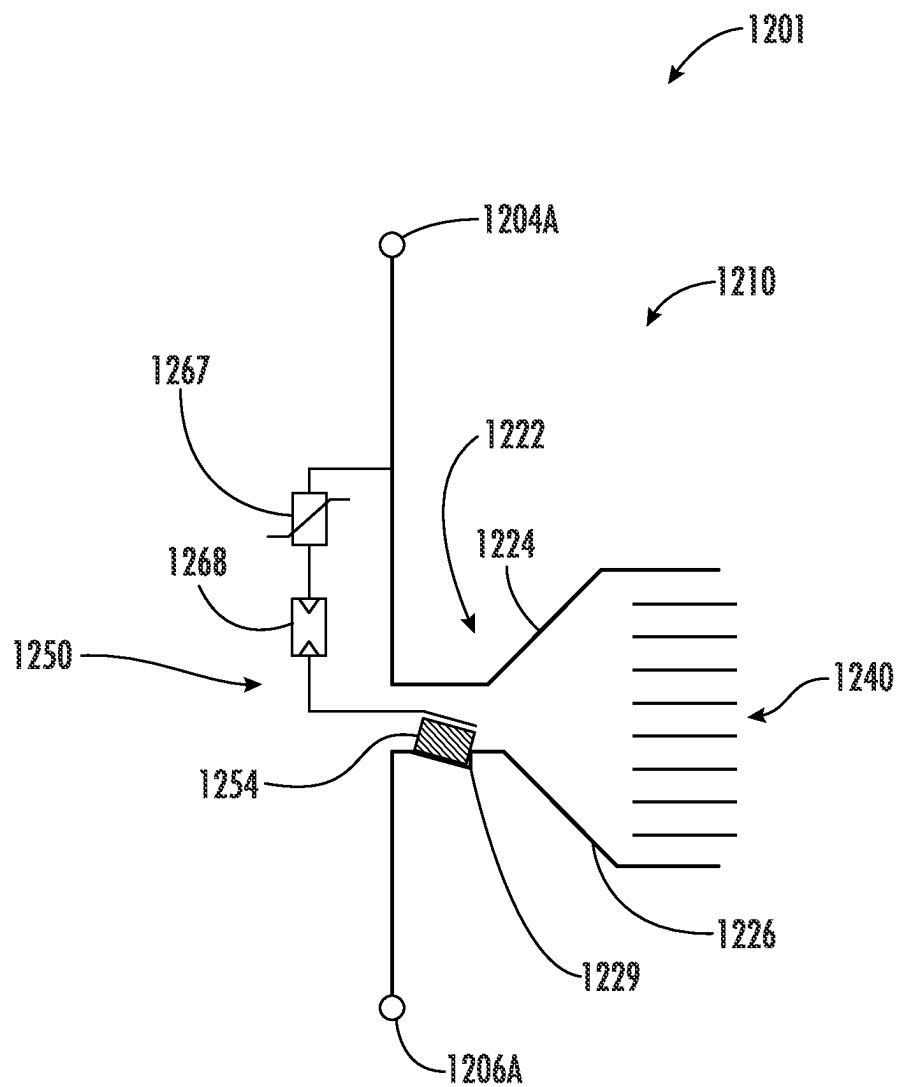


FIG. 24

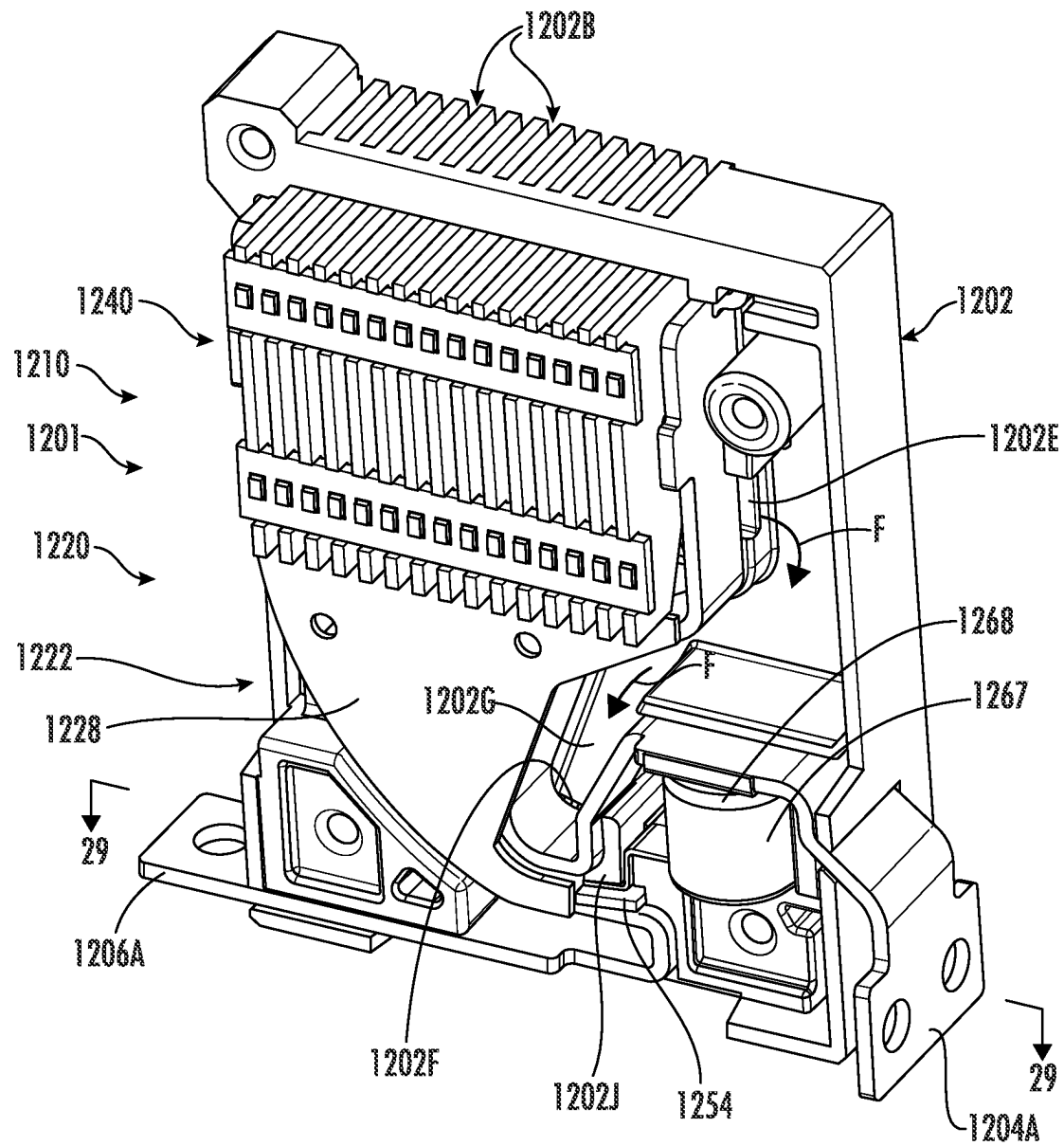


FIG. 25

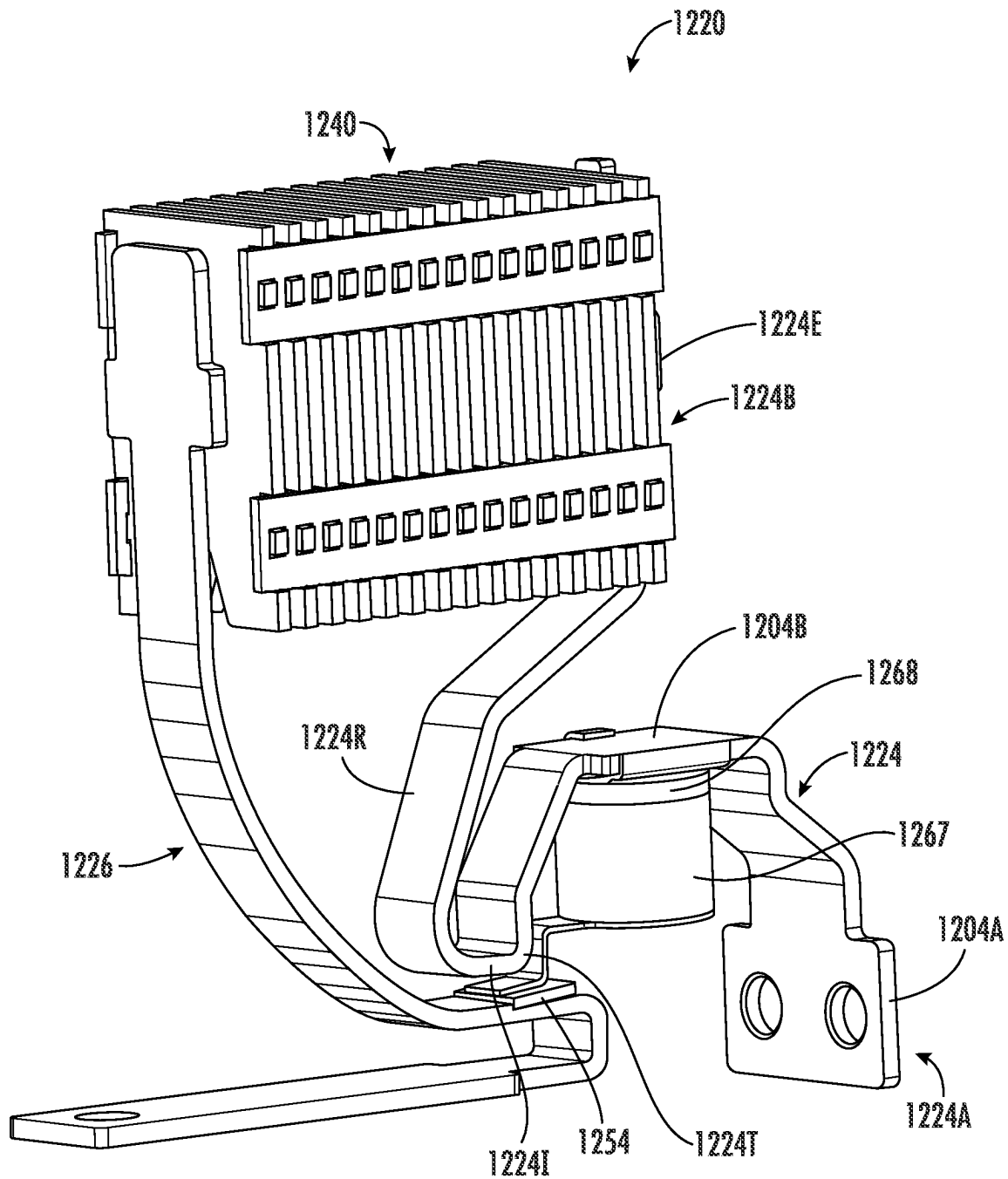
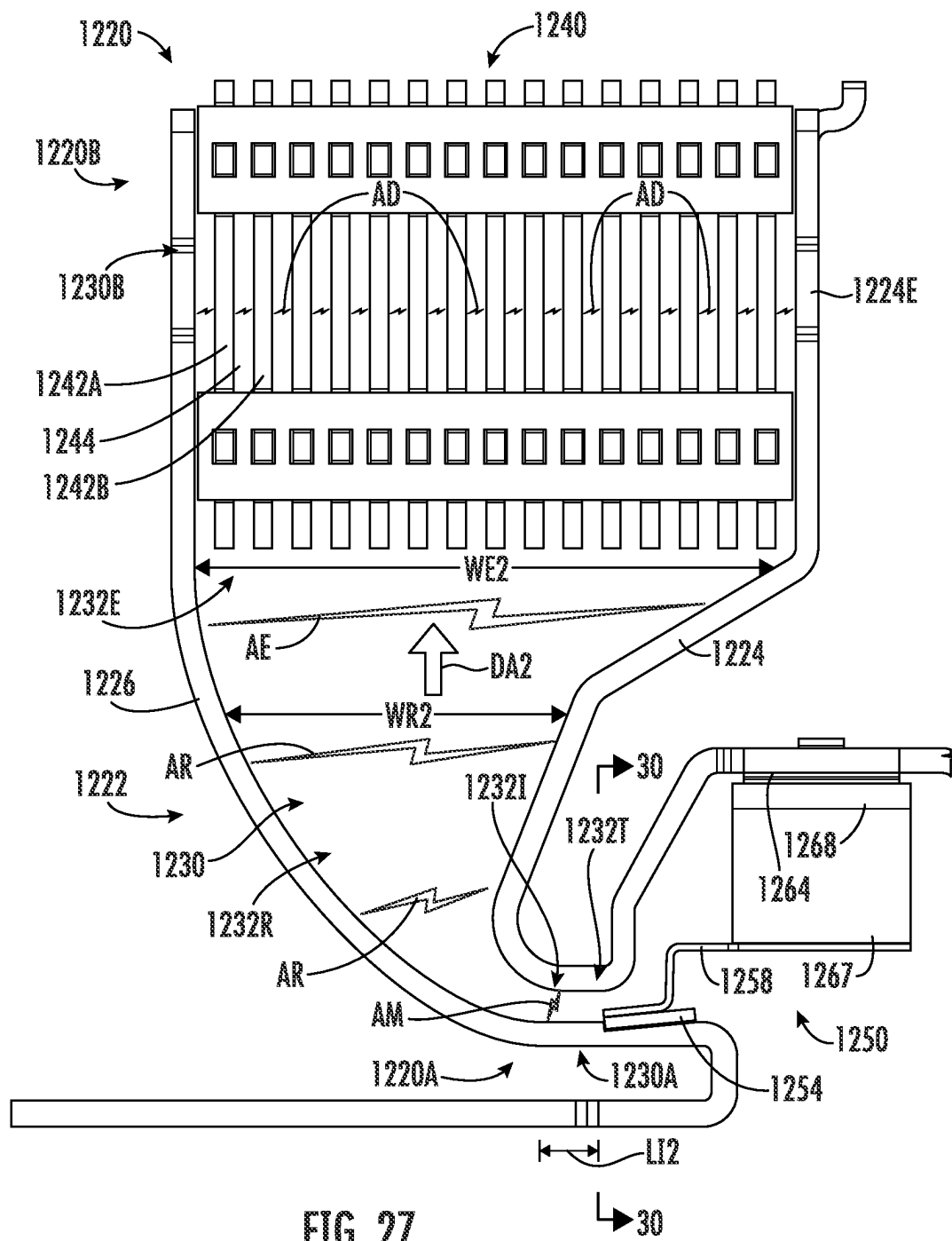
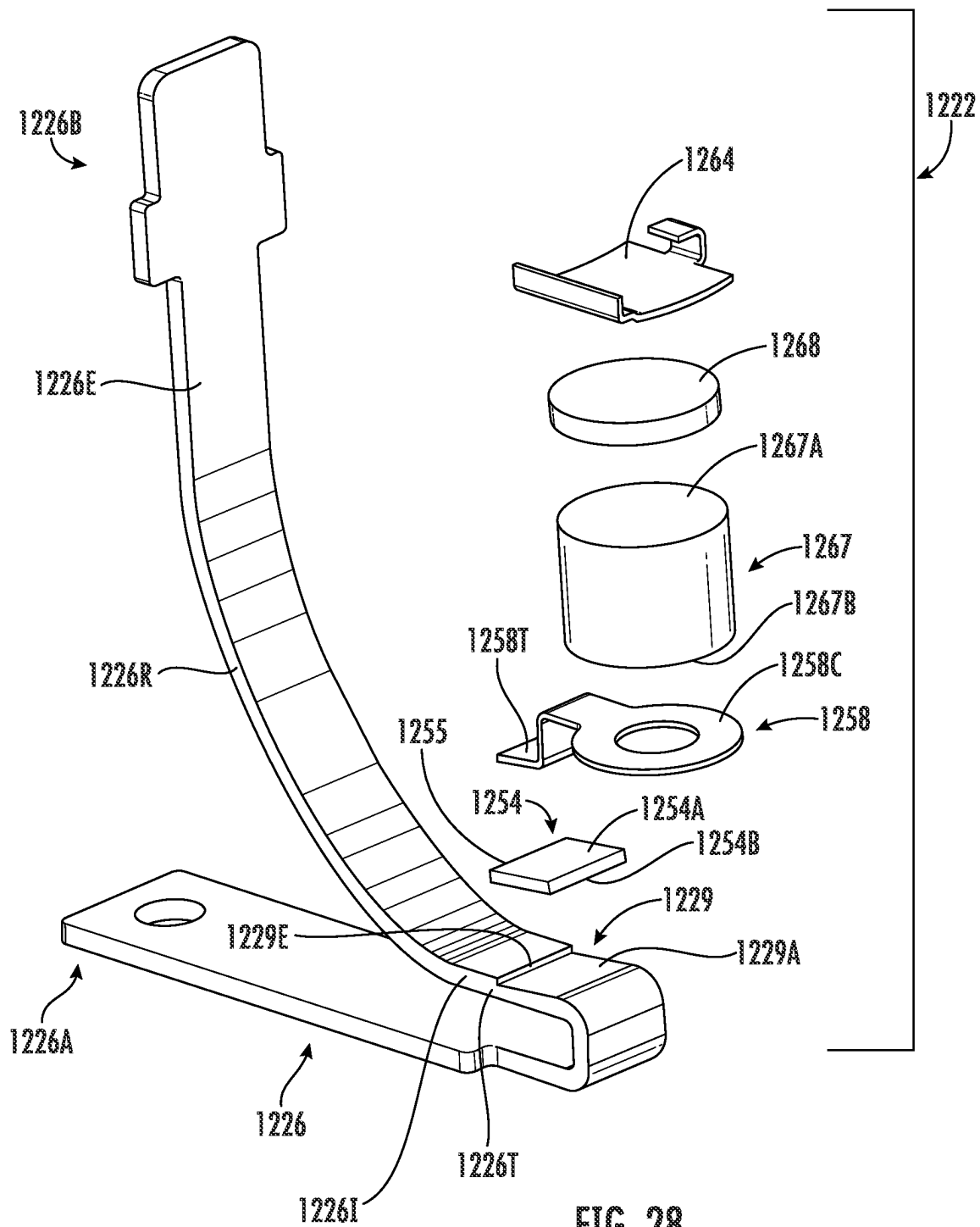


FIG. 26





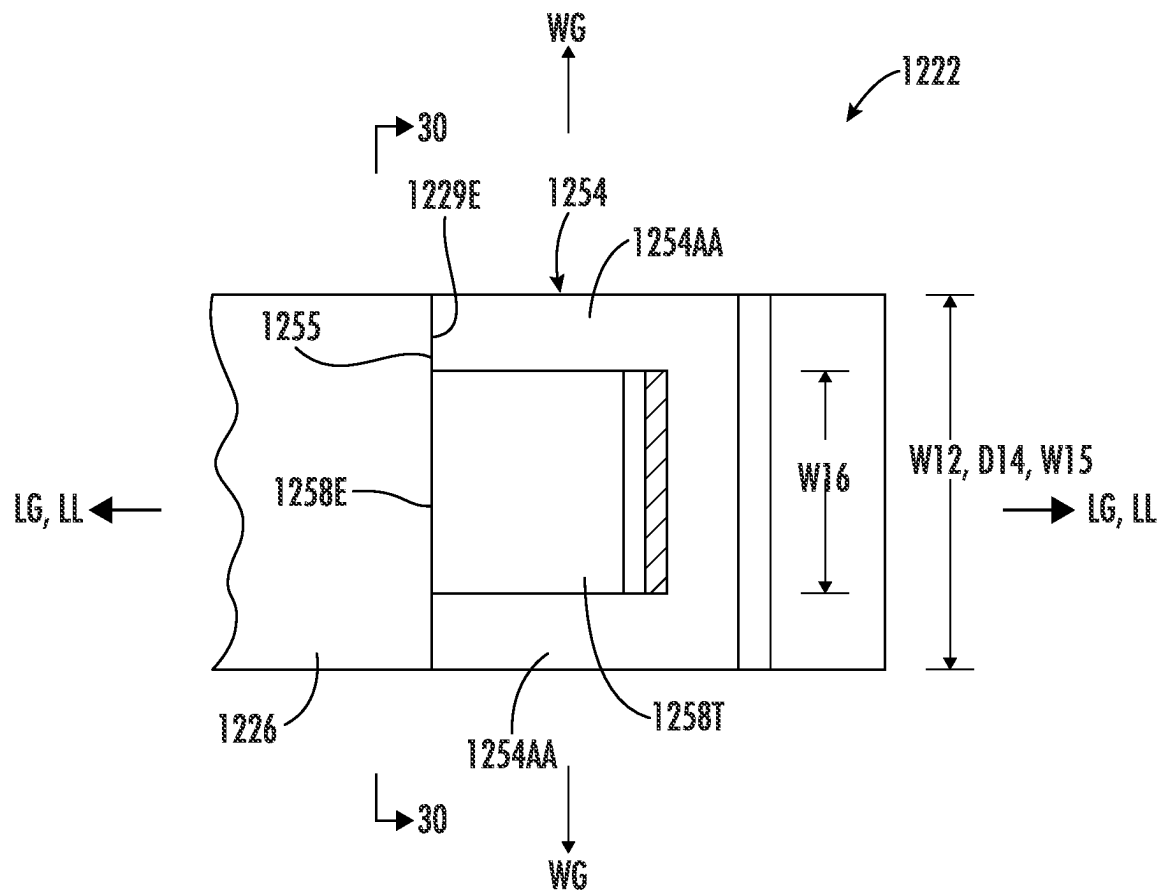


FIG. 29

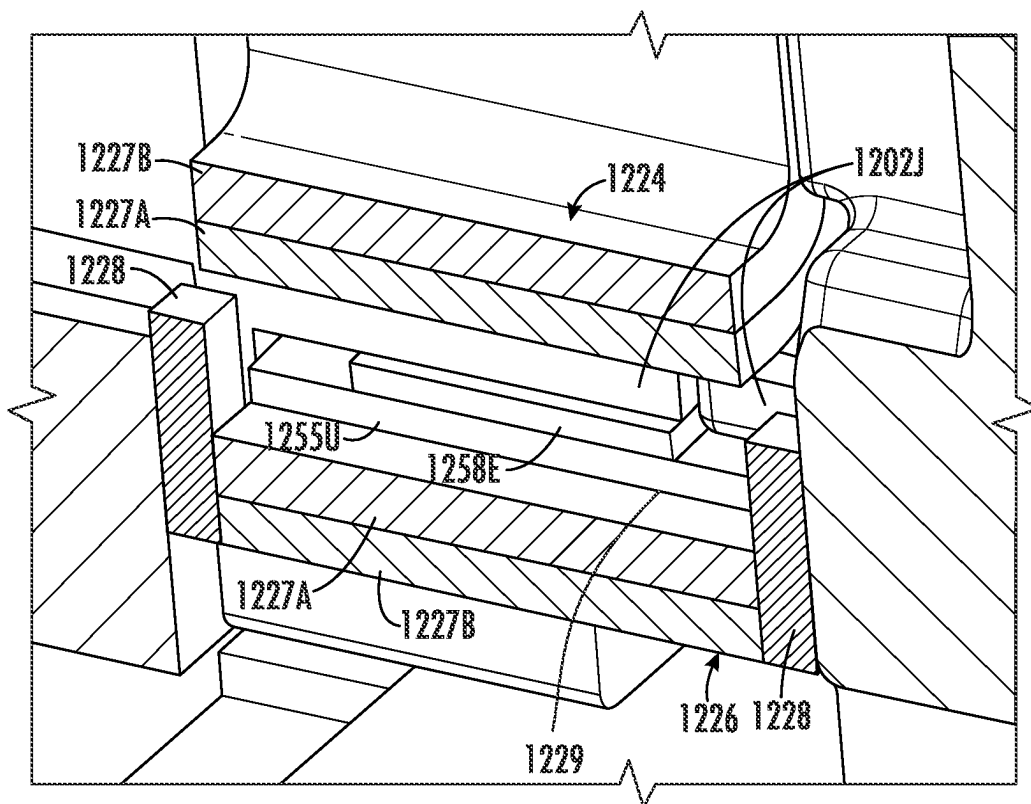


FIG. 30

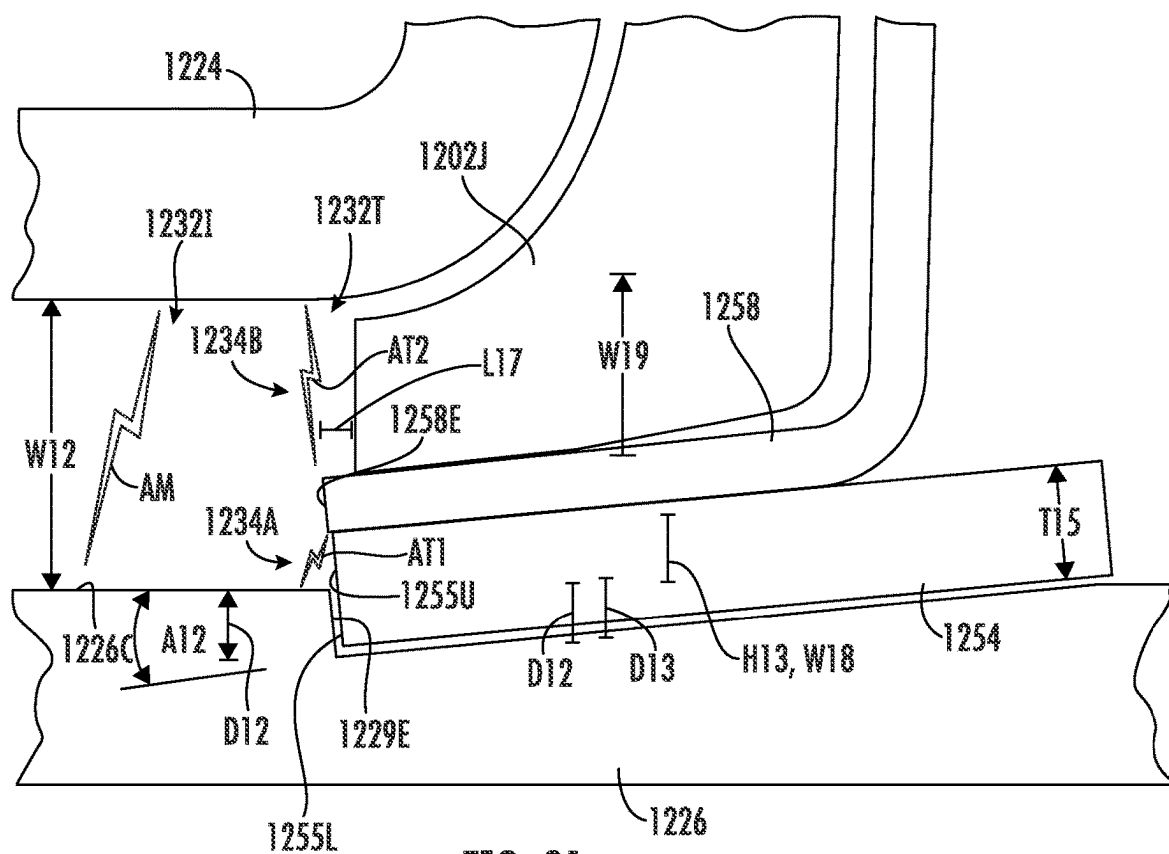
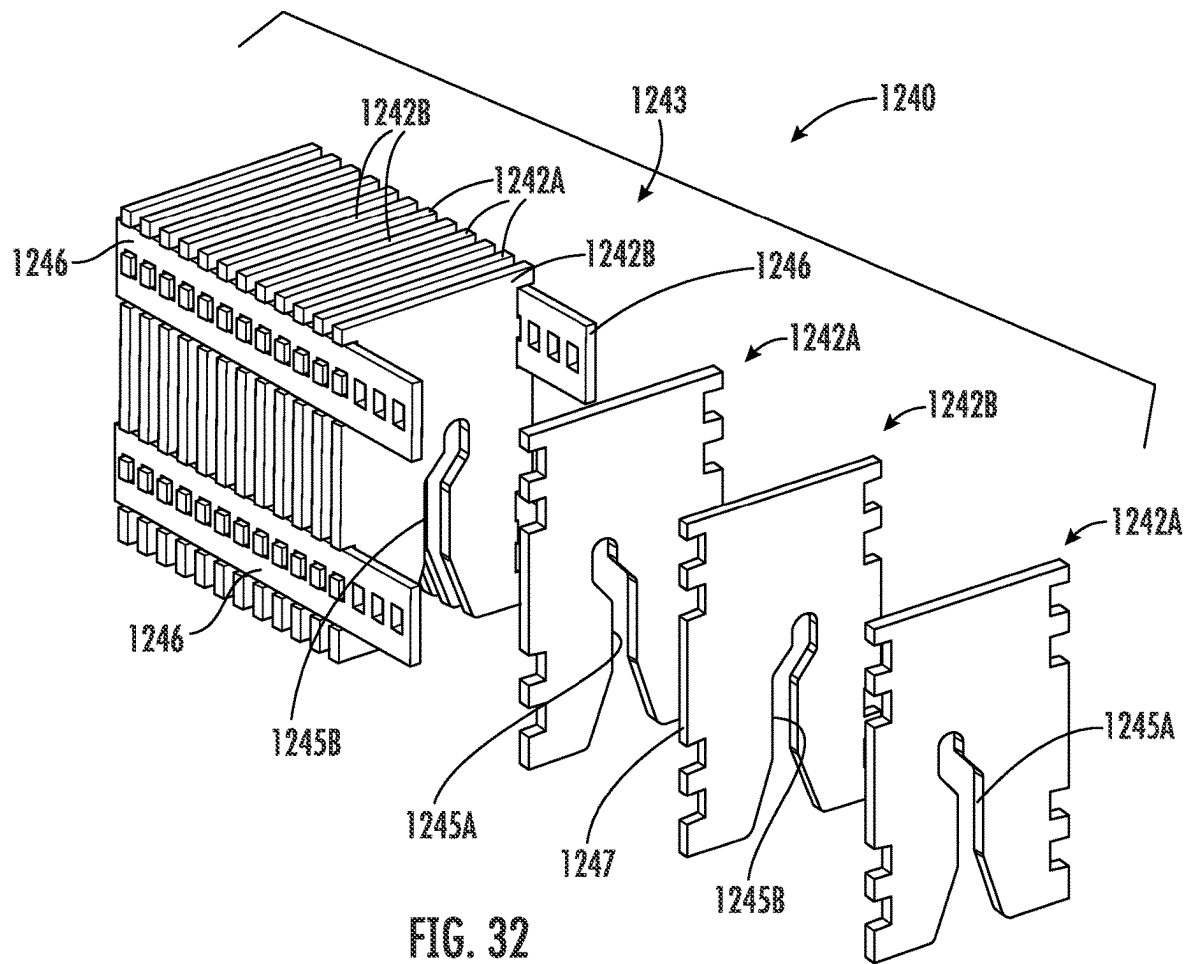


FIG. 31



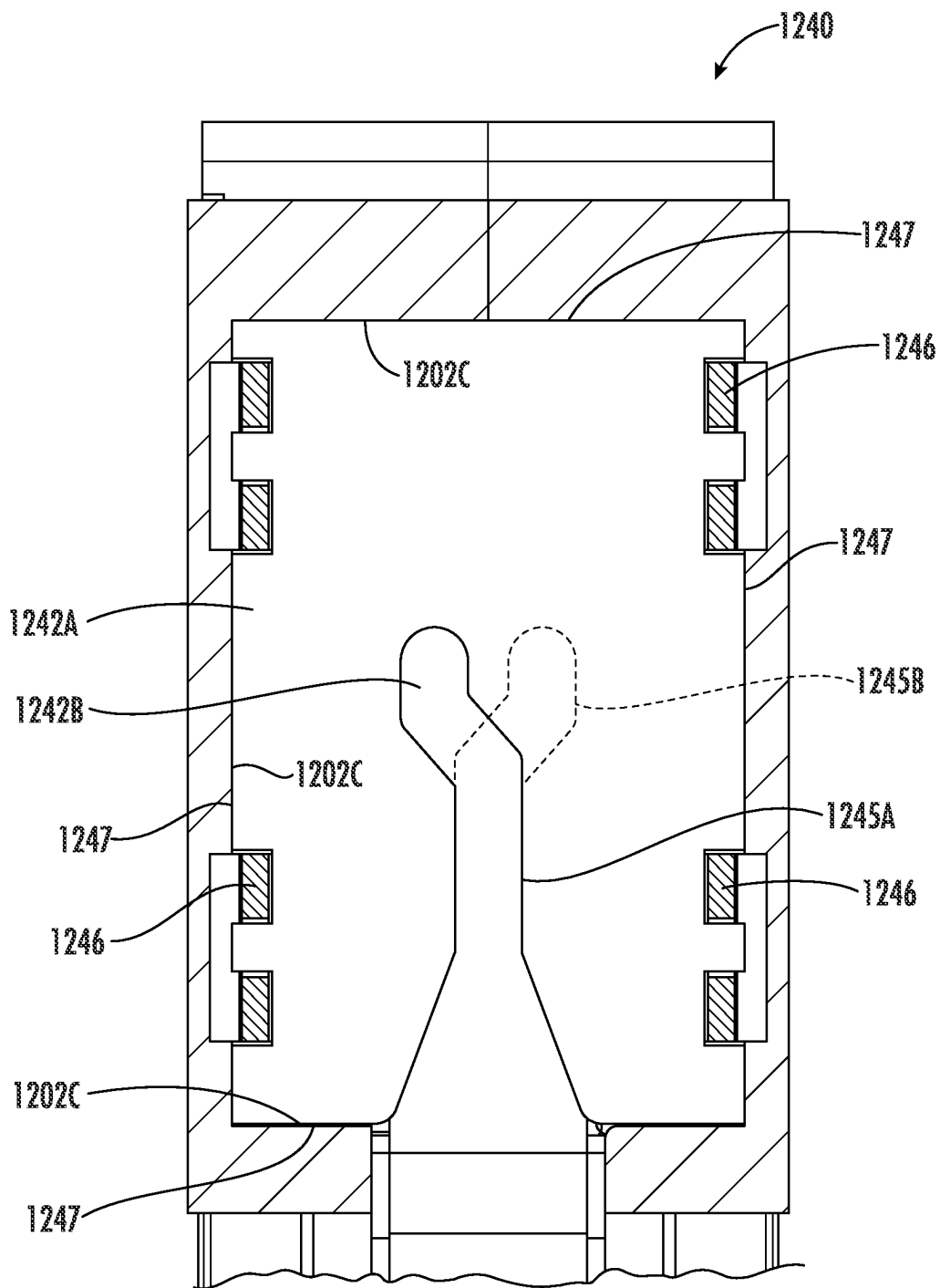


FIG. 33

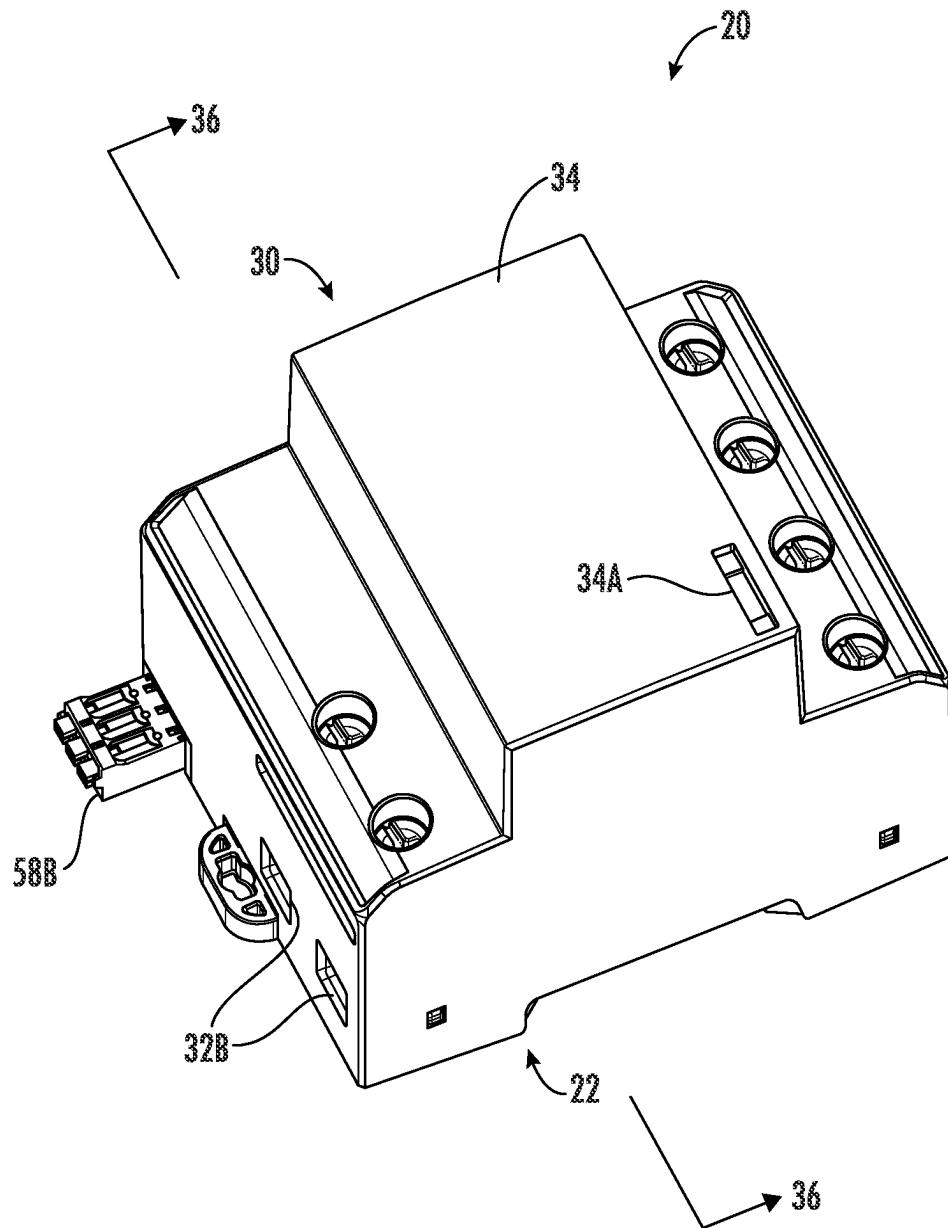


FIG. 34

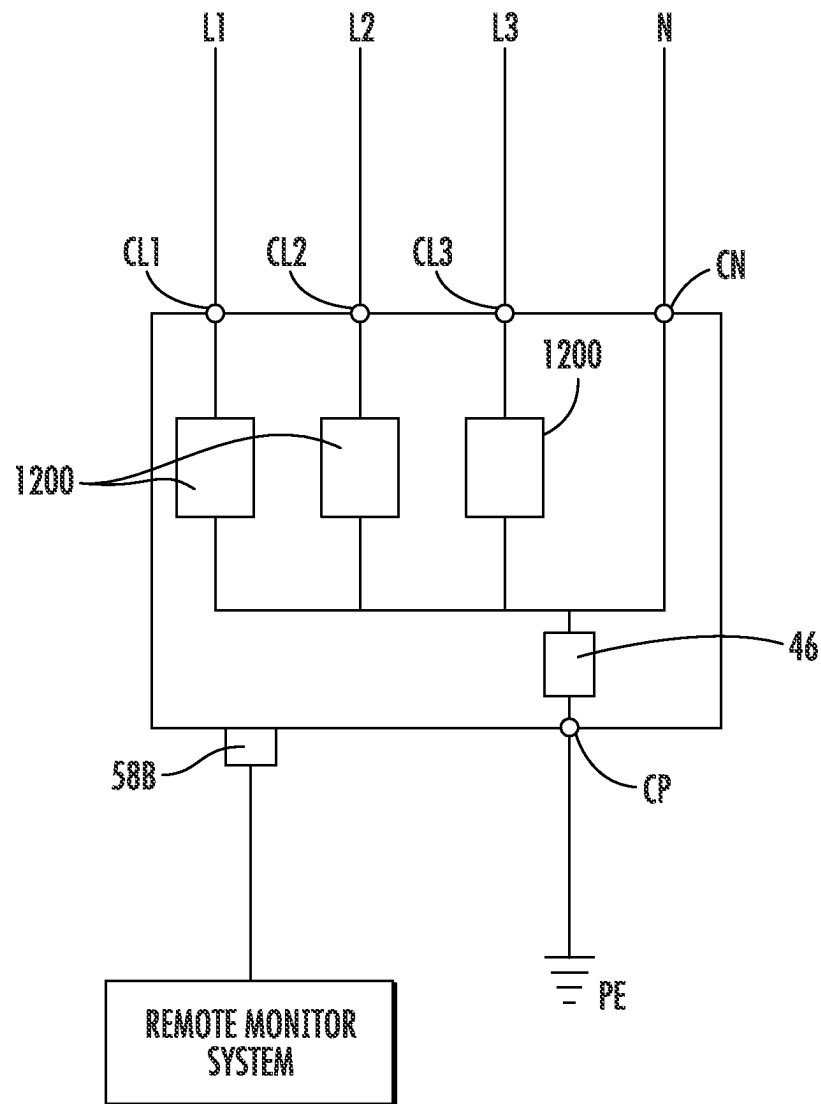


FIG. 35

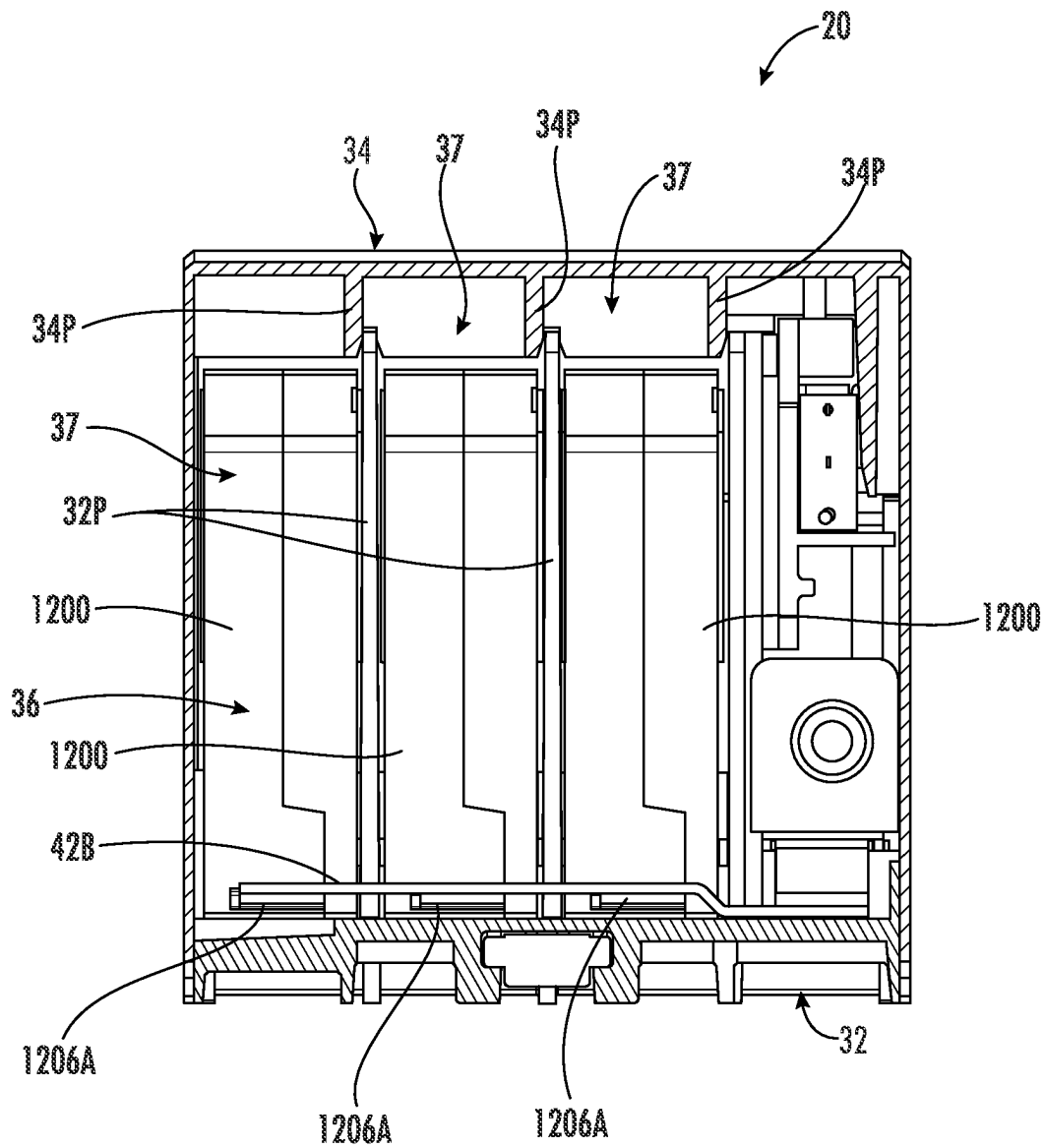


FIG. 36

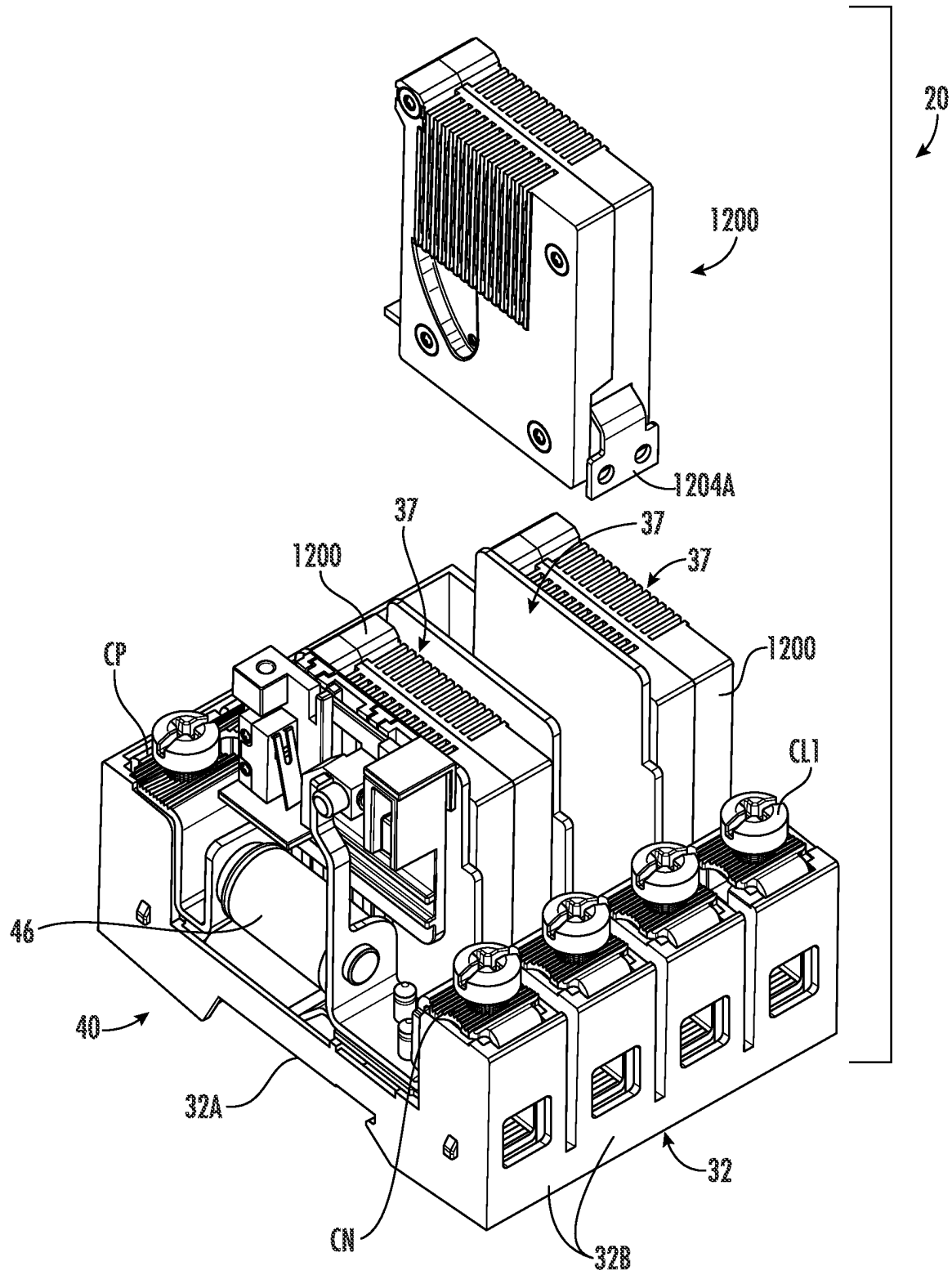


FIG. 37

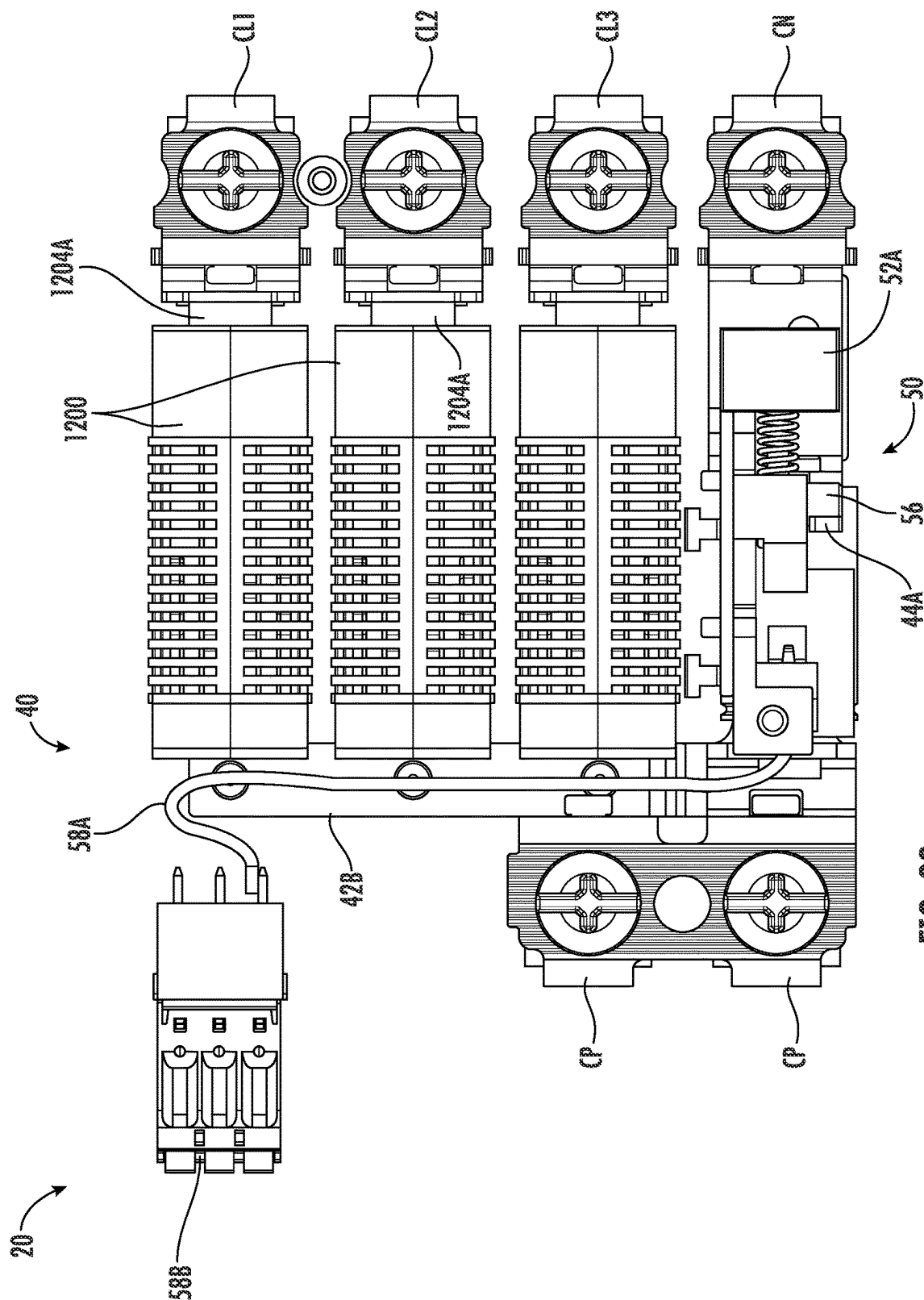


FIG. 38

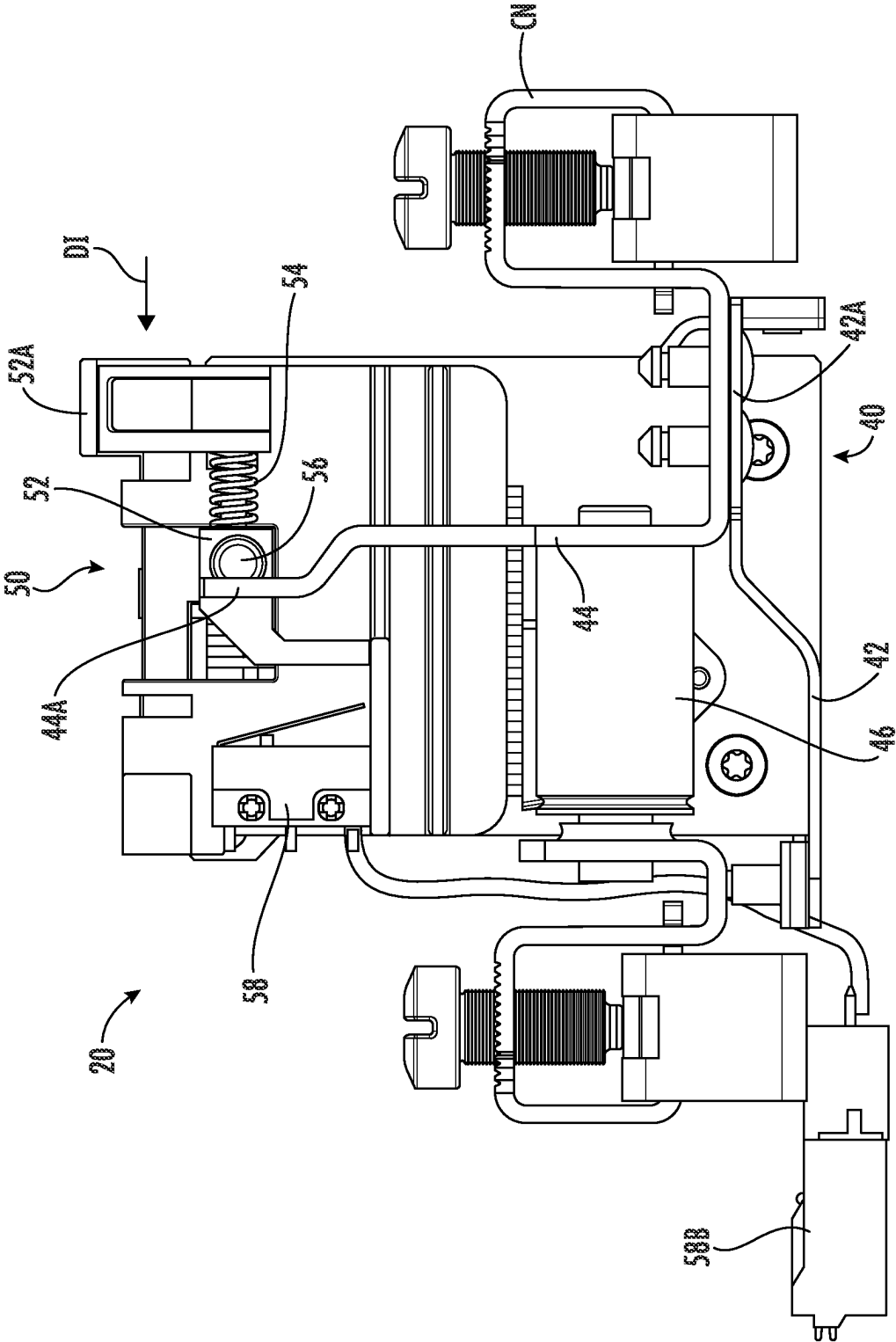


FIG. 39

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SURGE PROTECTIVE DEVICES**RELATED APPLICATION(S)**

The present application claims the benefit of and priority from U.S. Provisional Patent Application No. 63/493,026, filed Mar. 30, 2023, U.S. Provisional Patent Application No. 63/375,591, filed Sep. 14, 2022, and U.S. Provisional Patent Application No. 63/375,588, filed Sep. 14, 2022, the disclosures of which are incorporated herein by reference.

FIELD

The present invention relates to surge protective devices (SPDs).

BACKGROUND

Frequently, excessive voltage or current is applied across or through service lines that deliver power to residences and commercial and institutional facilities. Such excess voltage or current spikes (transient overvoltages and surge currents) may result from lightning strikes, for example. The above events may be of particular concern in telecommunications distribution centers, hospitals and other facilities where equipment damage caused by overvoltages and/or current surges is not acceptable and resulting downtime may be very costly.

Typically, sensitive electronic equipment may be protected against transient overvoltages and surge currents using surge protective devices (SPDs). For example, an overvoltage protection device may be installed at a power input of equipment to be protected, which is typically protected against overcurrents when it fails. Typical failure mode of an SPD is a short circuit. The overcurrent protection typically used is a combination of an internal thermal disconnect to protect the SPD from overheating due to increased leakage currents and an external fuse to protect the SPD from higher fault currents. Different SPD technologies may avoid the use of the internal thermal disconnect because, in the event of failure, they change their operation mode to a low ohmic resistance.

SPDs may use one or more active voltage switching/limiting components, such as a varistor or gas discharge tube, to provide overvoltage protection. These active voltage switching/limiting components may degrade at a rapid pace as they approach the end of their operational lifespans, which may result in their exhibiting continuous short circuit behavior.

SUMMARY

According to some embodiments, a surge protective device includes a first electrical terminal, a second electrical terminal, and an overvoltage protection circuit connected between the first electrical terminal and the second electrical terminal. The overvoltage protection circuit includes a spark gap assembly between the first electrical terminal and the second electrical terminal. The spark gap assembly includes a first spark gap (SG) electrode and a second SG electrode defining a spark gap therebetween, and a trigger circuit operative to ignite a main electric arc between the first and second SG electrodes across the spark gap. The trigger circuit includes a groove defined in the second SG electrode, and a trigger member disposed in the groove. The trigger member is operative to assist formation of a trigger arc.

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According to some embodiments, the trigger member is a semiconductive trigger member.

In some embodiments, the semiconductive trigger member is formed of a semiconductive ceramic.

In some embodiments, the semiconductive ceramic is selected from the group consisting of zinc oxide, barium titanate, and silicon carbide.

According to some embodiments, the trigger circuit includes a trigger varistor, a trigger gas discharge tube, and/or a trigger resistive element in electrical series with the trigger member and in electrical parallel with the spark gap.

According to some embodiments, the trigger circuit includes a trigger electrode in electrical contact with the trigger member and connecting the trigger member in electrical parallel with first SG electrode.

In some embodiments, the trigger circuit includes a trigger varistor, a trigger gas discharge tube, and/or a trigger resistive element in electrical series with the trigger electrode and in electrical parallel with first SG electrode.

According to some embodiments, the spark gap assembly is a horn spark gap assembly.

In some embodiments, the trigger member is located at a first end of the spark gap, and the spark gap assembly includes an arc chute located at an opposing second end of the spark gap.

According to some embodiments, the surge protective device of claim 1 includes an active voltage-switching/limiting component in electrical series with the spark gap assembly.

In some embodiments, the active voltage-switching/limiting component includes a varistor or a gas discharge tube.

In some embodiments, the trigger circuit has a trigger threshold flashover voltage for initiating electrical flashover between the first and second SG electrodes that is less than a threshold flashover voltage that would initiate electrical flashover between the first and second SG electrodes in the absence of the trigger circuit.

According to some embodiments, the surge protective device is a surge protective device module including a spark gap module housing, and the spark gap assembly is disposed in the spark gap module housing.

In some embodiments, the trigger member is located at a first end of the spark gap, and the spark gap module housing includes an arc gas recirculation channel configured to direct a flow of arc gas from a second end of the spark gap opposite the first end of the spark gap to an ignition region of the spark gap adjacent the trigger member.

In some embodiments, the spark gap module housing includes an intake port configured to direct a flow ambient air into an ignition region of the spark gap adjacent the trigger member.

According to some embodiments, a surge protective device includes a first electrical terminal, a second electrical terminal, and an overvoltage protection circuit connected between the first electrical terminal and the second electrical terminal. The overvoltage protection circuit includes a spark gap assembly between the first electrical terminal and the second electrical terminal. The spark gap assembly includes a first spark gap (SG) electrode and a second SG electrode defining a spark gap therebetween, and a trigger circuit operative to ignite a main electric arc between the first and second SG electrodes across the spark gap. The trigger circuit includes a semiconductive trigger member operative to assist formation of a trigger arc. The semiconductive trigger member is formed of a semiconductive ceramic.

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In some embodiments, the semiconductive ceramic is selected from the group consisting of zinc oxide, barium titanate, and silicon carbide.

In some embodiments, the trigger circuit includes a trigger varistor, a trigger gas discharge tube, and/or a trigger resistive element in electrical series with the trigger member and in electrical parallel with the spark gap.

According to some embodiments, the trigger circuit includes a trigger electrode in electrical contact with the trigger member and connecting the trigger member in electrical parallel with first SG electrode.

In some embodiments, the trigger circuit includes a trigger varistor, a trigger gas discharge tube, and/or a trigger resistive element in electrical series with the trigger electrode and in electrical parallel with first SG electrode.

According to some embodiments, the spark gap assembly is a horn spark gap assembly.

In some embodiments, the trigger member is located at a first end of the spark gap, and the spark gap assembly includes an arc chute located at an opposing second end of the spark gap.

According to some embodiments, the surge protective device includes an active voltage-switching/limiting component in electrical series with the spark gap assembly.

In some embodiments, the active voltage-switching/limiting component includes a varistor or a gas discharge tube.

In some embodiments, the trigger circuit has a trigger threshold flashover voltage for initiating electrical flashover between the first and second SG electrodes that is less than a threshold flashover voltage that would initiate electrical flashover between the first and second SG electrodes in the absence of the trigger circuit.

According to some embodiments, the surge protective device is a surge protective device module including a spark gap module housing, and the spark gap assembly is disposed in the spark gap module housing.

In some embodiments, the trigger member is located at a first end of the spark gap, and the spark gap module housing includes an arc gas recirculation channel configured to direct a flow of arc gas from a second end of the spark gap opposite the first end of the spark gap to an ignition region of the spark gap adjacent the trigger member.

In some embodiments, the spark gap module housing includes an intake port configured to direct a flow ambient air into an ignition region of the spark gap adjacent the trigger member.

In some embodiments, the spark gap assembly is a horn spark gap assembly and the spark gap assembly includes an arc chute. The arc chute includes a set of deion plates including: a plurality of first deion plates each having a first slot having a first profile; and a plurality of second deion plates each having a second slot having a second profile different than the first profile. The first and second deion plates are arranged in spaced apart relation and in alternating series along a chute axis such that the first and second deion plates define a series of arc chute spark gaps extending along the chute axis.

In some embodiments, the overvoltage protection circuit includes an active voltage-switching/limiting component, the spark gap assembly is in electrical series with the active voltage-switching/limiting component between the first electrical terminal and the second electrical terminal, the spark gap assembly is a horn spark gap assembly, and the spark gap assembly includes an arc chute.

According to some embodiments, a surge protective device includes a first electrical terminal, a second electrical terminal, and an overvoltage protection circuit connected

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between the first electrical terminal and the second electrical terminal. The overvoltage protection circuit includes a spark gap assembly including a horn spark gap assembly, and an arc chute. The arc chute includes a set of deion plates including a plurality of first deion plates each having a first slot having a first profile, and a plurality of second deion plates each having a second slot having a second profile different than the first profile. The first and second deion plates are arranged in spaced apart relation and in alternating series along a chute axis such that the first and second deion plates define a series of arc chute spark gaps extending along the chute axis.

In some embodiments, the first and second slots cross one another along the chute axis.

According to some embodiments, a surge protective device includes a first electrical terminal, and a second electrical terminal, an overvoltage protection circuit connected between the first electrical terminal and the second electrical terminal. The overvoltage protection circuit includes an active voltage-switching/limiting component, and a spark gap assembly in electrical series with the active voltage-switching/limiting component between the first electrical terminal and the second electrical terminal. The spark gap assembly includes a horn spark gap assembly, an arc chute, and a spark gap trigger circuit operative to ignite the horn spark gap assembly.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which form a part of the specification, illustrate embodiments of the present invention.

FIG. 1 is a block diagram that illustrates an electrical power supply circuit including a surge protective device (SPD) module in accordance with some embodiments.

FIG. 2 is a schematic view of an SPD circuit and module in accordance with some embodiments.

FIG. 3 is a perspective view of the SPD module in accordance with the schematic view of FIG. 2.

FIG. 4 is a perspective view of the SPD module of FIG. 3 with a portion of the housing removed.

FIG. 5 is a fragmentary, perspective view of the SPD module of FIG. 3.

FIG. 6 is a fragmentary, side view of the SPD module of FIG. 3.

FIG. 7 is a fragmentary, cross-sectional view of the SPD module of FIG. 3 taken along the line 7-7 of FIG. 6.

FIG. 8 is an exploded, fragmentary, perspective view of a spark gap assembly of the SPD module of FIG. 3.

FIG. 9 is a fragmentary, cross-sectional view of the SPD module of FIG. 3 taken along the line 9-9 of FIG. 7.

FIG. 10 is an enlarged, fragmentary, side view of an SPD module according to further embodiments including an electrically resistive trigger member.

FIG. 11 is an enlarged, fragmentary, side view of an SPD module according to further embodiments including an MOV trigger member.

FIG. 12 is perspective view of the MOV trigger member of the SPD module of FIG. 11.

FIG. 13 is an enlarged, fragmentary, side view of an SPD module according to further embodiments including a semiconductive trigger member.

FIG. 14 is an enlarged, fragmentary, side view of an SPD module according to further embodiments including an electrically resistive trigger member.

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FIG. 15 is an enlarged, fragmentary, side view of an SPD module according to further embodiments including a MOV trigger member.

FIG. 16 is an enlarged, fragmentary, side view of an SPD module according to further embodiments including an electrically resistive trigger member.

FIG. 17 is a fragmentary, cross-sectional view of the SPD module of FIG. 16 taken along the line 17-17 of FIG. 16.

FIG. 18 is an enlarged, fragmentary, perspective view of an SPD module according to further embodiments.

FIG. 19 is a fragmentary, side view of the SPD module of FIG. 18.

FIG. 20 is an enlarged, fragmentary, side view of an SPD module according to further embodiments.

FIG. 21 is an enlarged, fragmentary, side view of an SPD module according to further embodiments.

FIG. 22 is an enlarged, fragmentary, side view of an SPD module according to further embodiments.

FIG. 23 is a perspective view of an SPD module according to further embodiments.

FIG. 24 is a schematic view of an SPD circuit of the SPD module of FIG. 23.

FIG. 25 is a fragmentary, perspective view of the SPD module of FIG. 23.

FIG. 25 is a fragmentary, perspective view of the SPD module of FIG. 23.

FIG. 26 is a fragmentary, perspective view of a spark gap assembly of the SPD module of FIG. 23.

FIG. 27 is a fragmentary, side view of the spark gap assembly of FIG. 26.

FIG. 28 is an exploded, fragmentary, perspective view of the spark gap assembly of FIG. 26.

FIG. 29 is a fragmentary, cross-sectional view of the spark gap assembly of FIG. 26 taken along the line 29-29 of FIG. 25.

FIG. 30 is a fragmentary, cross-sectional view of the SPD module of FIG. 23 taken along the line 30-30 of FIG. 29.

FIG. 31 is a fragmentary, side view of the SPD module of FIG. 23.

FIG. 32 is an exploded, perspective view of a deion chamber of the SPD module of FIG. 23.

FIG. 33 is a fragmentary, cross-sectional view of the SPD module of FIG. 23 taken along the line 33-33 of FIG. 23.

FIG. 34 is a perspective view of an SPD assembly according to further embodiments, wherein the SPD module includes the SPD module of FIG. 23.

FIG. 35 is a block diagram that illustrates an electrical power supply circuit including the SPD assembly of FIG. 34.

FIG. 36 is a cross-sectional view of the SPD assembly of FIG. 34 taken along the line 36-36 of FIG. 34.

FIG. 37 is an exploded, perspective view of the SPD assembly of FIG. 34.

FIG. 38 is a fragmentary, front view of the SPD assembly of FIG. 34.

FIG. 39 is a fragmentary, side view of the SPD assembly of FIG. 34.

DETAILED DESCRIPTION OF EMBODIMENTS

The present invention now will be described more fully hereinafter with reference to the accompanying drawings, in which illustrative embodiments of the invention are shown. In the drawings, the relative sizes of regions or features may be exaggerated for clarity. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather,

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these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art.

It is noted that aspects described with respect to one embodiment may be incorporated in different embodiments although not specifically described relative thereto. That is, all embodiments and/or features of any embodiments can be implemented separately or combined in any way and/or combination. Moreover, other apparatus, methods, and systems according to the inventive concept will be or become apparent to one with skill in the art upon review of the following drawings and detailed description. It is intended that all such additional apparatus, methods, and/or systems be included within this description, be within the scope of the present inventive subject matter, and be protected by the accompanying claims.

As used herein, “monolithic” means an object that is a single, unitary piece formed or composed of a material without joints or seams. Alternatively, a unitary object can be a composition composed of multiple parts or components secured together at joints or seams.

With reference to FIGS. 2-9, a surge protective device (SPD) in the form of an SPD unit or SPD module 100 according to some embodiments is shown therein. The SPD module 100 includes an SPD electrical circuit 101 according to some embodiments.

In some embodiments, the SPD electrical circuit 101 or the SPD module 100 is provided, installed and used as a component in a protection circuit of a power supply circuit 10 as shown in FIG. 1, for example. In the power supply circuit 10, the SPD electrical circuit 101 is in electrical parallel across sensitive equipment. The SPD module 100 is designed to protect the sensitive equipment from overvoltages and current surges. The SPD module 100 may also be connected to the power source via an upstream second fuse or circuit breaker 12.

The SPD module 100 includes a module housing 102, a first or phase electrical terminal electrode 104, a second or ground electrical terminal electrode 106, and an overvoltage protection circuit (OPC) 110. The OPC 110 is disposed in the housing 102, and is electrically connected between the terminals 104 and 106 to form the SPD electrical circuit 101.

The housing 102 may be formed of any suitable electrically insulating material (e.g., an insulating polymer).

The first terminal electrode 104 includes a terminal contact or terminal 104A and a GDT contact 104B. The second terminal electrode 106 includes a terminal contact or terminal 106A and an electrode contact 106B.

The OPC 110 includes an active voltage-switching or active voltage limiting component 112 (referred to herein as a “voltage-switching/limiting component”) and a spark gap assembly 120. In some embodiments and as illustrated in the figures, the voltage-switching/limiting component 112 is a gas discharge tube (GDT).

In some embodiments, the GDT 112 has a rating of at least 12.5 kA of lightning impulse current.

In other embodiments, the OPC 110 may include an active voltage-switching or active voltage limiting component (referred to herein as a “voltage-switching/limiting component”) other than a GDT in place of the GDT 112 or in addition to the GDT 112.

In some embodiments, the OPC 110 is a varistor-based overvoltage protection circuit and the voltage-switching/limiting component 112 is a varistor. In some embodiments, the voltage-switching/limiting component 112 is a metal oxide varistor (MOV). For example, in some embodiments the OPC 110 is a varistor-based SPD as disclosed in U.S.

Pat. No. 8,743,525 to Xepapas et al., the disclosure of which is incorporated herein by reference.

The voltage-switching/limiting component **112** may also be another type of voltage-switching/limiting surge protective device. Other types of voltage-switching/limiting component **112** that may form, or form a part of, the OPC **110** may include spark gap devices, multi-cell GDTs (e.g., as disclosed in U.S. Pat. No. 10,685,805 to Rozman and U.S. Pat. No. 10,186,842 to Rozman, the disclosures of which are incorporated herein by reference), diodes, or thyristors.

In some embodiments, only a single voltage-switching/limiting component **112** is included in OPC **110**. In some embodiments, the OPC **110** includes or consists of only the active voltage-switching/limiting component(s) **112**, the spark gap assembly **120** and associated electrical connections, if any.

The OPC **110** may include a plurality of voltage-switching/limiting components **112**. The OPC **110** may include one or more voltage-switching/limiting components **110** in combination with other electrical components. In some embodiments, the OPC **110** includes multiple varistors (connected in electrical parallel or series between the module terminals), multiple GDTs (e.g., connected in electrical series), and/or both varistor(s) and GDT(s) (e.g., connected in electrical series with the varistor(s)), and/or other circuit elements, such as resistors, inductors, or capacitors.

The GDT **112** includes opposed electrical contact terminals **112A**, **112B**.

The spark gap assembly **120** includes a spark gap subassembly or horn spark gap assembly **122**, a spark gap (SG) trigger circuit **150**, and a deion chamber or arc chute **140**. The SG trigger circuit **150** is located at an inner end **120A** of the spark gap assembly **120** and the arc chute **140** is located at an opposing outer end **120B** of the spark gap assembly **120**.

The horn spark gap assembly **122** includes a first spark gap (SG) electrode **124**, a second spark gap (SG) electrode **126**, a spark gap **130**, and laterally opposed side or containment walls **128**. The spark gap **130** is defined between the electrodes **124**, **126**.

The first SG electrode **124** (FIG. 8) has an inner end **124A** and an opposing outer end **124B**. The first SG electrode **124** includes (extending sequentially from the inner end **124A** to the outer end **124B**) a contact section **124C**, a trigger section **124T**, an ignition section **124I**, a running section **124R**, and an extinguishing section **124E**.

The second SG electrode **126** (FIG. 8) has an inner end **126A** and an opposing outer end **126B**. The second SG electrode **124** includes (extending sequentially from the inner end **126A** to the outer end **126B**) a contact section **126C**, a trigger section **126T**, an ignition section **126I**, a running section **126R**, and an extinguishing section **126E**.

The spark gap **130** extends from an inner end **130A** and to an opposing outer end **130B**. With reference to (FIGS. 6 and 9), the spark gap **130** includes (extending sequentially from the inner end **130A** to the outer end **130B**) an ignition region **132I**, a running region **132R**, and an extinguishing region **132E**. The spark gap **130** expands, widens or flares outward in a direction DA from the end **130A** to the end **130B**. As a result, the spark gap **130** has a first width WI in the ignition region **132I**, a second width WR in the running region **132R**, and a third width WE in the extinguishing region **132E**. Width WR is great than width WI, and width WE is greater than width WR. Width WR varies and expands along the length of the running region **132R**. In some embodiments, the spark gap **130** expands smoothly from the

ignition region **132I** to the extinguishing region **132E**. In some embodiments, the SG electrodes **124**, **126** each have a curved profile.

In some embodiments, the spark gap width WI (FIG. 9; i.e., the separation distance between the SG electrodes **124**, **126**) is at least 0.5 mm and, in some embodiments, is at least 1 mm. In some embodiments, the spark gap width WI is in the range of from about 0.5 mm to 2 mm.

In some embodiments, the spark gap width WE (FIG. 9) is in the range of from about 20 mm to 40 mm. In some embodiments, the spark gap width WE is in the range of from about 1000 to 4000 percent greater than the spark gap width WI.

The SG electrodes **124**, **126** may be formed of any suitable electrically conductive metal. In some embodiments, the SG electrodes **124**, **126** are formed of copper, copper-steel bimetal, or any other combination of metals in bimetal composition.

The containment walls **128** may be formed of any suitable material. In some embodiments, the containment walls **128** are formed of ceramic, polymer or plastic, insulating paper, ferromagnetic material insulated by paper, plastic or ceramic, or ferromagnetic material coated with a polymer or plastic.

With reference to FIGS. 5 and 9, the arc chute **140** is located at the outer end **130B** of the horn spark gap assembly **122** at, in or proximate the extinguishing region **132E**. The arc chute **140** includes a set **143** or series of electrically conductive arc, chute or deion electrodes or plates **142** supported by electrically insulating supports **146**. The deion plates **142** are separated by deion plate arc gaps **144**.

The SG trigger circuit **150** includes the ignition region **132I** of the spark gap **130** and a semiconductive trigger member **154**. The semiconductive member **154** is positioned in or adjacent the gap **132I** and interposed between the gap electrode trigger sections **124T** and **126T**. The semiconductive trigger member **154** is connected in electrical series between the SG electrodes **124**, **126**.

The semiconductive trigger member **154** extends from a first end face **154A** to an opposing second end face **154B**. The first end face **154A** electrically engages or contacts the electrode trigger section **124T**. The second end face **154B** electrically engages or contacts the electrode trigger section **126T**.

The semiconductive trigger member **154** may be formed of any suitable semiconductive material.

In some embodiments, the semiconductive trigger member **154** is formed of a composition including a mixture of a polymeric material (e.g., a rubber or a plastic) as a nonconductive matrix and an electrically conductive filler. In some embodiments, the nonconductive matrix includes silicone. In some embodiments, the conductive filler is graphite powder. In some embodiments, the conductive filler is expanded graphite powder.

The conductive filler may be a material other than graphite powder having a relatively high secondary emission, such as beryllium oxide (BeO), magnesium oxide (MgO), or gallium phosphide (GaP).

In some embodiments, the semiconductive member **154** is formed of a semiconductive ceramic. In some embodiments, the semiconductive member **1254** is formed of ZnO (zinc oxide) or BaTiO₃ (Barium titanate) or SiC (silicon carbide), with different dopants (oxides, metals).

The semiconductive member **154** can be rigid or flexible depending on the polymer matrix. In some embodiments, the

semiconductive member **154** has a porous structure. In some embodiments, the semiconductive member **154** has an amorphous structure.

In some embodiments, the semiconductive material of the semiconductive member **154** has a specific electrical resistance in the range of from 1 to 10,000 Ωcm , and a dielectric constant in the range of from 1 to 10,000. The semiconductive material may have a positive or negative temperature coefficient.

In some embodiments, the semiconductive trigger member **154** has a thickness **T3** (FIG. 7) of at least 0.5 mm and, in some embodiments, of at least 1 mm. In some embodiments, the thickness **T3** is in the range of from about 0.5 mm to 2 mm.

With reference to FIG. 6, the terminal **104A** is electrically connected to the GDT terminal **112A** by the GDT contact **104B**. The SG electrode **124** is electrically connected to the GDT terminal **112B** by the contact section **124C**. The terminal **106A** is electrically connected to the SG electrode **126** by the contact section **126C**. Under some conditions, as discussed below, the SG electrode **124** is electrically connected to the SG electrode **126** by electric arcing to complete the electrical circuit from the terminal **104** to the terminal **106**.

The SG electrodes **124**, **126** and the semiconductive member **154** define gap open regions or volumes in the ignition region **132I** adjacent or around the semiconductive member **154** (i.e., around the volume of the gap ignition region **132I** filled or occupied by the semiconductive member **154**). Opposed regions of the SG electrodes **124**, **126** in the ignition region **132I** are exposed (i.e., are not covered by the semiconductive member **154**). The ignition region **132I** serves as a trigger gap.

The spark gap **130** at the gap ignition region **132I** has a prescribed threshold flashover voltage. In some embodiments, the prescribed threshold flashover voltage is less than 1500V. In some embodiments, the prescribed threshold flashover voltage is in the range of from about 300V to 1500V. When a voltage is applied across the SG electrodes **124**, **126** that is less than the threshold flashover voltage, the applied voltage will not ignite or initiate arc flashover between the SG electrodes **124**, **126**. As discussed below, when a voltage is applied across the SG electrodes **124**, **126** that is greater than or equal to the threshold flashover voltage, the applied voltage will ignite or initiate arc flashover between the SG electrodes **124**, **126**.

In some embodiments, the length **LI** (FIG. 9) of the ignition region **132I** is in the range of from about 1 mm to 5 mm.

In some embodiments, the height **DI** (FIG. 7; i.e., the lateral dimension of the spark gap **130** and the width of the SG electrodes **124**, **126**) of the ignition region **132I** is in the range of from about 3 mm to 8 mm.

The SPD module **100** may operate as follows in service.

As described, the SPD circuit **101** includes a high surge current rated GDT **112** in electrical series with a horn spark gap assembly **122**. The horn spark gap assembly **122** is provided with a trigger element **154**.

According to some embodiments of the inventive concept, the SPD module **100** is configured to operate under two different conditions: 1) normal (stand by) operation; and 2) an overvoltage or current surge event in which the SPD module **100** is designed to shunt an SPD surge impulse current to ground.

The SPD module **100** is designed to shunt an SPD surge impulse current to ground in response to an overvoltage or current surge event.

The SPD module **100** is configured to operate in three alternative modes: 1) a standby mode; 2) a surge current mode; and 3) a follow current extinguishing mode.

The terminal **104** is electrically connected to the Line (L) of the circuit **10**, and the terminal **106** is electrically connected to the Ground (G) of the circuit **10** (FIG. 1).

During normal operation, the OPC **110** practically acts as an insulator. The voltage applied across the GDT **112** and across the spark gap **130** is insufficient to initiate current through the GDT **112** or to initiate a spark across the spark gap **130**.

When an overvoltage or current surge event (e.g., a transient power surge) applies a surge impulse current to the SPD circuit **101**, the OPC **110** will temporarily go to a low impedance state (e.g., effectively becoming a short circuit). The voltage-switching/limiting component **112** (e.g., GDT) of the OPC **110** is designed to shunt the surge impulse current associated with such events to ground to protect sensitive equipment. The SPD surge impulse current may be on the order of tens of kA, but will typically last only a short duration (in the range of from about tens of microseconds to a few milliseconds).

During the surge event, the voltage applied across the ignition region **132I** of the spark gap **130** by the surge event exceeds the prescribed threshold flashover voltage of the ignition region **132I**. In response, the voltage across the ignition region **132I** initiates electrical arc flashover or trigger arc AT (FIG. 9) across the spark gap **130** in the ignition region **132I**. That is, electrical arcing AT is generated between the SG electrode ignition sections **124I** and **126I**.

The initiation of the trigger arc AT is assisted by the semiconductive trigger member **154** in response to the overvoltage developed across the ignition section **124I** and the ignition section **126I**. At the beginning of the surge current, current conduction occurs through the bulk of the semiconductive member body and along the exterior surface **154C** of the semiconductive trigger member **154**. Very quickly thereafter (e.g., within less than 1 microsecond), the flashover of the trigger arc AT occurs so that most of the surge current is bypassed through the arc column(s) established between the SG electrodes **124**, **126**. By diverting the current around the semiconductive trigger member **154**, degradation of the semiconductive trigger member **154** is prevented or reduced.

Shortly after the start of the trigger arc AT (e.g., within microseconds), the current flow triggers an electric arc flashover or surge arc AM (as illustrated in FIG. 9) across the spark gap **130**. The SG trigger circuit **150** thereby ignites the horn spark gap assembly **122** and the full surge current is thereby conducted through the spark gap assembly **122** via the surge arc AM.

The GDT **112** and the spark gap assembly **120** are thereby each triggered changing their states to low impedance. The surge current flows from the terminal **104A** to the terminal **106A** through the GDT **112** and the spark gap **130**, thereby diverting the surge current to ground. During the surge event, the arcing or arc column AM across the spark gap **130** may be located entirely or mainly in the arc ignition region **132I** or partially in the running region **132R**.

The SPD **100** then enters its follow current mode. Once the surge current is diverted to the ground, the GDT **112** and the spark gap **122** remain electrically conductive, causing additional follow current from the system to flow into the SPD **100**. This causes the arc column to move in an arc migration direction DA (from the end **130A** toward the end **130B**) along the arc running region **132R** of the spark gap

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122 to the arc extinguishing region 132E. FIG. 9 illustrates the arc column in the arc running region 132R as an electric arc AR, and the arc column in the arc extinguishing region 132E as an electric arc AE.

In the arc extinguishing region 132E, the follow current is redirected to (is conducted to, or flashes over to or arcs to) the deion plates 142 from the SG electrode 124. The current then flows through the set 143 of deion plates to the SG electrode 126 by arcing AD between the deion plates 143.

Each deion plate arc AD creates a voltage drop between the SG electrodes 124, 126 and the terminals 104, 106. These voltage drops add together increasing the voltage between the SG electrodes 124, 126 to relatively high values, until the voltage across the arc chute 140 is higher than the power system voltage. The voltage drop developed on the arc chute 140 opposes the mains voltage thus the follow current is rapidly reduced and eventually extinguished.

These voltage drops also reduce the voltage across the spark gap 122 to relatively low values, until the voltage across the spark gap 122 is less than the ignition voltage necessary to sustain the arcing between the SG electrodes 124, 126. The voltage across the spark gap 122 is then also less than necessary to trigger flashover between the ignition sections 124I, 126I or to conduct current through or along the surface of the semiconductive trigger member 154. That is, at this lower voltage, the semiconductive trigger member 154 operates as an electrical insulator between the SG electrodes 124, 126. The spark gap assembly 122 is thereby opened at the spark gap 130 and the follow current through the spark gap assembly 122 and the SPD module 100 is cut off or interrupted. The SPD module 100 returns to its standby mode.

Thus, it will be appreciated that the electrical arc flashover AT triggers the electric arc flashover AM, which during follow current expands and propagates to the arc chute 140. The expansion and propagation are driven by electromagnetic and acoustical forces. Once the arc AE enters the arc chute 140, it is divided or split into a plurality of smaller arcs in the arc chute 140 and cooled down until the arc is eventually extinguished. This causes the SPD module 100 to return to its high impedance (standby) mode.

The semiconductive trigger member 154 functions as a spark gap trigger that assists in initiating the flashover between the SG electrodes 124, 126. The spark gap initiation region 132I is the closest distance between the SG electrodes 124, 126 and creates a small chamber that serves as the arc initiation area. In this chamber, the spark gap 130 has the dimensions WI, DI, and LI. The minimum separation distance WI between the electrodes 124, 126 at the trigger member 154 can prevent melting and soldering of the electrodes 124, 126 during surges. Because the semiconductive trigger member 154 is provided, the threshold flashover voltage of the gap 130 is less than the voltage that would be required to initiate the flashover across the gap 130 in the absence of the semiconductive trigger member 154. By reducing the minimum required flashover voltage, the semiconductive trigger member 154 lowers the protection level of the SPD electrical circuit 101.

The horn shape of the spark gap 130 facilitates the path of the arc AM. The geometry of the SG electrodes guides the arc AM from the ignition region 132I to the extinguishing region 132E. The containment walls 128 firmly restrict or contain the path of the arcing in the lateral directions DL (FIG. 7).

The GDT 112 is provided in front of the trigger member 154 because the trigger member 154 has some resistance and

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thus will conduct the current on nominal voltage. By including the GDT 112, the OPC 110 is made leakage current free (and, thus, the SPD module 100 is leakage free). Additionally, the GDT 112 takes or absorbs some of the energy from surges and helps the trigger circuit 150 to operate more effectively.

With reference to FIG. 10, an SPD module 200 and SPD circuit 201 according to further embodiments in shown therein. The SPD module 200 and circuit 201 are constructed in the same manner as the SPD module 100 and circuit 101, except as follows. For the purpose of explanation, the numerals used to reference parts of the SPD module 100 in FIGS. 2-9 are likewise used to designate the same parts in FIG. 10.

The SPD module 200 includes an SG trigger circuit 250 constructed in the same manner as the SG trigger circuit 150, except that an electrically resistive trigger member 254 is provided in place of the semiconductive trigger member 154.

The end faces 254A, 254B of the resistive trigger member 254 may be in direct contact with the trigger sections 124T, 126T of the SG electrodes 124, 126. Alternatively, the end faces 254A, 254B may be directly electrically connected to the trigger sections 124T, 126T by intervening metal layers.

The trigger member 254 may be formed of any suitable material. In some embodiments, the trigger member 254 is formed of a mixture including metal oxide powder, glass powder and/or graphite powder that is sintered. In some embodiments, the resistive material forming the trigger member 254 is a sintered body of ZnO, MgO with the addition of Al₂O₃ in a suitable percentage, or a sintered mixture of metal powder with SiO₂.

The SPD module 200 can be used and may operate in the manner described for the SPD module 100. In this case, the resistive trigger member 254 serves in place of the semiconductive member 154 to trigger the flashover AT.

With reference to FIGS. 11 and 12, an SPD module 300 and SPD circuit 301 according to further embodiments in shown therein. The SPD module 300 and circuit 301 are constructed in the same manner as the SPD module 100 and circuit 101, except as follows. For the purpose of explanation, the numerals used to reference parts of the SPD module 100 in FIGS. 2-9 are likewise used to designate the same parts in FIG. 11.

The SPD module 300 includes an SG trigger circuit 350 constructed in the same manner as the SG trigger circuit 150, except that a metal oxide varistor (MOV) trigger member 354 (FIG. 12) is provided in place of the semiconductive trigger member 154.

The end faces 354A, 354B of the MOV trigger member 354 may be in direct contact with the trigger sections 124T, 126T of the SG electrodes 124, 126. Alternatively, the end faces 354A, 354B may be directly electrically connected to the trigger sections 124T, 126T by intervening metal layers 355 (e.g., thin metallization layers).

The MOV trigger member 354 may be formed of any suitable material. The MOV trigger member material may be any suitable material conventionally used for varistors, namely, a material exhibiting a nonlinear resistance characteristic with applied voltage. Preferably, the resistance becomes very low when a prescribed voltage is exceeded. The varistor material may be a doped metal oxide or silicon carbide, for example. Suitable metal oxides include zinc oxide compounds.

The SPD module 300 can be used and may operate in the manner described for the SPD module 100. In this case, the

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MOV trigger member 354 serves in place of the semiconductive member 154 to trigger the flashover AT.

In some embodiments, the MOV trigger member 354 includes an MOV body 354C (formed of a varistor material as described above) and an electrically insulating passivation layer 354D surrounding portions of the MOV body 354C. A surface region 354E of the MOV body 354C is not covered by the passivation layer 354D. The surface region 354E provides a surface region for flashover of the trigger arc AT. In some embodiments, the surface region 354E is a relatively narrow region on a side wall of the MOV body 354C and the remainder of the side wall is covered by the passivation layer 354D. The passivation layer 354D may be formed of glass, for example.

With reference to FIG. 13, an SPD module 400 and SPD circuit 401 according to further embodiments in shown therein. The SPD module 400 and circuit 401 are constructed in the same manner as the SPD module 100 and circuit 101, except as follows. For the purpose of explanation, the numerals used to reference parts of the SPD module 100 in FIGS. 2-9 are likewise used to designate the same parts in FIG. 13.

The SPD module 400 includes an SG trigger circuit 450 including a semiconductive trigger member 454 constructed in the same manner as the semiconductive trigger member 154. The SG trigger circuit 450 is constructed in the same manner as the SG trigger circuit 150, except that an electrical insulator layer or film 456 contacts the electrode trigger section 126T and is interposed between the semiconductive trigger member 454 and the SG electrode 126. The first end face 454A of the trigger member 454 electrically engages the electrode trigger section 124T. The second end face 454A of the trigger member 454 engages the insulator layer 456. The trigger member 454 is electrically insulated from the electrode 126 by the insulator layer 456.

The insulator layer 456 may be formed of any suitable electrically insulating material. In some embodiments, the insulator layer 456 is formed of a polymeric material (e.g., a rubber or a plastic). In some embodiments, the insulator layer 456 is formed of polyimide (e.g., KAPTON film) or ceramic.

In some embodiments, the insulator layer 456 has a thickness T4 in the range of from about 0.1 mm to 0.3 mm.

The SPD module 400 can be used and may operate in the manner described for the SPD module 100. In this case, the MOV trigger member 454 serves in place of the semiconductive member 154 to trigger the flashover AT.

With reference to FIG. 14, an SPD module 500 and SPD circuit 501 according to further embodiments in shown therein. The SPD module 500 and circuit 501 are constructed in the same manner as the SPD module 200 and circuit 201, except as follows. For the purpose of explanation, the numerals used to reference parts of the SPD module 100 in FIGS. 2-9 are likewise used to designate the same parts in FIG. 14.

The SPD module 500 includes an SG trigger circuit 550 including a resistive trigger member 554 constructed as described for the resistive trigger member 254. The SG trigger circuit 550 is constructed in the same manner as the SG trigger circuit 250, except that an electrical insulator layer or film 556 contacts the electrode trigger section 126T and is interposed between the trigger member 554 and the SG electrode 126. The first end face 554A of the trigger member 554 electrically engages the electrode trigger section 124T. The second end face 554A of the trigger member

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554 engages the insulator layer 556. The trigger member 554 is electrically insulated from the electrode 126 by the insulator layer 556.

With reference to FIG. 15, an SPD module 600 and SPD circuit 601 according to further embodiments in shown therein. The SPD module 600 and circuit 601 are constructed in the same manner as the SPD module 300 and circuit 301, except as follows. For the purpose of explanation, the numerals used to reference parts of the SPD module 100 in FIGS. 2-9 are likewise used to designate the same parts in FIG. 15.

The SPD module 600 includes an SG trigger circuit 650 including an MOV trigger member 654 constructed in the same manner as the MOV trigger member 354. The SG trigger circuit 650 is constructed in the same manner as the SG trigger circuit 350, except that the SG trigger circuit 650 further includes a resistive material layer 660 and metal MOV electrodes 662A and 662B.

The end faces 654A and 654B of the MOV trigger member 654 are in contact with respective ones of the metal MOV electrodes 662A and 662B. The MOV electrodes 362A, 362B may be discrete metal films or thin metallization layers, for example.

The resistive material layer 660 is formed of an electrically resistive material as described above for the resistive trigger member 254. The resistive material layer 660 may be formed of a hard material or a polymer. In some embodiments, the resistive material layer 660 is porous.

The resistive material layer 660 is interposed between and electrically connects the MOV end face 654A and the electrode trigger section 624T. The MOV electrode 662B is interposed between the MOV end face 654B is electrically connected to the electrode trigger section 626T by the MOV electrode 662B.

In some embodiments, the resistive material layer 660 has a thickness T5 in the range of from about 0.1 mm to 0.5 mm.

In some embodiments, the MOV trigger member 654 has a thickness T6 in the range of from about 0.5 mm to 1.5 mm.

The SPD module 600 can be used and may operate in the manner described for the SPD module 100. In this case, the MOV trigger member 654 serves to trigger the flashover AT as described for the MOV trigger member 354. The resistive material layer 660 enables low triggering or flashover current and voltage levels, while the MOV trigger member 654 prevents and cuts off any continuation currents through the resistive film 660 after the surge arc AM is initiated.

With reference to FIG. 16, an SPD module 700 and SPD circuit 701 according to further embodiments in shown therein. The SPD module 700 and circuit 701 are constructed in the same manner as the SPD module 200 and circuit 201, except as follows. For the purpose of explanation, the numerals used to reference parts of the SPD module 100 in FIGS. 2-9 are likewise used to designate the same parts in FIG. 16.

The SPD module 700 includes an SG trigger circuit 750 constructed in the same manner as the SG trigger circuit 150, except that an electrically resistive trigger member 754 is provided in place of the resistive trigger member 254.

The resistive trigger member 754 includes an electrically nonconducting substrate 764 (e.g., ceramic) and a layer 760 of an electrical resistive material bonded to a face of the substrate 764 extending between the SG electrode trigger sections 124T, 126T. The resistive material layer 760 contacts each of the SG trigger sections 124T, 126T.

The resistive material layer 760 may be formed of an electrically resistive material as described above for the resistive trigger member 200. In some embodiments, the

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resistive material layer **760** may be porous. The resistive material layer **760** may be applied as a coating to the substrate **764**.

A discontinuity or gap **760C** is defined in the resistive material layer **760**. The gap **760C** may be formed by cutting or etching the resistive material layer **760**, for example.

In some embodiments and as illustrated in FIG. **17**, the gap **760C** is a slot or gap that extends fully through the thickness of the resistive material layer **760** to the substrate **764**. In other embodiments, the gap **760C** is a groove, slot or gap that extends only partially through the thickness of the resistive material layer **760** (i.e., the thickness of the resistive material layer **760** is reduced).

While only one gap **760C** is shown in the embodiment of FIGS. **16** and **17**, in other embodiments the resistive material layer **760** may be provided with multiple gaps **760C**.

In some embodiments, each gap **760C** has a width **W7** (FIG. **17**) of less than 0.2 mm.

The SPD module **700** can be used and may operate in the manner described for the SPD module **100**. In this case, the resistive trigger member **754** serves to trigger the flashover AT as described for the resistive trigger member **254**. The resistive material layer **760** enables low triggering or flash-over current and voltage levels.

With reference to FIGS. **18** and **19**, an SPD module **800** and SPD circuit **801** according to further embodiments in shown therein. The SPD module **800** and circuit **801** are constructed in the same manner as the SPD module **100** and circuit **101**, except as follows. For the purpose of explanation, the numerals used to reference parts of the SPD module **100** in FIGS. **2-9** are likewise used to designate the same parts in FIGS. **18** and **19**.

The SPD module **800** includes an SG trigger circuit **850** provided in place of the SG trigger circuit **150**. The SG trigger circuit **850** includes a trigger MOV member **868**, a trigger electrode **866**, and an electrically resistive film or layer **854**.

The resistive layer **854** surrounds a tip portion **866A** of the trigger electrode **866**. The resistive material layer **854** may be applied as a coating to the trigger electrode **866**. In some embodiments, the resistive material layer **854** fully surrounds the tip portion **866A**. In other embodiments, a portion or portions of the tip portion **866A** are not covered by the resistive material layer **854**.

The resistive layer **854** separates the tip portion **866A** from the SG electrodes **124**, **126**. The resistive layer **854** also spaces the trigger electrode **866** away from the SG electrode **126**.

In some embodiments, the resistive material layer **854** has a thickness **T8** in the range of from about 0.1 mm to 0.5 mm.

In some embodiments, the spacing distance **W8** between the trigger electrode **866** and the SG electrode **126** is in the range of from about 0.1 mm to 0.5 mm.

The SPD module **800** can be used and may operate in the manner described for the SPD module **100**. In this case, the resistive material layer **854** enables low triggering or flash-over current and voltage levels, while the trigger MOV member **868** prevents and cuts off any continuation currents through the resistive film **854** after the surge arc AM is initiated.

With reference to FIG. **20**, an SPD module **900** and SPD circuit **901** according to further embodiments in shown therein. The SPD module **900** and circuit **901** are constructed in the same manner as the SPD module **800** and circuit **801**, except as follows. For the purpose of explanation,

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the numerals used to reference parts of the SPD module **100** in FIGS. **2-9** are likewise used to designate the same parts in FIG. **20**.

The SPD module **900** includes an SG trigger circuit **950**. The SG trigger circuit **950** includes a trigger MOV member **968**, a trigger electrode **966**, and an electrically resistive film or layer **954**.

The SPD module **900** is constructed in the same manner as the SG trigger circuit **850** except that the resistive layer **954** is interposed between the trigger electrode **966** and the SG electrode **126** but does not surround a tip portion of the trigger electrode **966**. In some embodiments, the resistive layer **954** is a coated onto the SG electrode **126** and the trigger electrode **966** is placed against the resistive layer **954**.

The SPD module **900** can be used and may operate in the manner described for the SPD module **800**. The resistive material layer **954** enables low triggering or flashover current and voltage levels, while the trigger MOV member **968** prevents and cuts off any continuation currents through the resistive film **954** after the surge arc AM is initiated.

With reference to FIG. **21**, an SPD module **1000** and SPD circuit **1001** according to further embodiments in shown therein. The SPD module **1000** and circuit **1001** are constructed in the same manner as the SPD module **800** and circuit **801**, except as follows. For the purpose of explanation, the numerals used to reference parts of the SPD module **100** in FIGS. **2-9** are likewise used to designate the same parts in FIG. **21**.

The SPD module **1000** includes an SG trigger circuit **1050**. The SG trigger circuit **1050** includes a trigger MOV member **1068**, a trigger electrode **1066**, and an electrically resistive trigger member **1054**.

The resistive trigger member **1054** can be formed of a material as described for the resistive trigger member **254**.

In some embodiments, the resistive trigger member **1054** has a thickness **T9** in the range of from about 0.1 mm to 0.5 mm.

In some embodiments, the spacing distance **W9** between the trigger electrode **1066** and the SG electrode **126** is in the range of from about 0.1 mm to 0.5 mm.

The SPD module **1000** can be used and may operate in the manner described for the SPD module **800**. The resistive trigger member **1054** operates as described for the resistive trigger member **254**. The resistive trigger member **1054** enables low triggering or flashover current and voltage levels, while the trigger MOV member **1068** prevents and cuts off any continuation currents through or along the resistive trigger member **1054** after the surge arc AM is initiated.

With reference to FIG. **22**, an SPD module **1100** and SPD circuit **1101** according to further embodiments in shown therein. The SPD module **1100** and circuit **1101** are constructed in the same manner as the SPD module **100** and circuit **101**, except as follows. For the purpose of explanation, the numerals used to reference parts of the SPD module **100** in FIGS. **2-9** are likewise used to designate the same parts in FIG. **22**.

The SPD module **1100** includes an SG trigger circuit **1150**. The SG trigger circuit **1150** includes a trigger MOV member **1168**, a trigger electrode **1066**, and an electrically semiconductive trigger member **1154**.

The semiconductive trigger member **1154** can be formed of a material as described for the semiconductive trigger member **154** (e.g., a polymeric semiconductive material).

In some embodiments, the semiconductive trigger member **1154** has a thickness **T10** in the range of from about 0.1 mm to 0.5 mm.

In some embodiments, the spacing distance **W10** between the trigger electrode **1166** and the SG electrode **126** is in the range of from about 0.1 mm to 0.5 mm.

The SPD module **1100** can be used and may operate in the manner described for the SPD module **800**. The semiconductive trigger member **1154** operates as described for the resistive trigger member **154**. The semiconductive trigger member **1154** enables low triggering or flashover current and voltage levels, while the trigger MOV member **1154** prevents and cuts off any continuation currents through or along the semiconductive trigger member **1154** after the surge arc AM is initiated.

While the trigger circuits **850**, **950**, **1050** and **1150** have been described including trigger MOV members **868**, **968**, **1068**, **1168**, in other embodiments the trigger circuit may include other voltage-switching/limiting components in addition to or in place of the MOV member. For example, the trigger circuit may include a trigger GDT in place of the trigger MOV or in electrical series with the trigger MOV and in electrical parallel with the SG electrode **124**.

With reference to FIGS. **23-33**, an SPD module **1200** and SPD electrical circuit **1201** (FIG. **24**) according to further embodiments is shown therein. The SPD module **1200** and circuit **1201** are constructed in the same manner as the SPD module **100** and circuit **101**, except as follows. The SPD module **1200** and circuit **1201** may be installed and used as a component in a protection circuit of a power supply circuit **10** as shown in FIG. **1**, for example.

The SPD module **1200** includes a module housing **1202** and an overvoltage protection circuit (OPC) **1210**. The OPC **1210** is disposed in the housing **1202**, and is electrically connected between electrical terminals **1204A** and **1206A** to form the SPD electrical circuit **1201**.

The OPC **1210** includes a spark gap assembly **1220**, a first electrode member **1204**, and a second electrode member **1206**. Portions of the electrode members **1204**, **1206** form parts of the spark gap assembly **1220**.

The first electrode member **1204** includes the terminal **1204A**, a bridge portion **1204B**, and a first main or spark gap (SG) electrode **1224**. The second electrode member **1206** includes the terminal **1206A** and a second main or SG electrode **1226**.

The spark gap assembly **1220** includes a spark gap subassembly or horn spark gap assembly **1222**, a spark gap (SG) trigger circuit **1250**, and a deion chamber or arc chute **1240**. The SG trigger circuit **1250** is located at an inner end **1220A** of the spark gap assembly **1220** and the arc chute **1240** is located at an opposing outer end **1220B** of the spark gap assembly **1220**.

The horn spark gap assembly **1222** includes the first SG electrode **1224**, the second SG electrode **1226**, a spark gap **1230**, and laterally opposed side or containment walls **1228**. The spark gap **1230** is defined between the electrodes **1224**, **1226**.

The first SG electrode **1224** (FIG. **26**) has an inner end **1224A** and an opposing outer end **1224B**. The first SG electrode **1224** includes (extending sequentially from the inner end **1224A** to the outer end **1224B**) a trigger section **1224T**, an ignition section **1224I**, a running section **1224R**, and an extinguishing section **1224E**.

The second SG electrode **1226** (FIG. **28**) has an inner end **1226A** and an opposing outer end **1226B**. The second SG electrode **1226** includes (extending sequentially from the inner end **1226A** to the outer end **1226B**), a trigger section

1226T, an ignition section **1226I**, a running section **1226R**, and an extinguishing section **1226E**.

The second SG electrode **1226** further includes a channel or groove **1229** defined therein. The groove **1229** defines and extends along a groove longitudinal axis LG-LG and along a groove lateral axis WG-WG transverse or perpendicular to the groove longitudinal axis LG-LG. The groove longitudinal axis LG-LG is parallel to the lengthwise axis L-L of the second SG electrode **1226** and the ignition region **1232I**.

The groove **1229** includes a base wall **1229A** and an end wall **1229E** at the distal end of the base wall **1229A**.

In some embodiments and as illustrated, the base wall **1229A** slopes or angles an angle **A12** (FIG. **31**) relative to the inner surface **1226C** of the second SG electrode **1226** in the direction toward the ignition region **1232I**. That is, the base wall **1229A** slopes outwardly relative to the spark gap **1230** (or inwardly with respect to the thickness of the second SG electrode **1226**). In some embodiments, the angle **A12** is at least 5 degrees. In some embodiments, the angle **A12** is in the range of 1 to 10 degrees.

The groove **1229** extends laterally across the width of the trigger section **1226T** (i.e., transverse or perpendicular to the length of the second SG electrode **1226**). In some embodiments, the groove **1229** extends across the full width of the trigger section **1226T**.

In some embodiments, the groove **1229** has a depth **D12** (FIG. **31**) at the end wall **1229E** in the range of from 0.1 times the trigger member thickness **T15** to 1 times the trigger member thickness **T15**.

In some embodiments, the groove **1229** has a width **W12** (FIG. **29**) at the end wall **1229E** in the range of from 2 mm to 10 mm. The width **W12** can be the same as the width of the electrode **1226** or less.

The spark gap **1230** is an air gap. The spark gap **1230** extends from an inner end **1230A** and to an opposing outer end **1230B**. With reference to FIG. **27**, the spark gap **1230** includes (extending sequentially from the inner end **1230A** to the outer end **1230B**) a trigger region **1232T**, an ignition region **1232I**, a running region **1232R**, and an extinguishing region **1232E**. The spark gap **1230** expands, widens or flares outward in a direction **DA2** from the end **1230A** to the end **1230B**. As a result, the spark gap **1230** has a first width **W12** (FIG. **31**) in the ignition region **1232I** and the trigger region **1232T**, a second width **WR2** (FIG. **27**) in the running region **1232R**, and a third width **WE2** (FIG. **27**) in the extinguishing region **1232E**. Width **WR2** is great than width **W12**, and width **WE2** is greater than width **WR2**. Width **WR2** varies and expands along the length of the running region **1232R**. In some embodiments, the spark gap **1230** expands smoothly from the ignition region **1232I** to the extinguishing region **1232E**. In some embodiments, the SG electrodes **1224**, **1226** each have a curved profile.

In some embodiments, the ignition region spark gap width **W12** (FIG. **31**; i.e., the separation distance between the SG electrodes **1224**, **1226**) is at least 1 mm and, in some embodiments, is at least 0.5 mm. In some embodiments, the spark gap width **W12** is in the range of from about 0.5 mm to 3 mm.

In some embodiments, the length **LI2** (FIG. **27**) of the ignition region **1232I** is in the range of from about 0 mm to 7 mm.

In some embodiments, the depth **D14** (FIG. **29**; i.e., the lateral dimension of the spark gap **1230** and the width of the SG electrodes **1224**, **1226**) of the ignition region **1232I** is in the range of from about 2 mm to 10 mm.

In some embodiments, the extinguishing region spark gap width **WE2** (FIG. **27**) is in the range of from about 15 mm

to 50 mm. In some embodiments, the spark gap width WE2 is in the range of from about 500 to 5000 percent greater than the spark gap width WI2.

The SG electrodes 1224, 1226 may be formed of any suitable electrically conductive metal. In some embodiments, the SG electrodes 1224, 1226 are formed of copper, copper-steel bimetal, or any other combination of metals in bimetal composition.

In some embodiments and as illustrated in FIG. 30, each of the SG electrodes 1224, 1226 has a bimetal construction including an inner layer 1227A of a first metal and an outer layer 1227B of a second metal different than the first metal. In some embodiments, the inner layers 1227A are formed of copper and the outer layers 1227B are formed of steel. In some embodiments, each layer 1227A, 1227B runs the full length of the electrode 1224, 1226.

The containment walls 1228 may be formed of any suitable material. In some embodiments, the containment walls 1228 are formed of ceramic, polymer or plastic, insulating paper, ferromagnetic material insulated by paper, plastic or ceramic, or ferromagnetic material coated with a polymer or plastic.

With reference to FIGS. 27, 32 and 33, the arc chute 1240 is located at the outer end 1230B of the horn spark gap assembly 1222 at, in or proximate the extinguishing region 1232E. The arc chute 1240 includes a set 1243 or series of electrically conductive arc, chute or deion electrodes or plates 1242A, 1242B supported by electrically insulating supports 1246. The deion plates 1242A, 1242B are separated by deion plate arc gaps 1244. The deion plates 1242A, 1242B exposed side and end faces 1247.

The arc chute 1240 differs from the arc chute 140 in that the arc chute 1240 includes alternating first deion plates 1242A and second deion plates 1242B having different configurations from one another, as shown in FIGS. 32 and 33. More particularly, the plates 1242A each have a slot 1245A defined therein having a first shape or profile. The plates 1242B each have a slot 1245B defined therein having a second shape or profile that is different from the first profile.

The housing 1202 may be formed of any suitable electrically insulating material (e.g., an insulating polymer). The module housing 1202 includes a series of vent slots 1202B defined therein. The module housing 1202 further includes a first arc gas recirculation port 1202E, a second arc gas recirculation port 1202F, an arc gas recirculation channel 1202G, and opposed intake ports 1202H defined therein.

The supports 1246 hold the deion plates 1242A, 1242B together as a single unit, and the unit is seated in the housing 1202 such that features of the housing 1202 positively position the unit and the plates 1242A, 1242B relative to the housing 1202. The deion plates 1242A, 1242B are thereby positioned such that each deion plate arc gap 1244 is aligned with a respective set of the vent slots 1202B (i.e., a front vent slot and a pair of opposed side vent slots).

In some embodiments, the deion plate side and end edges 1247 are positioned substantially flush with the mating or facing surfaces 1202C of the module housing 1202.

The SG trigger circuit 1250 includes the spark gap trigger region 1232T, the ignition region 1232I, a semiconductive trigger member 1254, a trigger electrode 1258, a trigger varistor (e.g., MOV) 1268, a trigger GDT 1267, and an electrically conductive (e.g., copper) contact spring 1264. In alternative embodiments, a resistor is used in place of the MOV 1268.

The trigger electrode 1258 (FIG. 28) may be formed of any suitable electrically conductive metal (e.g., copper). The

trigger electrode 1258 includes a contact portion 1258C, a trigger portion 1258T, and a terminal end 1258E.

The trigger GDT 1267 has opposed contact terminals 1267A, 1267B.

The semiconductive member 1254 is positioned in the groove 1229 adjacent the trigger gap 1232T and interposed between the SG electrodes 1224, 1226.

The MOV 1268, the GDT 1267, the trigger electrode 1258, and the semiconductive member 1254 are connected in electrical series between the electrode 1224 and the electrode 1226 as represented schematically in FIG. 24. The contact spring 1264 serves to take up manufacturing tolerances.

The semiconductive trigger member 1254 (FIGS. 28-31) extends from a first end face 1254A to an opposing second end face 1254B. The first end face 1254A electrically engages or contacts the trigger electrode 1258T. The second end face 1254B electrically engages or contacts the electrode trigger section 1226T.

The semiconductive trigger member 1254 may be formed of any suitable semiconductive material. In some embodiments, the semiconductive member 1254 is formed of a material as described herein for the semiconductive member 154.

In some embodiments, the semiconductive member 1254 is formed of a semiconductive ceramic. In some embodiments, the semiconductive member 1254 is formed of ZnO (zinc oxide) or BaTiO₃ (Barium titanate) or SiC (silicon carbide), with different dopants (oxides, metals).

In some embodiments, the semiconductive material of the semiconductive member 1254 has a specific electrical resistance in the range of from 1 to 10,000 Ωcm , and a dielectric constant in the range of from 1 to 10,000. The semiconductive material may have a positive or negative temperature coefficient.

In some embodiments, the semiconductive trigger member 1254 has a thickness T15 (FIG. 31) of at least 0.2 mm and, in some embodiments, of at least 0.5 mm. In some embodiments, the thickness T15 is in the range of from about 0.2 mm to 1.5 mm.

The width W16 (FIG. 29) of the trigger portion 1258T is less than the width W15 (FIG. 29) of the trigger member 1254 so that lateral strips of portions 1254AA of the first end face 1254A are exposed and face opposing surfaces of the SG electrode 1224.

The semiconductive trigger member 1254 is seated in the angled groove 1229. The trigger member 1254 has a terminal side face 1255 adjacent the end wall 1229E of the groove 1229. As shown in FIG. 31, a first portion 1255L of the terminal side face 1255 is embedded in the groove 1229 and second portion 1255U of the terminal side face 1255 projects inwardly beyond the inner surface 1226A of the electrode 1226 toward the electrode 1224. In some embodiments, the terminal end 1258E of the trigger electrode 1258 is coterminous with the terminal side face 1255.

In some embodiments, the height H13 (FIG. 31) of the second portion 1255U above the inner surface 1226A is at least 0.05 mm or 0.1 times the thickness T15 of the trigger member 1254. In some embodiments, the height H13 is in the range of from about 0 to 1 mm or 1 times the thickness T15.

In some embodiments, the depth D13 (FIG. 31) of the first portion 1255L into the electrode 1226 is at least 0.3 mm. In some embodiments, the depth D13 is at least 0.5 times the thickness T15 of the trigger member 1254. In some embodiments, the depth D13 is in the range of from about 0 to the thickness T15 of the trigger member 1254.

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With reference to FIGS. 25 and 30, the housing 1202 includes electrically insulating features 1202J interposed between the trigger electrode 1258 and the first SG electrode 1224. In some embodiments and as illustrated, the semiconductive trigger member 1254 and the trigger electrode 1258 project forwardly beyond the housing features 1202J so that an exposed portion of the trigger electrode 1258 faces the first SG electrode 1224 and the end faces 1254E, 1258E are located a projection distance L17 from the features 1202J.

With reference to FIG. 31, the trigger region 1232T of the spark gap 1230 includes a first trigger spark gap 1234A and a second trigger spark gap 1234B. The first trigger spark gap 1234A is defined between the second SG electrode 1226 and the terminal end 1258E of the trigger electrode 1258. The second trigger spark gap 1234B (i.e., an air gap) is defined between the first SG electrode 1224 and the terminal end 1258E of the trigger electrode 1258.

In some embodiments, the first trigger spark gap 1234A has a width W18 (FIG. 31) of at least 0.1 mm and, in some embodiments, of at least 1 mm. In some embodiments, the width W18 is in the range of from about 0 mm to 1 mm. In some embodiments, the width W18 is in the range of from 0 mm to the thickness T15.

In some embodiments, the second trigger spark gap 1234B has a width W19 (FIG. 31) of at least 0.5 mm and, in some embodiments, of at least 1 mm. In some embodiments, the width W19 is in the range of from about 0.5 mm to 1.5 mm. In some embodiments, the width W19 is in the range of 0.2 times the width W12 of the spark gap 1230 in the ignition region 1232I to 0.8 times the width W12.

The widths W18, W19 are each less than the width W12 of the ignition region 1232I.

The spark gap 1230 has a prescribed threshold flashover voltage. The prescribed threshold flashover voltage is the minimum voltage that will cause a flashover between the first SG electrode 1224 and the second SG electrode 1226, wherein this flashover between the SG electrodes 1224, 1226 is initiated by the trigger circuit 1250 at the trigger region 1232T (as discussed in more detail below). In some embodiments, the prescribed trigger threshold flashover voltage is less than 1500V. In some embodiments, the prescribed threshold flashover voltage is in the range of from about 300V to 1500V. When a voltage is applied across the terminals 1204A, 1206A that is less than the threshold flashover voltage, the applied voltage will not ignite or initiate arc flashover between the SG electrodes 1224, 1226. As discussed below, when a voltage is applied across the terminals 1204A, 1206A that is greater than or equal to the threshold flashover voltage, the applied voltage will ignite or initiate arc flashover between the SG electrodes 1224, 1226.

The SPD module 1200 may operate as follows in service.

According to some embodiments of the inventive concept, the SPD module 1200 is configured to operate under two different conditions: 1) normal (stand by) operation; and 2) an overvoltage or current surge event in which the SPD module 1200 is designed to shunt an SPD surge impulse current to ground.

The SPD module 1200 is designed to shunt an SPD surge impulse current to ground in response to an overvoltage or current surge event.

The SPD module 1200 is configured to operate in three alternative modes: 1) a standby mode; 2) a surge current mode; and 3) a follow current extinguishing mode.

The terminal 1204A is electrically connected to the Line (L) of the circuit 10, and the terminal 1206A is electrically connected to the Ground (G) of the circuit 10 (FIG. 1).

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During normal operation, the OPC 1210 practically acts as an insulator. The voltage applied across the spark gap 1230 is insufficient to initiate a spark across the spark gap 1230. Also, the voltage applied is insufficient to initiate current through the trigger MOV 1268 and the trigger GDT 1267 of the trigger circuit.

When an overvoltage or current surge event (e.g., a transient power surge) applies a surge impulse current to the SPD circuit 1201, the OPC 1210 will temporarily go to a low impedance state (e.g., effectively becoming a short circuit). The OPC 1210 is designed to shunt the surge impulse current associated with such events to ground to protect sensitive equipment. The SPD surge impulse current may be on the order of tens of kA, but will typically last only a short duration (in the range of from about tens of microseconds to a few milliseconds).

During the surge event, the voltage applied across the trigger region 1232T of the spark gap 1230 and the trigger components/member (trigger GDT 1268, trigger MOV 1267 and semiconductive member 1254) by the surge event exceeds the prescribed threshold flashover voltage of the trigger region 1232T assisted by the trigger circuit 1250. In response, the voltage across the trigger region 1232T initiates a first electrical arc flashover or trigger arc AT1 (FIG. 31) across the first trigger spark gap 1234A. That is, first arcing AT1 is generated between the trigger electrode 1258 and the second SG electrode 1226 at section 1226T.

The initiation of the first trigger arc AT1 is assisted by the semiconductive trigger member 1254 in response to the overvoltage developed across the first trigger spark gap 1234A. At the beginning of the surge current, the trigger MOV 1268 and the trigger GDT 1267 change their states to low impedance and current conduction occurs therethrough. At the beginning of the surge current, current conduction occurs through the bulk of the semiconductive member body and along the exterior surface (including the terminal side face 1255) of the semiconductive trigger member 1254. Very quickly thereafter (e.g., within less than 1 microsecond), the flashover of the first trigger arc AT1 occurs so that most of the surge current is bypassed through the arc column(s) established between the second SG electrode 1226 and the trigger electrode 1258. By diverting the current around the semiconductive trigger member 1254, degradation of the semiconductive trigger member 1254 is prevented or reduced.

Once formed, the first trigger arc AT1 quickly (e.g., within less than 1 microsecond) propagates to generate a second electrical arc flashover or trigger arc AT2 (FIG. 31) across the second trigger spark gap 1234B. That is, second arcing AT2 is generated between the trigger electrode 1258 and the first SG electrode 1224 at section 1224T.

Shortly after the start of the second trigger arc AT2 (e.g., within microseconds), the current flow triggers an electric arc flashover or surge arc AM (as illustrated in FIGS. 27 and 31) across the spark gap 1230 in the ignition region 1232I. The SG trigger circuit 1250 thereby ignites the horn spark gap assembly 1222 and the full surge current is thereby conducted through the spark gap assembly 1222 via the surge arc AM.

Once the surge arc AM is established, the voltage drop of the surge arc AM is less than knee voltage of the trigger MOV 1267. Thus, the trigger MOV 1267 reduces current through the trigger member 1254 below the level necessary to sustain the trigger GDT 1268 in conduction. Thus, any remaining current through the trigger member 1254 is cutoff.

The spark gap assembly 1220 is thereby triggered changing its state to low impedance. The surge current flows from

the terminal 1204A to the terminal 1206A through the spark gap 1230, thereby diverting the surge current to ground. During the surge event, the arcing or arc column AM across the spark gap 1230 may be located entirely or mainly in the arc ignition region 1232I or partially in the running region 1232R.

Once the surge arc AM is active, the current no longer flows through the trigger MOV 1268 and the trigger GDT 1267.

The SPD 1200 then enters its follow current mode. Once the surge current is diverted to the ground, the spark gap 1222 remains electrically conductive, causing additional follow current from the system to flow into the SPD 1200. This causes the arc column to move in an arc migration direction DA (from the end 1230A toward the end 1230B) along the arc running region 1232R of the spark gap 1222 to the arc extinguishing region 1232E. FIG. 27 illustrates the arc column in the arc running region 1232R as an electric arc AR, and the arc column in the arc extinguishing region 1232E as an electric arc AE.

In the arc extinguishing region 1232E, the follow current is redirected to (is conducted to, or flashes over to or arcs to) the deion plates 1242 from the SG electrode 1224. The current then flows through the set 1243 of deion plates 1242A, 1242B to the SG electrode 1226 by arcing AD between the deion plates 1242A, 1242B (FIG. 27).

Each deion plate arc creates a voltage drop between the SG electrodes 1224, 1226 and the terminals 1204A, 1206A. These voltage drops add together increasing the voltage between the SG electrodes 1224, 1226 to relatively high values, until the voltage across the arc chute 1240 is higher than the power system voltage. The voltage drop developed on the arc chute 1240 opposes the mains voltage thus the follow current is rapidly reduced and eventually extinguished.

These voltage drops also reduce the voltage across the spark gap 1222 to relatively low values, until the voltage across the spark gap 1222 is less than the ignition voltage necessary to sustain the arcing between the SG electrodes 1224, 1226. The voltage across the spark gap 1222 is then also less than necessary to trigger flashover in the trigger region 1232T and the ignition region 1232I. The spark gap assembly 1222 is thereby opened at the spark gap 1230 and the follow current through the spark gap assembly 1222 and the SPD module 1200 is cut off or interrupted. The SPD module 1200 returns to its standby mode.

Thus, it will be appreciated that the electrical arc flashover AT1, AT2 triggers the electric arc flashover AM, which during follow current expands and propagates to the arc chute 1240. The expansion and propagation are driven by electromagnetic and acoustical forces. Once the arc AE enters the arc chute 1240, it is divided or split into a plurality of smaller arcs in the arc chute 1240 and cooled down until the arc is eventually extinguished. This causes the SPD module 1200 to return to its high impedance (standby) mode.

The trigger circuit 1250 (including the semiconductive trigger member 1254) functions as a spark gap trigger that assists in initiating the flashover between the SG electrodes 1224, 1226. The trigger circuit 1250 permits a minimum separation distance W12 between the electrodes 1224, 1226 that is large enough to prevent melting and soldering of the electrodes 1224, 1226 during surges. Because the trigger circuit 1250 is provided, the threshold flashover voltage of the gap 1230 is less than the voltage that would be required to initiate the flashover across the gap 1230 in the absence of the trigger circuit 1250 and the semiconductive trigger

member 1254. By reducing the minimum required flashover voltage, the trigger circuit 1250 and the semiconductive trigger member 1254 lower the protection level of the SPD electrical circuit 1201.

The horn shape of the spark gap 1230 facilitates the path of the arc AM. The geometry of the SG electrodes guides the arc AM from the trigger region 1232T to the extinguishing region 1232E. The containment walls 1228 firmly restrict or contain the path of the arcing in the lateral directions.

The MOV 1268 and GDT 1267 are provided in front of the trigger member 1254 because the trigger member 1254 has some resistance and thus will conduct the current on nominal voltage. By including MOV 1268 and GDT 1267, the OPC 110 is made leakage current free (and, thus, the SPD module 1200 is leakage free).

Additionally, the MOV 1268 and GDT 1267 prevent and cut off any continuation currents through or along the semiconductive trigger member 1254 after the surge arc AM is initiated.

While the trigger circuit 1250 has been described including a trigger MOV 1268 and a trigger GDT 1267, in other embodiments the trigger circuit may include other voltage-switching/limiting components. For example, the alternative trigger circuit may include the trigger MOV 1268 without a trigger GDT or may include the trigger GDT 1267 without a trigger MOV.

The groove 1229 ensures that the flashover AT1 always occurs on the front end or side (i.e., on or proximate face 1229E) since the electrical path is longer on the back side of the trigger member 1254. Also, the groove 1229 prevents the movement of the trigger member 1254 in the "front" or forward direction (i.e., in the direction toward the ignition region 1232I).

The cross-over pattern of the slots 1245A, 1245B of the deion plates 1242A, 1242B influences the movement or path of the individual arcs AD between the adjacent plates 1242A, 1242B so that it forces the arcs AD into a spiral "tornado" or helical path. The arc AD between the first two plates 1242A, 1242B goes left, the arc AD between the second and third plates 1242B, 1242A goes right, the arc AD between the third and fourth plates 1242A, 1242B goes left, and so forth. As a result, the paths of the arcs AD and the combined lengths of the arcs AD are made longer. Also, this prevents or inhibits re-ignitions and rebounding of the arc back to the running area 1232R of the spark gap 1230.

The SPD module 1200 also provides management for gases generated by arcing events. As discussed above, the gaps 1244 between the deion plates 1243 are each aligned with a respective set of exhaust slots 1202B in the module housing 1202. At least a portion of the pressurized arc gases generated by the arc AM flows between the deion plates 1242A, 1242B and the out through the exhaust slots 1202B. The arc gases are thereby expelled from the module housing 1202 so that back pressure on the arc gases in the spark gap 1220 is reduced or eliminated. Such back pressure may otherwise impede or interfere with the intended performance of the spark gap 1220.

As discussed above, in some embodiments, all of side and front the edge faces 1247 of the deion plates 1242A, 1242B fit substantially flush or in abutment with the facing surfaces of the plastic module housing 1202. This is needed or beneficial because otherwise an arc could flashover individual plates 1242A, 1242B creating one single long arc. That would diminish the follow through current extinguishing function of the spark gap assembly 1220. Adding insu-

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lating walls in tight contact with the plates **1242A**, **1242B** prevents this and gases are expelled without flashover across the arc chute **1240**.

During an arcing event, a portion of the arc gases generated thereby flow from the distal end **1230B** of the spark gap **1230** (e.g., in or adjacent the arc chute **1240**) back to the ignition region **1232I**. More particularly, pressurized gas **F** (FIG. **25**) flows sequentially through the arc gas recirculation port **1202E**, the recirculation channel **1202G** and the arc gas recirculation port **1202H** to the ignition region **1232I**. The recirculation can improve the follow through extinguishing behavior of the SPD module **1200**.

Ambient air is drawn into the ignition region **1232I** through the intake ports **1202H** by the negative pressure generated during an arcing event. The inrush of surrounding air into the spark gap can provide a pressure release that improves the follow current extinguishing behavior.

With reference to FIGS. **34-39**, an SPD assembly **20** according to some embodiments is shown therein. The SPD assembly **20** includes three of the SPD modules **1200** and a base module **22**. In other embodiments, the SPD assembly **20** includes SPD modules of other designs as disclosed herein (e.g., the SPD module **100**) in place of the SPD modules **1200**.

The SPD assembly **20** forms or embodies an SPD assembly circuit **41** as illustrated schematically in FIG. **35**. In FIG. **35**, the SPD assembly **20** is schematically shown installed in a representative system coupled with three power lines **L1**, **L2**, and **L3** (corresponding to a three-phase electrical power system), a neutral line **N**, and ground or protected earth **PE**.

The illustrated SPD assembly **20** is configured to be mounted on the DIN (Deutsches Institut für Normung e.V.) rail and to protect a three-phase system using a “3+1” protection configuration. However, other mounting and protection configurations may be provided in accordance with some embodiments of the technology. The illustrated SPD assembly **20** is configured to electrically and mechanically connect with electrical cables corresponding to lines **L1**, **L2**, **L3**, **N**, and **PE**, respectively.

The base module **22** includes a base housing **30**, a thermal indicator mechanism **50**, and a base electrical connection system **40**.

The base housing **30** includes a base member **32** and a cover **34**. The housing parts **32**, **34** may be formed of any suitable material or materials. In some embodiments, the housing parts **32**, **34** are formed of a rigid polymeric material or metal (e.g., aluminum). Suitable polymeric materials may include polyamide (PA), polypropylene (PP), polyphenylene sulfide (PPS), or ABS, for example.

The base member **32** includes a DIN rail receiver channel **32A**, cable ports **32B**, and integral partition walls **32P**.

The cover **34** includes partition walls **34P** and a front window **34A**.

When the cover **34** is mounted on the base member **32**, the parts **32**, **34** collectively define an internal cavity **36**. The partition walls **32P** and **34P** mate or align relative to one another to form three module subchambers **37** (FIGS. **36** and **37**). Each SPD module **1200** is seated or disposed in a respective one of the subchambers **37** (FIG. **38**).

The electrical connection system **40** (FIG. **38**) includes cable clamp connectors **CL1**, **CL2**, **CL3**, **CN**, **CP** (each positioned at respective cable port **32B**), a neutral busbar **42** and the GDT **46**. The neutral busbar **42** has a neutral connection end **42A** and a module connection section **42B**. The connector **CN** includes an extension leg **44** having a distal section **44A**.

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The terminals **1204A** of the SPD modules **1200** are each connected to a respective one of the connectors **CL1**, **CL2**, **CL3**. The terminals **1206A** of the SPD modules **1200** and neutral connector **CN** are each connected to neutral busbar **42**. The PE connector **CP** is connected to the neutral busbar **42** through the GDT **46**.

The thermal indicator mechanism **50** (FIG. **39**) includes the extension leg **44**, an indicator member or slider **52**, an indicator spring **54**, a melttable member or peg **56**, the window **34A**, an electrical switch **58** (e.g., a microswitch), and a remote signal connector **58B**. The remote signal connector **58B** is connected to the electrical switch **58** by a wire **58A**.

The melttable peg **56** is secured to the slider **52** (e.g., by friction fit). The spring **54** is compressed in a ready position (as shown in FIG. **39**) and tends to force the slider **52** in an indication direction **DI**. The melttable peg **56** bears against the distal section **44A** and the extension leg **44** acts as stop that prevents the slider **52** from moving in the direction **DI**.

The partition walls **32P**, **34P** partition the SPD modules **1200** in the subchambers **37** from one another. The partition walls **32P**, **34P** are interposed directly between the vent slots **1202B** of adjacent SPD modules **1200**. This partitioning prevents the occurrence of undesired short circuits between the electrical phases **L1**, **L2**, **L3** by hot gases expelled by the SPD modules **1200** during surge and follow current extinguishing events. It will be appreciated that the subchambers **37** may not be fluidly sealed from one another, but that the placement and configuration of the partition walls **32P**, **34P** will prevent gas flows between the subchambers **37** sufficient to create said undesired short circuits.

The thermal indicator mechanism **50** serves to provide a local alert and a remote alert in the event the SPD assembly **20** fails or reaches an end-of-life state. A failure of the SPD assembly **20** may occur when one of more of the SPD modules **1200** short-circuits or overheats. In this case, heat from the SPD module(s) **1200** is conducted to the melttable peg **56** through the terminal **1206A**, the neutral busbar **42**, and the extension leg **44**. Once sufficient heat has been transferred to the melttable peg **56**, the melttable peg **56** will melt or soften until it is no longer rigid enough to hold the slider **52** against the force of the spring **54**. The slider **52** is thereby released to translate in the direction **DI** from the ready position to an alerting position. The slider **52** presses the switch **58**, which generates a corresponding signal to the remote monitor connector **58B**. A remote monitoring system may be connected to the remote monitor connector **58B** to detect the failure condition. In the ready position, an indicator section **52A** of the slider **52** occupies the window **34A**. In the alerting position, the indicator section **52A** is displaced from the window **34A**, thereby providing a local, visual indication that the SPD assembly **20** has failed.

In other embodiments, the SPD module **1200** is modified to further include an active voltage-switching or active voltage-limiting component such as a varistor or GDT in electrical series with the spark gap assembly **1220**.

The terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. As

used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Like reference numbers signify like elements throughout the description of the figures.

It will be understood that, although the terms “first,” “second,” etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. Thus, a first element could be termed a second element without departing from the teachings of the inventive subject matter.

Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this inventive concept belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and this specification and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

Many alterations and modifications may be made by those having ordinary skill in the art, given the benefit of present disclosure, without departing from the spirit and scope of the invention. Therefore, it must be understood that the illustrated embodiments have been set forth only for the purposes of example, and that it should not be taken as limiting the invention as defined by the following claims. The following claims, therefore, are to be read to include not only the combination of elements which are literally set forth but all equivalent elements for performing substantially the same function in substantially the same way to obtain substantially the same result. The claims are thus to be understood to include what is specifically illustrated and described above, what is conceptually equivalent, and also what incorporates the essential idea of the invention.

What is claimed is:

1. A surge protective device comprising:
 - a first electrical terminal;
 - a second electrical terminal; and
 - an overvoltage protection circuit connected between the first electrical terminal and the second electrical terminal, the overvoltage protection circuit including:
 - a spark gap assembly between the first electrical terminal and the second electrical terminal, the spark gap assembly including:
 - a first spark gap (SG) electrode and a second SG electrode defining a spark gap therebetween; and
 - a trigger circuit operative to ignite a main electric arc between the first and second SG electrodes across the spark gap;
- wherein the trigger circuit includes:
 - a groove defined in the second SG electrode; and
 - a trigger member disposed in the groove, wherein the trigger member is operative to assist formation of a trigger arc; and
- wherein the spark gap assembly is a horn spark gap assembly.
2. The surge protective device of claim 1 wherein the trigger member is a semiconductive trigger member.
3. The surge protective device of claim 2 wherein the semiconductive trigger member is formed of a semiconductive ceramic.
4. The surge protective device of claim 3 wherein the semiconductive ceramic is selected from the group consisting of zinc oxide, barium titanate, and silicon carbide.

5. The surge protective device of claim 1 wherein the trigger circuit includes a trigger varistor, a trigger gas discharge tube, and/or a trigger resistive element in electrical series with the trigger member and in electrical parallel with the spark gap.

6. The surge protective device of claim 1 wherein the trigger circuit includes a trigger electrode in electrical contact with the trigger member and connecting the trigger member in electrical parallel with first SG electrode.

7. The surge protective device of claim 6 wherein the trigger circuit includes a trigger varistor, a trigger gas discharge tube, and/or a trigger resistive element in electrical series with the trigger electrode and in electrical parallel with first SG electrode.

8. The surge protective device of claim 1 including an active voltage-switching/limiting component in electrical series with the spark gap assembly.

9. The surge protective device of claim 8 wherein the active voltage-switching/limiting component includes a varistor or a gas discharge tube.

10. The surge protective device of claim 1 wherein the trigger circuit has a trigger threshold flashover voltage for initiating electrical flashover between the first and second SG electrodes that is less than a threshold flashover voltage that would initiate electrical flashover between the first and second SG electrodes in the absence of the trigger circuit.

11. The surge protective device of claim 1 wherein: the surge protective device is a surge protective device module including a spark gap module housing; and the spark gap assembly is disposed in the spark gap module housing.

12. The surge protective device of claim 11 wherein: the trigger member is located at a first end of the spark gap; and

the spark gap module housing includes an arc gas recirculation channel configured to direct a flow of arc gas from a second end of the spark gap opposite the first end of the spark gap to an ignition region of the spark gap adjacent the trigger member.

13. The surge protective device of claim 11 wherein the spark gap module housing includes an intake port configured to direct a flow ambient air into an ignition region of the spark gap adjacent the trigger member.

14. The surge protective device of claim 1 wherein: the spark gap assembly includes an arc chute; the arc chute includes a set of deion plates including: a plurality of first deion plates each having a first slot having a first profile; and a plurality of second deion plates each having a second slot having a second profile different than the first profile; and

the first and second deion plates are arranged in spaced apart relation and in alternating series along a chute axis such that the first and second deion plates define a series of arc chute spark gaps extending along the chute axis.

15. The surge protective device of claim 1 wherein: the overvoltage protection circuit includes an active voltage-switching/limiting component;

the spark gap assembly is in electrical series with the active voltage-switching/limiting component between the first electrical terminal and the second electrical terminal; and

the spark gap assembly includes an arc chute.

16. The surge protective device of claim 1 wherein: the trigger member is located at a first end of the spark gap; and

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the spark gap assembly includes an arc chute located at an opposing second end of the spark gap.

17. A surge protective device comprising:

a first electrical terminal;

a second electrical terminal; and

an overvoltage protection circuit connected between the first electrical terminal and the second electrical terminal, the overvoltage protection circuit including a spark gap assembly including:

a horn spark gap assembly; and

an arc chute;

wherein:

the arc chute includes a set of deion plates including:

a plurality of first deion plates each having a first slot having a first profile; and

a plurality of second deion plates each having a second slot having a second profile different than the first profile; and

the first and second deion plates are arranged in spaced apart relation and in alternating series along a chute axis such that the first and second deion plates define a series of arc chute spark gaps extending along the chute axis.

18. A surge protective device comprising:

a first electrical terminal;

a second electrical terminal; and

an overvoltage protection circuit connected between the first electrical terminal and the second electrical terminal, the overvoltage protection circuit including:

a spark gap assembly between the first electrical terminal and the second electrical terminal, the spark gap assembly including:

a first spark gap (SG) electrode and a second SG electrode defining a spark gap therebetween; and

a trigger circuit operative to ignite a main electric arc between the first and second SG electrodes across the spark gap;

wherein the trigger circuit includes:

a groove defined in the second SG electrode; and

a trigger member disposed in the groove, wherein the trigger member is operative to assist formation of a trigger arc; and

wherein the trigger circuit includes a trigger electrode in electrical contact with the trigger member and connecting the trigger member in electrical parallel with first SG electrode.

19. The surge protective device of claim **18** wherein the trigger circuit includes a trigger varistor, a trigger gas discharge tube, and/or a trigger resistive element in electrical series with the trigger electrode and in electrical parallel with first SG electrode.

20. The surge protective device of claim **18** wherein:

the trigger electrode is interposed between the trigger member and the first SG electrode;

the trigger electrode electrically connects the trigger member to the first electrical terminal in electrical parallel with the first SG electrode;

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the first SG electrode, the second SG electrode, and the trigger electrode define:

a first trigger spark gap between the second SG electrode and the trigger electrode; and

a second trigger spark gap between the trigger electrode and the first SG electrode; and

the spark gap assembly is configured to, in response to a surge impulse current, initiate a first trigger arc across the first trigger spark gap and thereafter a second trigger arc across the second trigger spark gap.

21. The surge protective device of claim **20** wherein:

the spark gap includes:

an ignition region; and

a trigger region including the first trigger spark gap and the second trigger spark gap and located between the trigger member and the ignition region;

the groove includes a base wall; and

the base wall slopes outwardly relative to the spark gap at an angle in a direction toward the ignition region.

22. A surge protective device comprising:

a first electrical terminal;

a second electrical terminal; and

an overvoltage protection circuit connected between the first electrical terminal and the second electrical terminal, the overvoltage protection circuit including:

a spark gap assembly between the first electrical terminal and the second electrical terminal, the spark gap assembly including:

a first spark gap (SG) electrode and a second SG electrode defining a spark gap therebetween; and

a trigger circuit operative to ignite a main electric arc between the first and second SG electrodes across the spark gap;

wherein the trigger circuit includes:

a groove defined in the second SG electrode; and

a trigger member disposed in the groove, wherein the trigger member is operative to assist formation of a trigger arc; and

wherein:

the surge protective device is a surge protective device module including a spark gap module housing; and

the spark gap assembly is disposed in the spark gap module housing.

23. The surge protective device of claim **22** wherein:

the trigger member is located at a first end of the spark gap; and

the spark gap module housing includes an arc gas recirculation channel configured to direct a flow of arc gas from a second end of the spark gap opposite the first end of the spark gap to an ignition region of the spark gap adjacent the trigger member.

24. The surge protective device of claim **22** wherein the spark gap module housing includes an intake port configured to direct a flow ambient air into an ignition region of the spark gap adjacent the trigger member.

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